

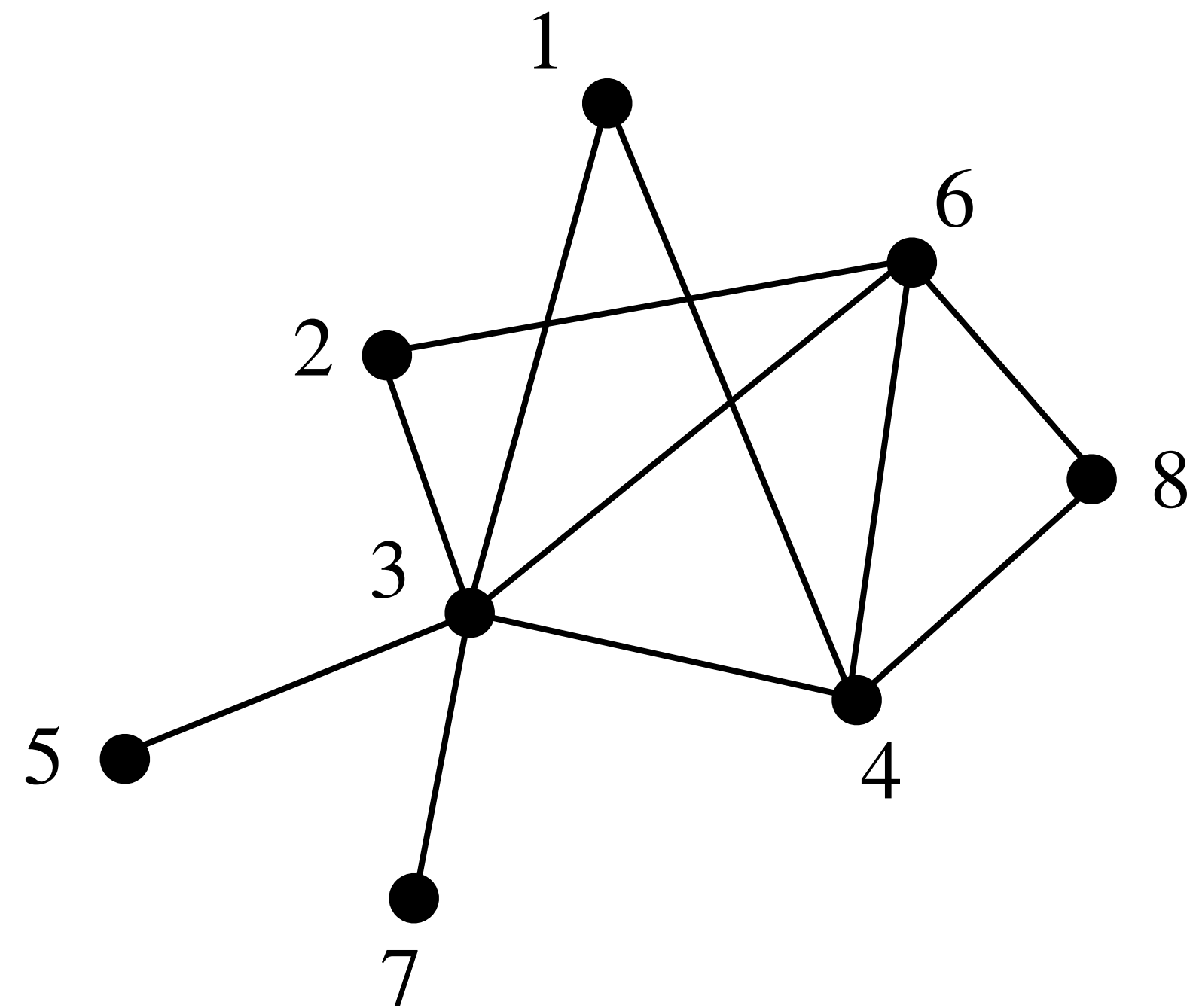
Lecture 18

Bipartite Maximum Matching

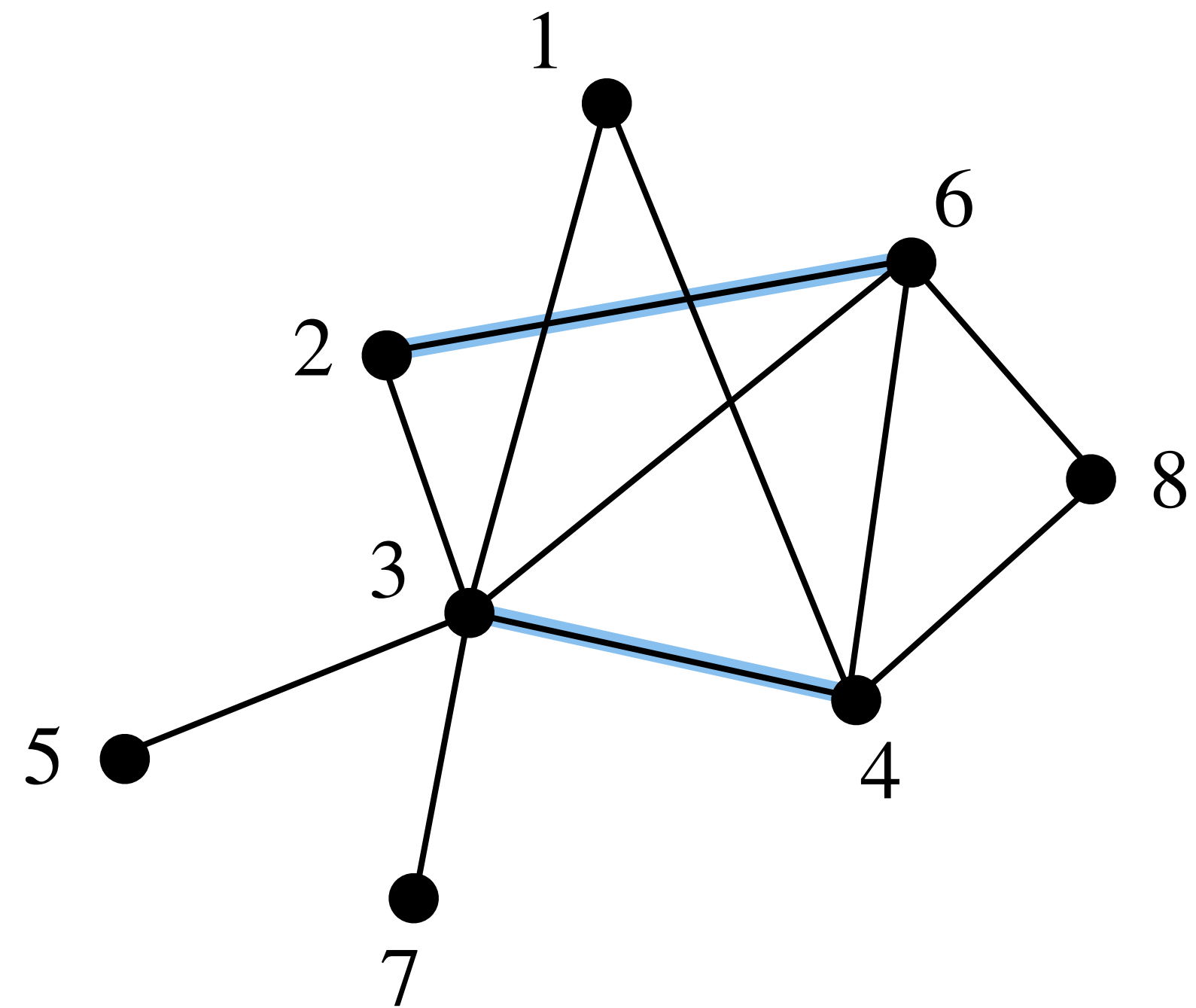
Source: Introduction to Algorithms, CLRS and Kleinberg & Tardos

Matching

Matching

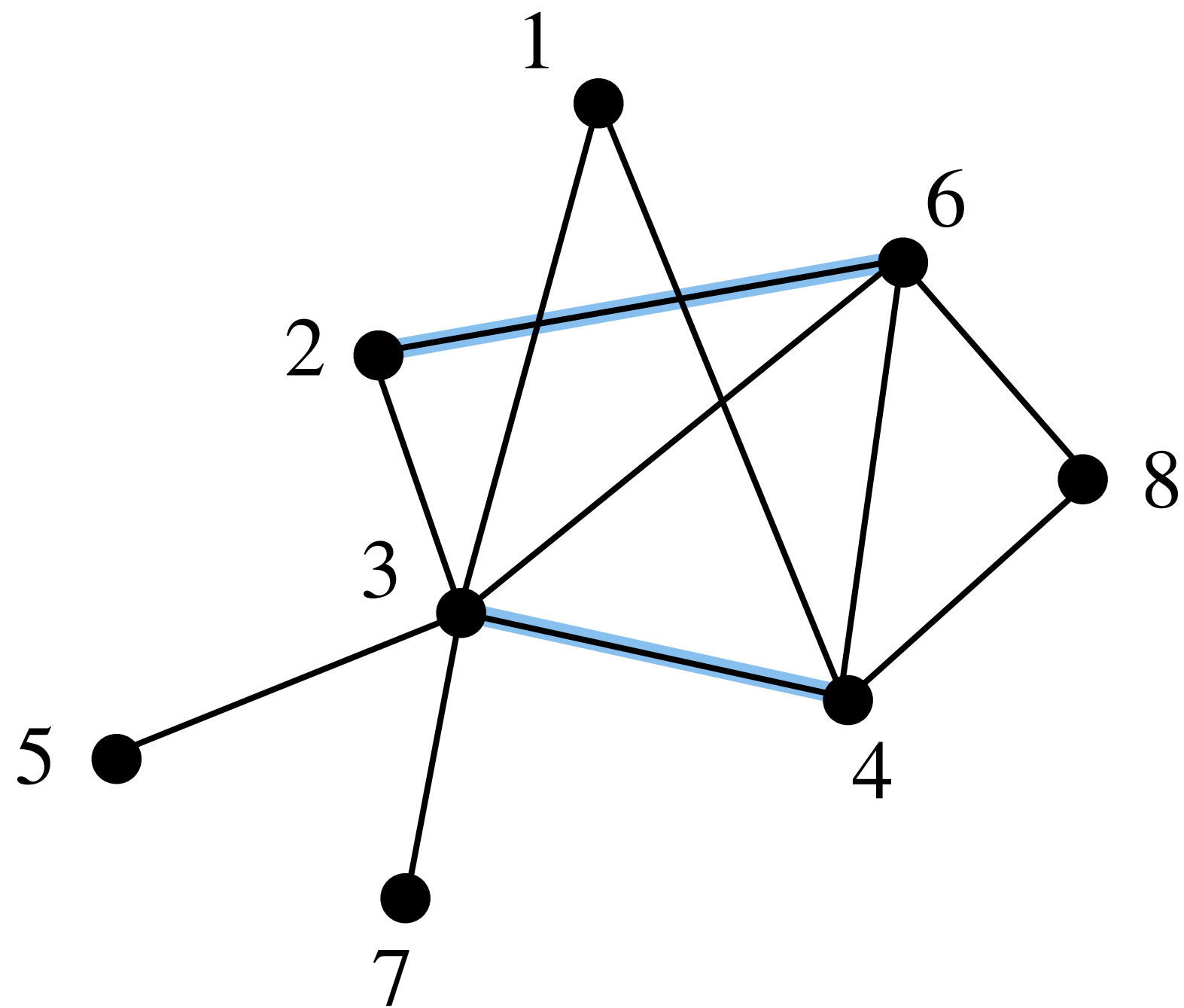


Matching



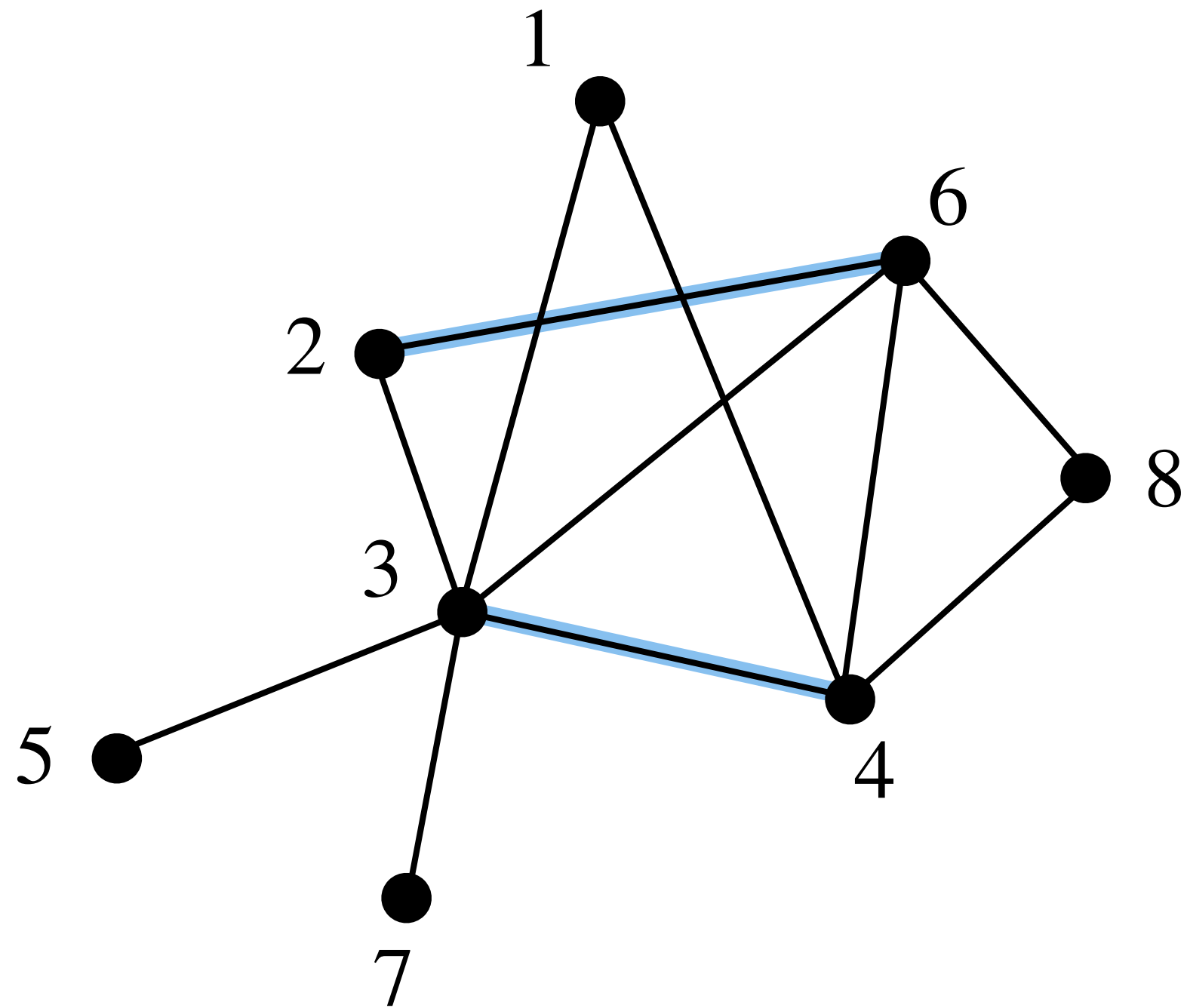
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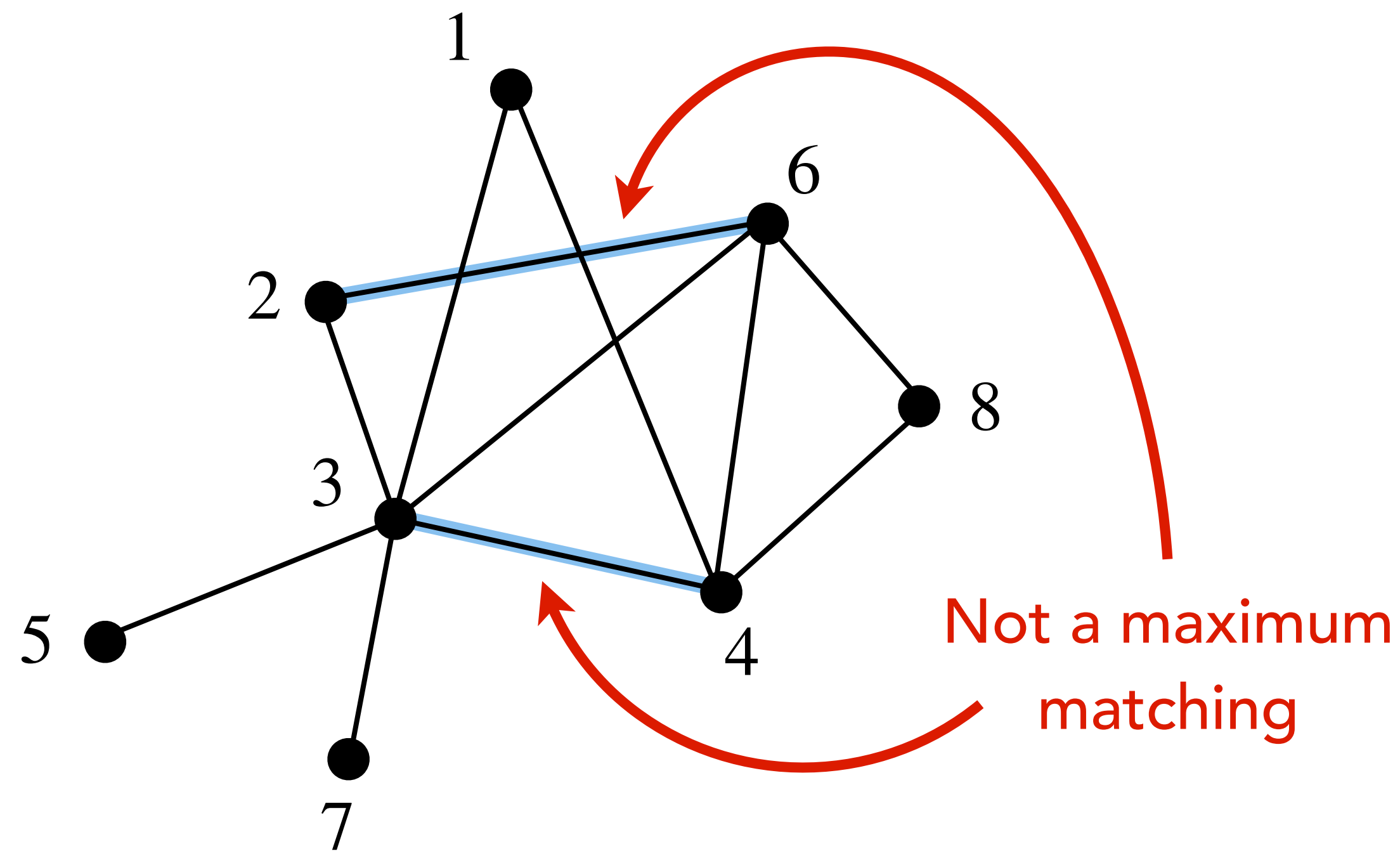
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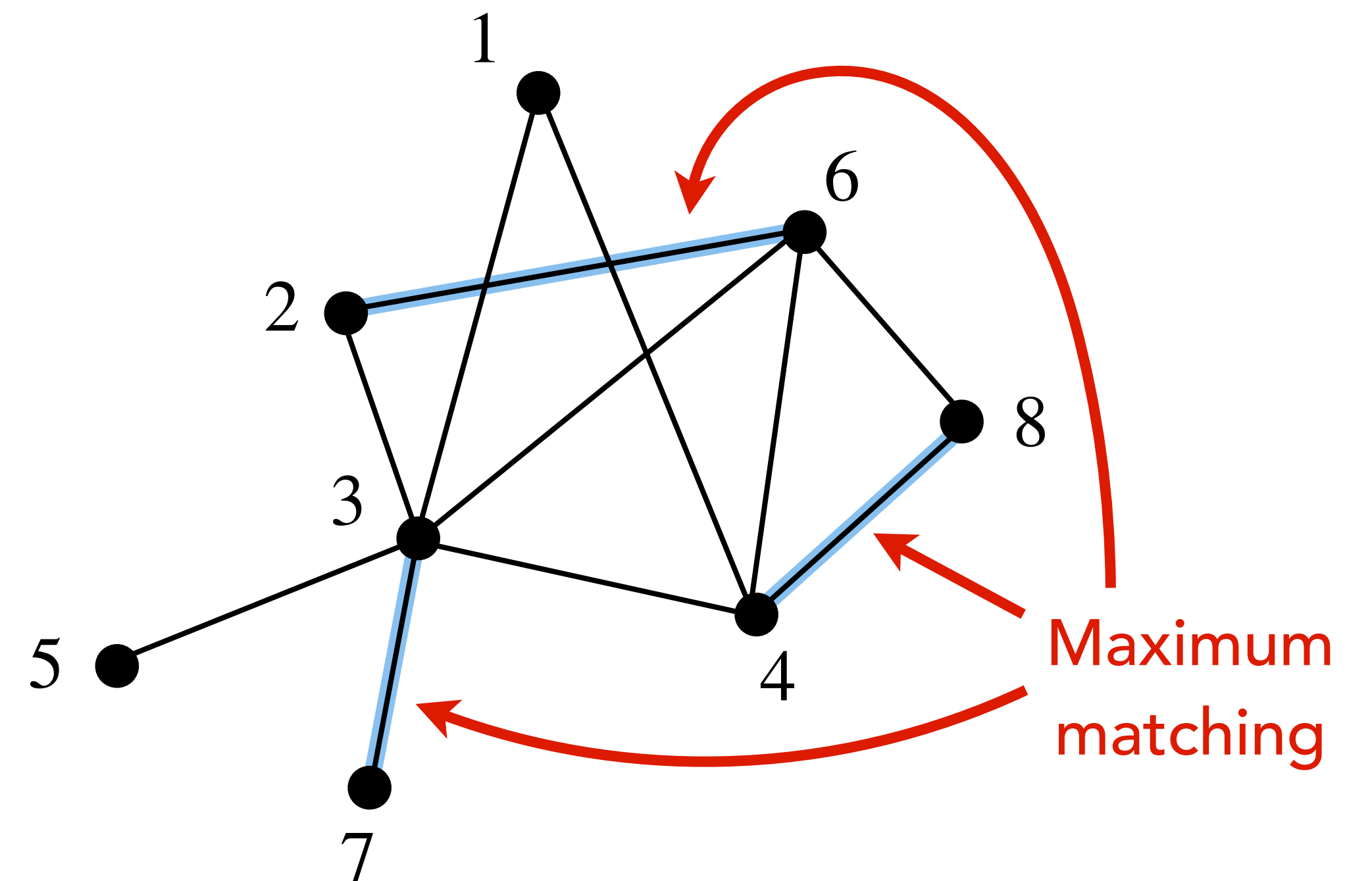
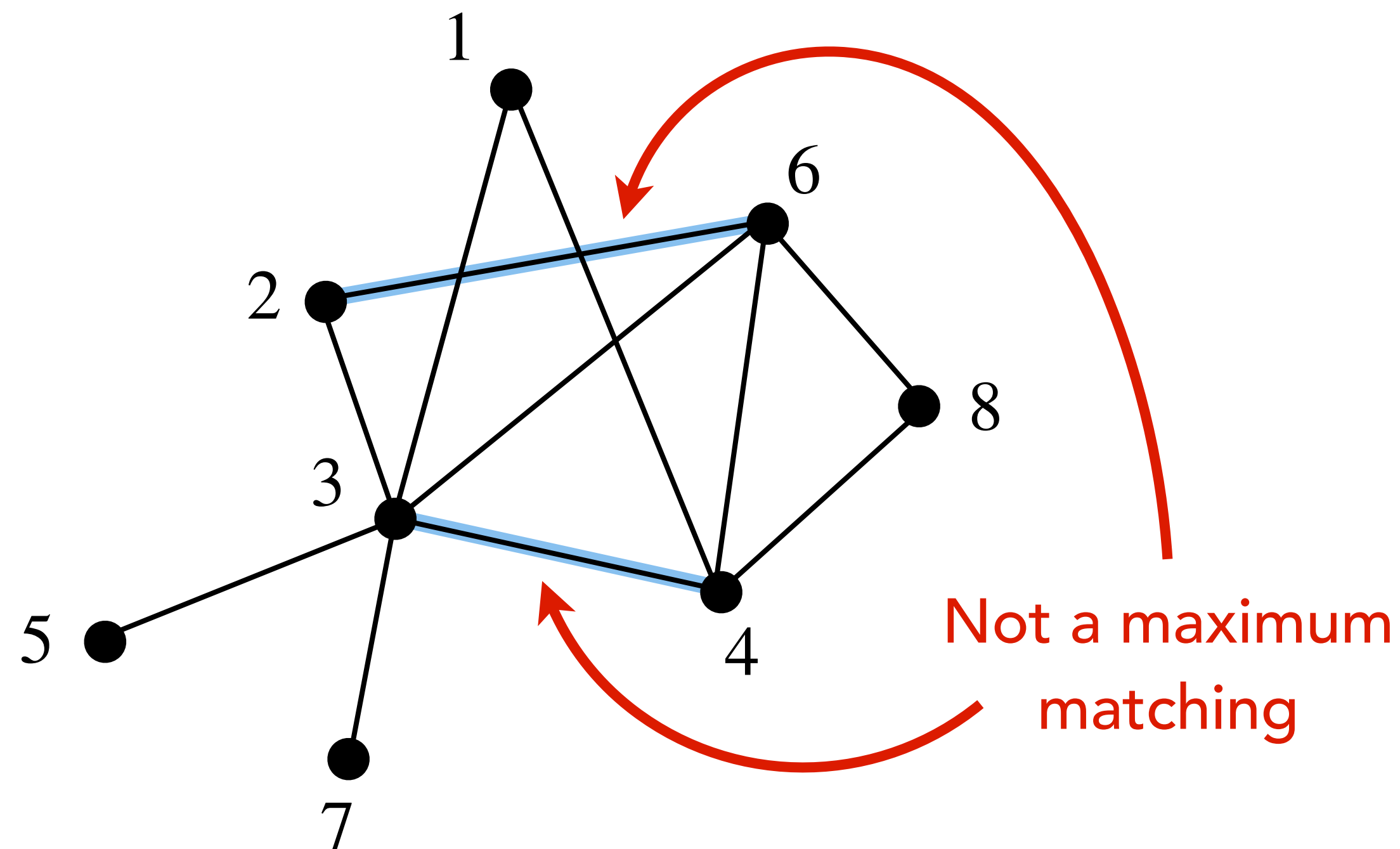
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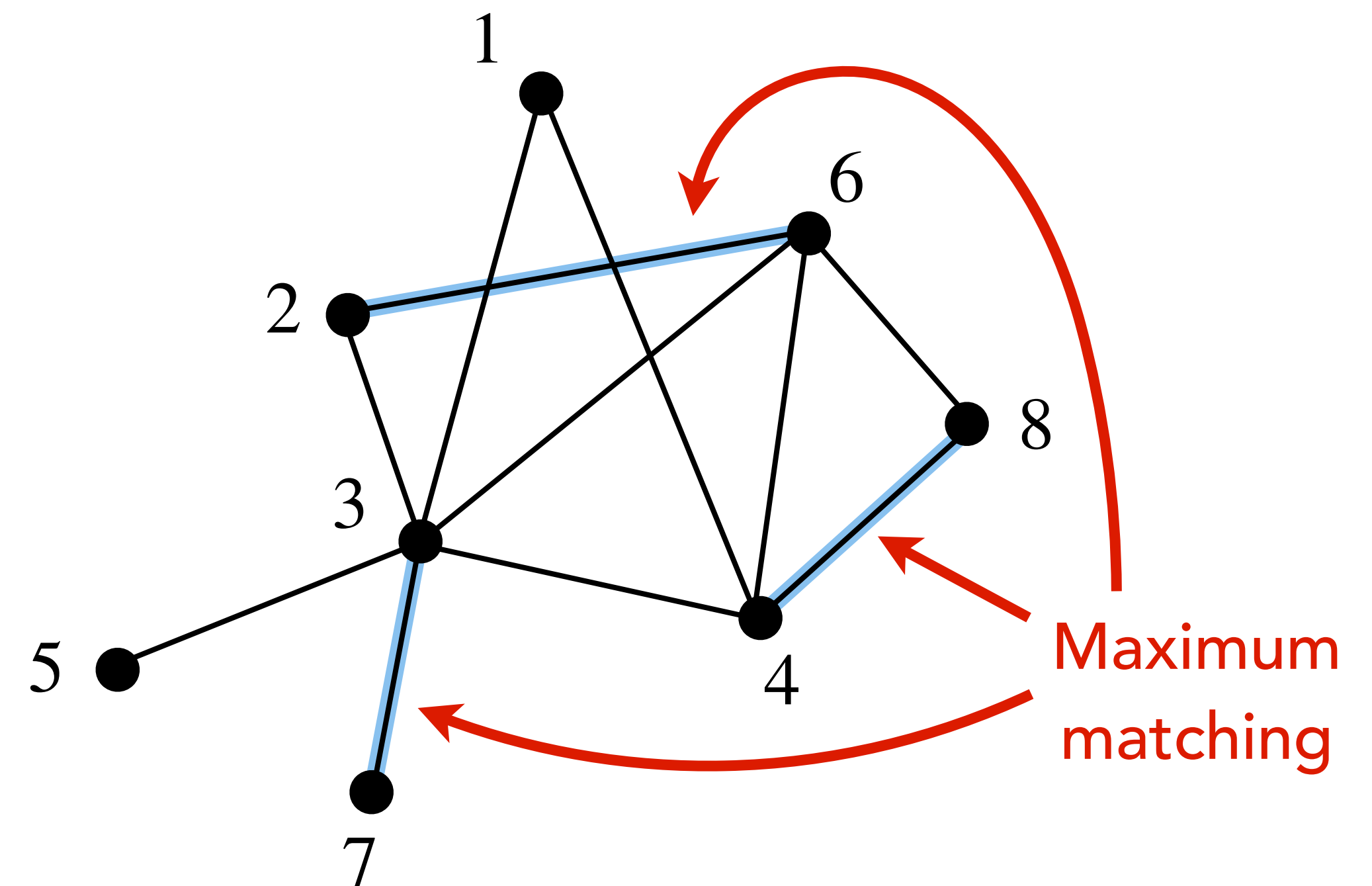
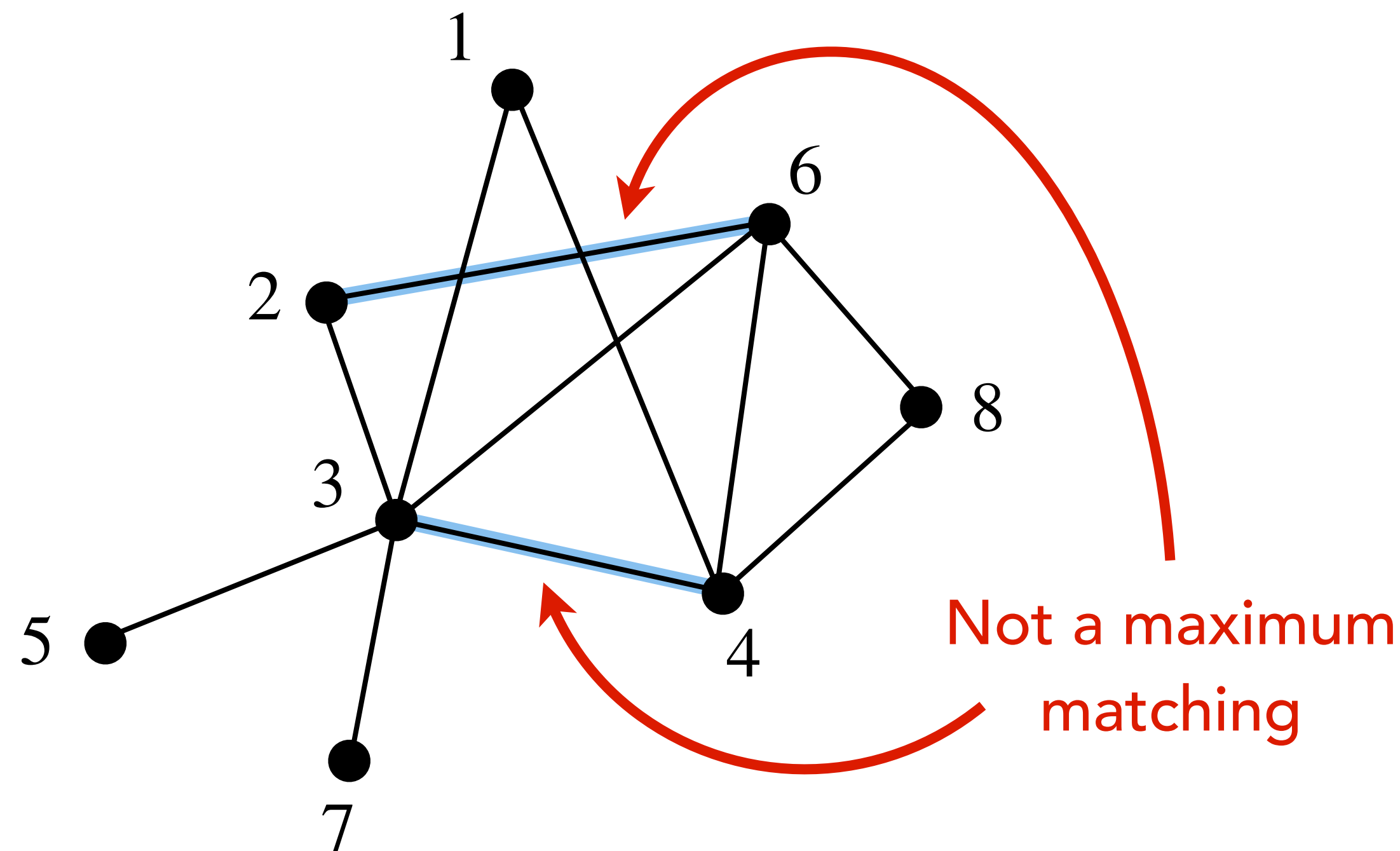
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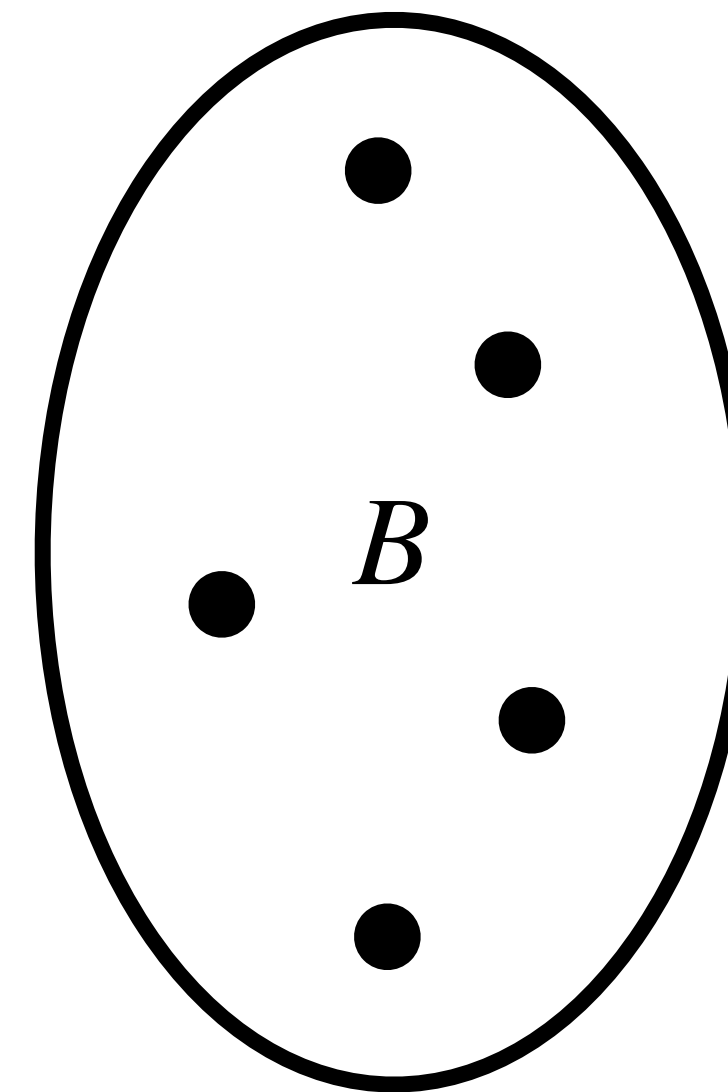
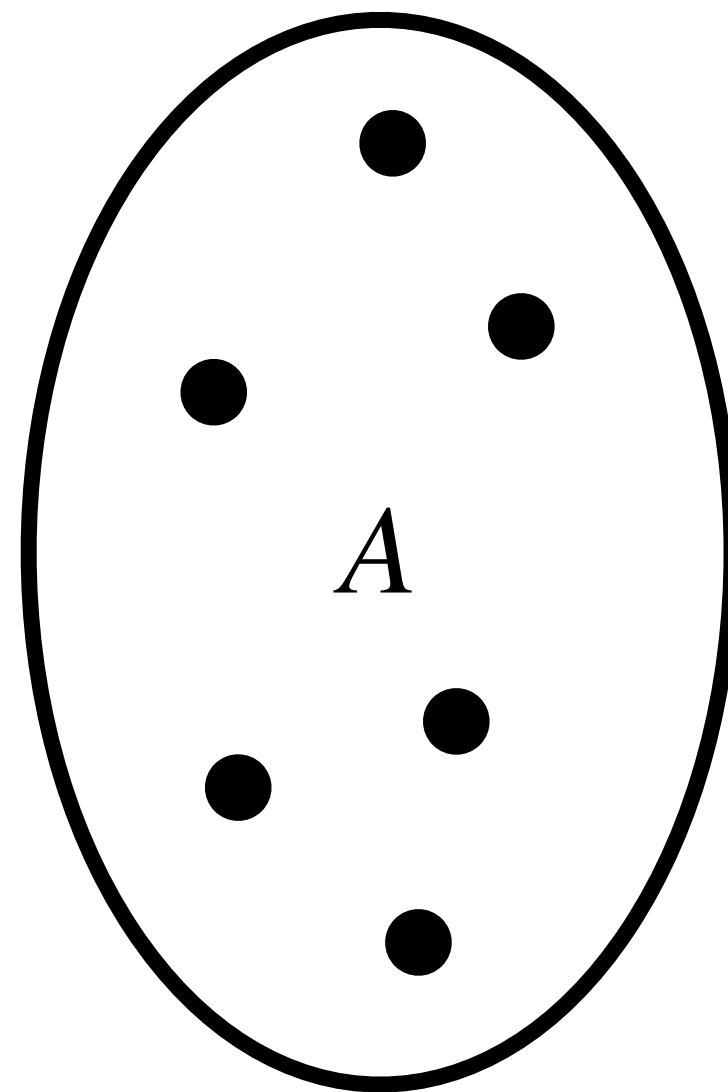
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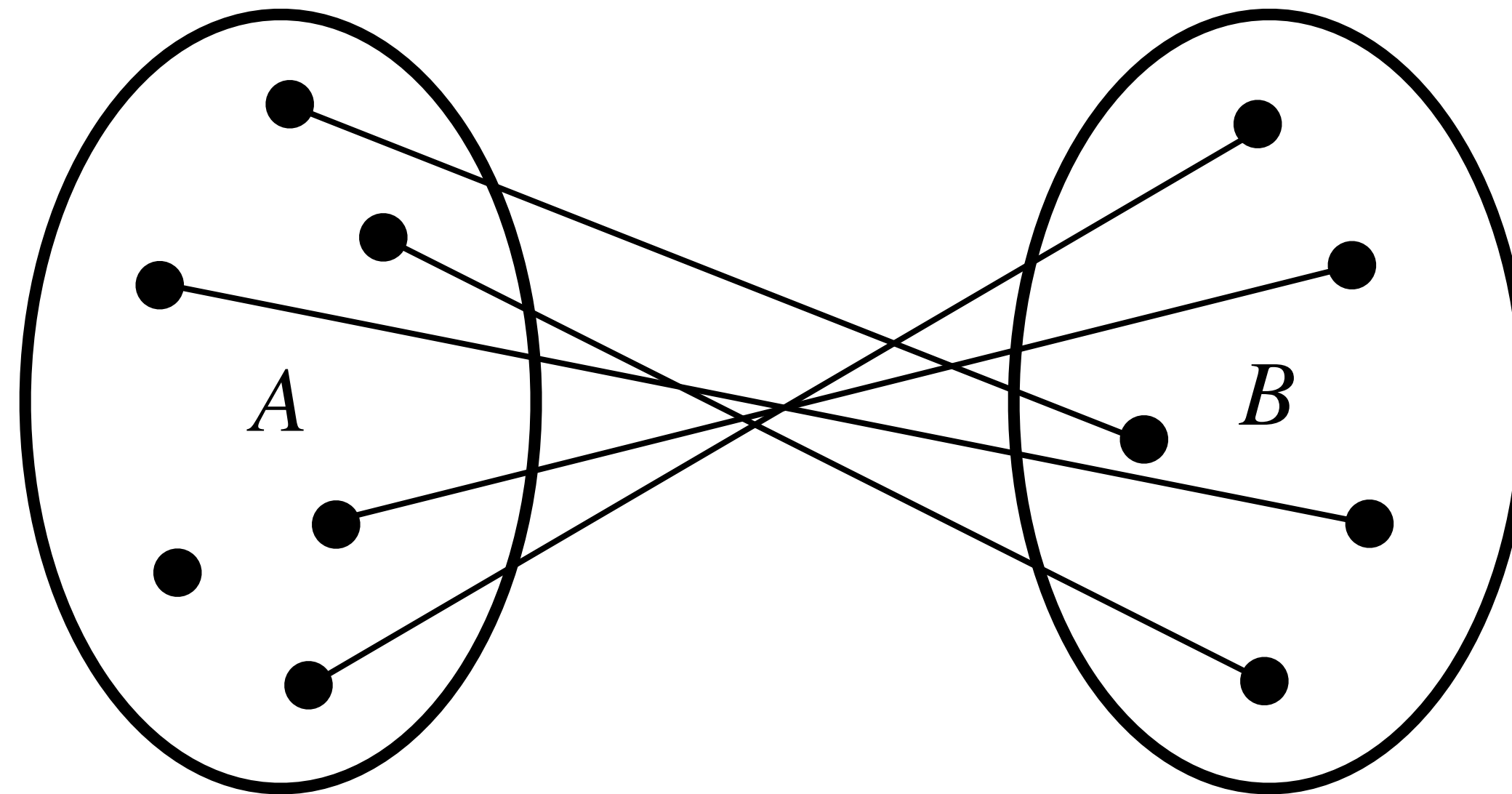
Defn: A matching M is **maximum** if there is no matching M' such that $|M| < |M'|$.

Bipartite Graphs

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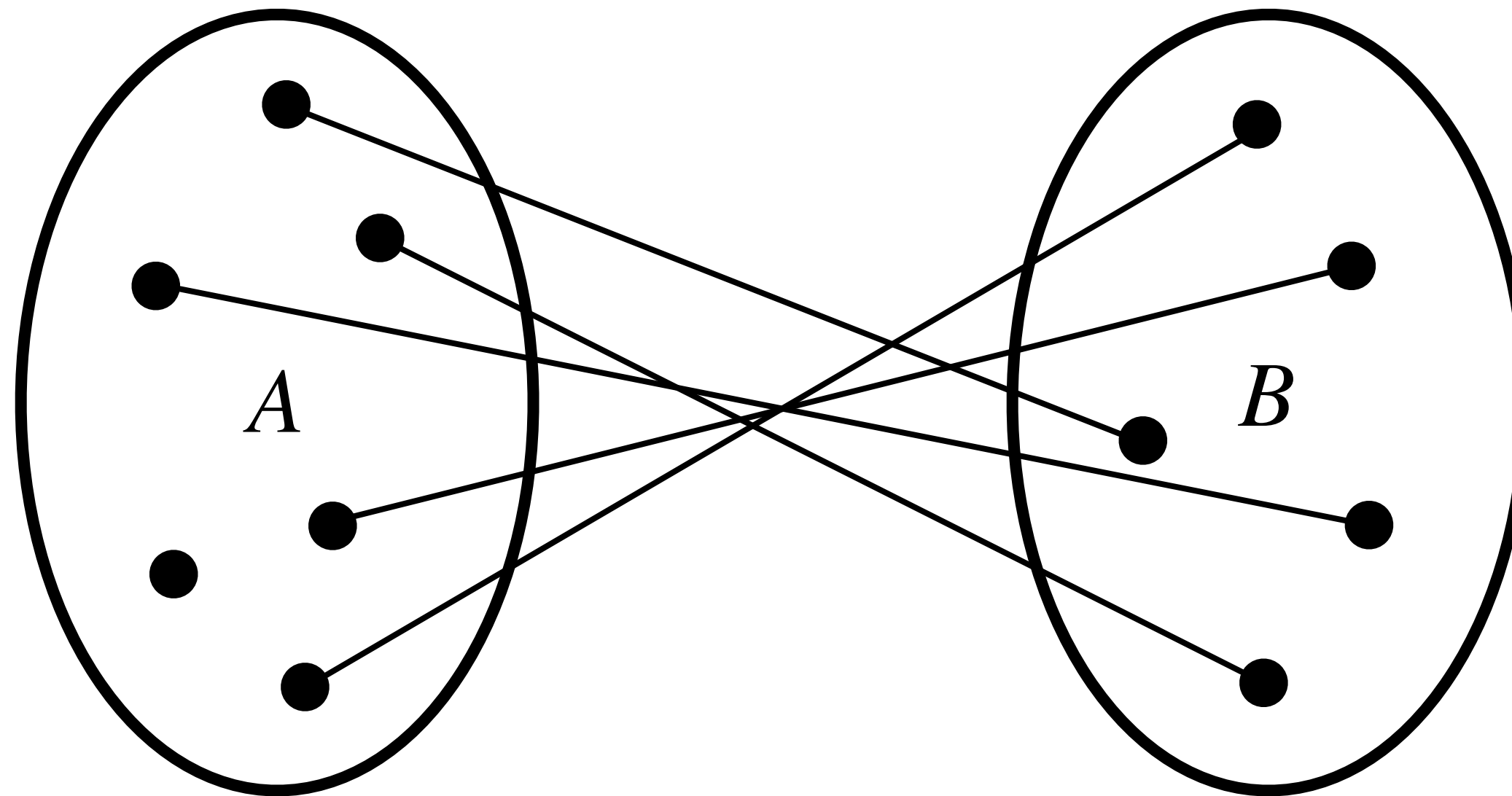


Bipartite Graphs



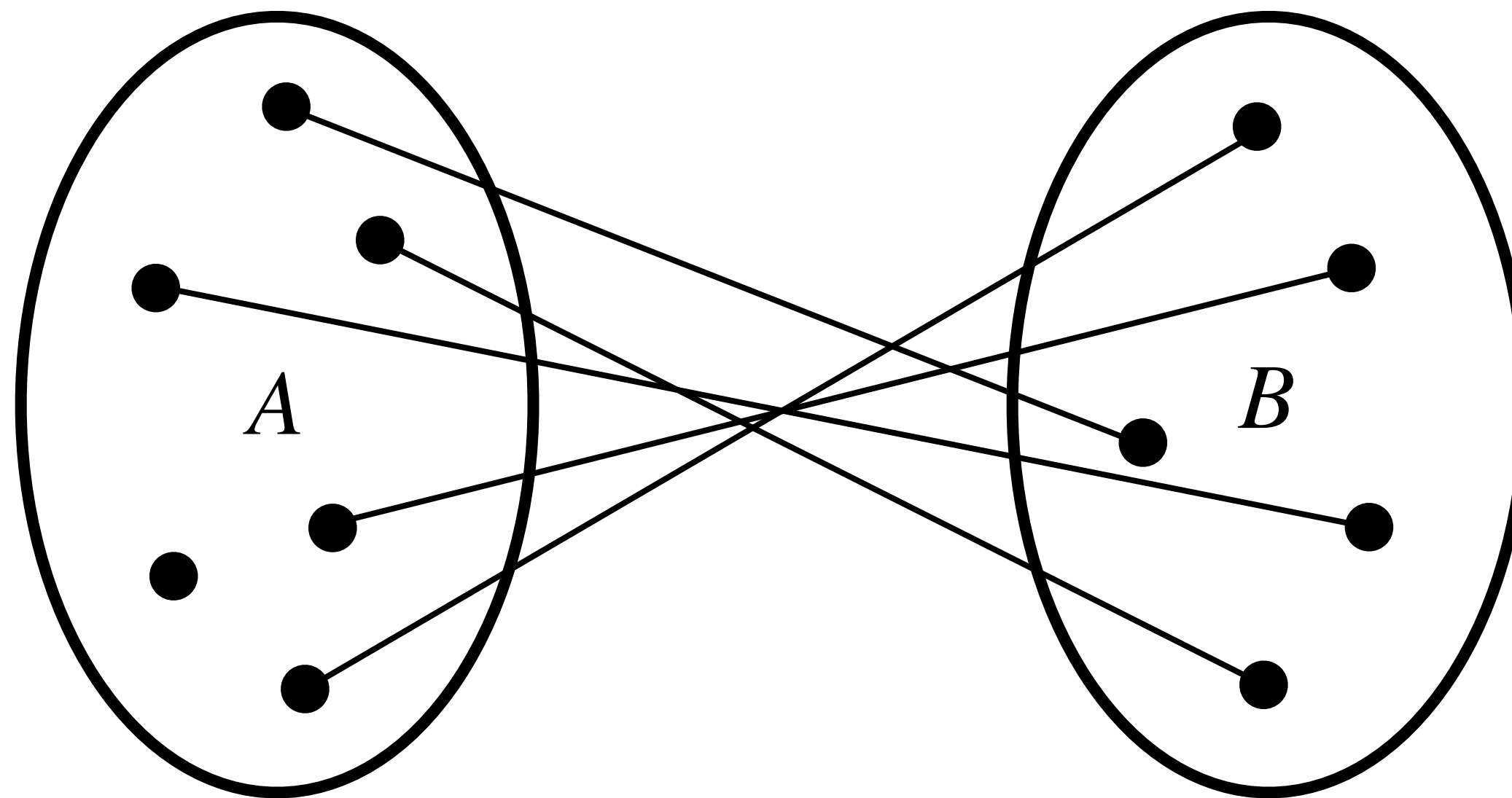
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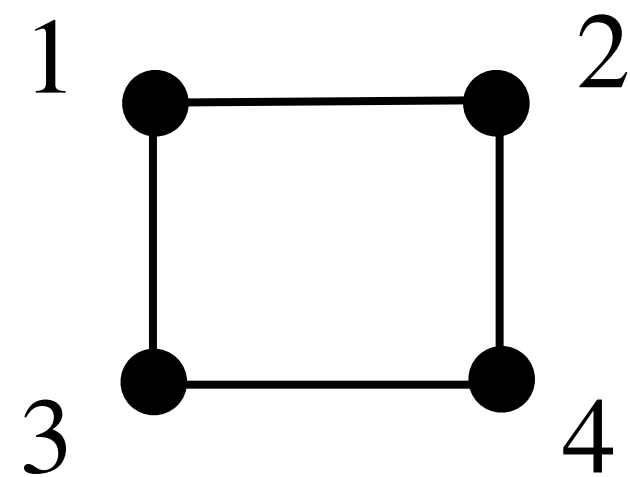
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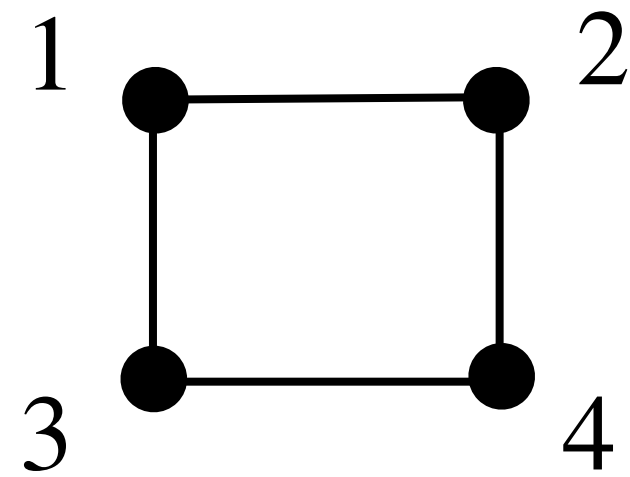
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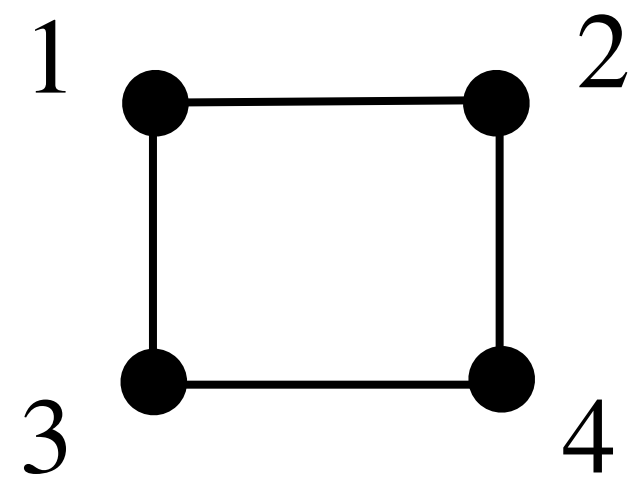


$$A = \{1,4\}, B = \{2,3\}$$

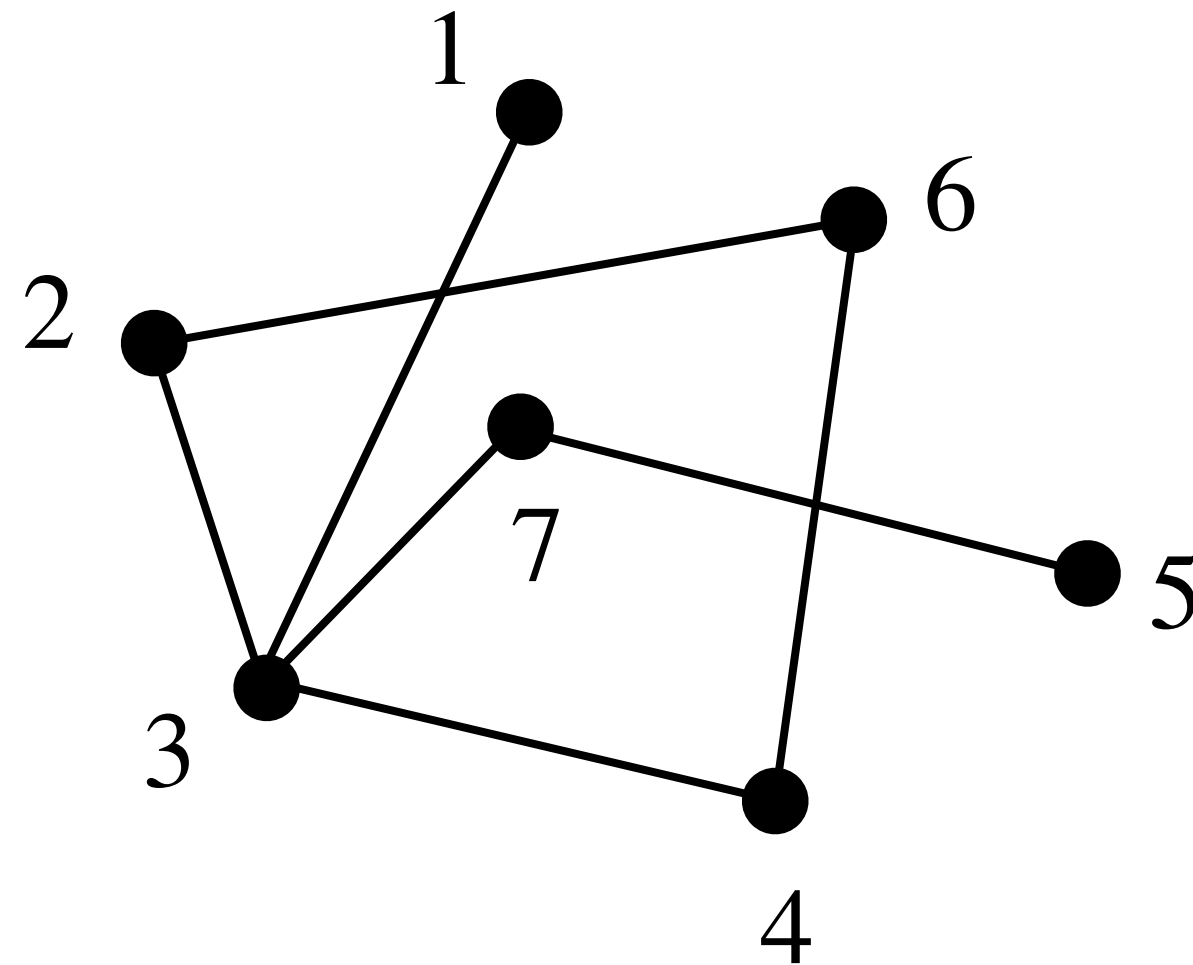
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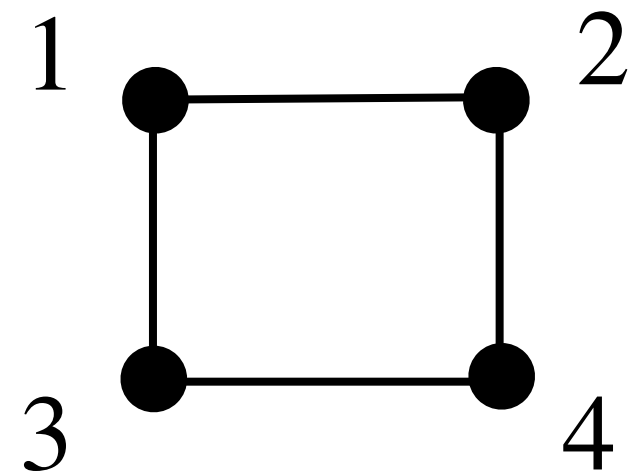
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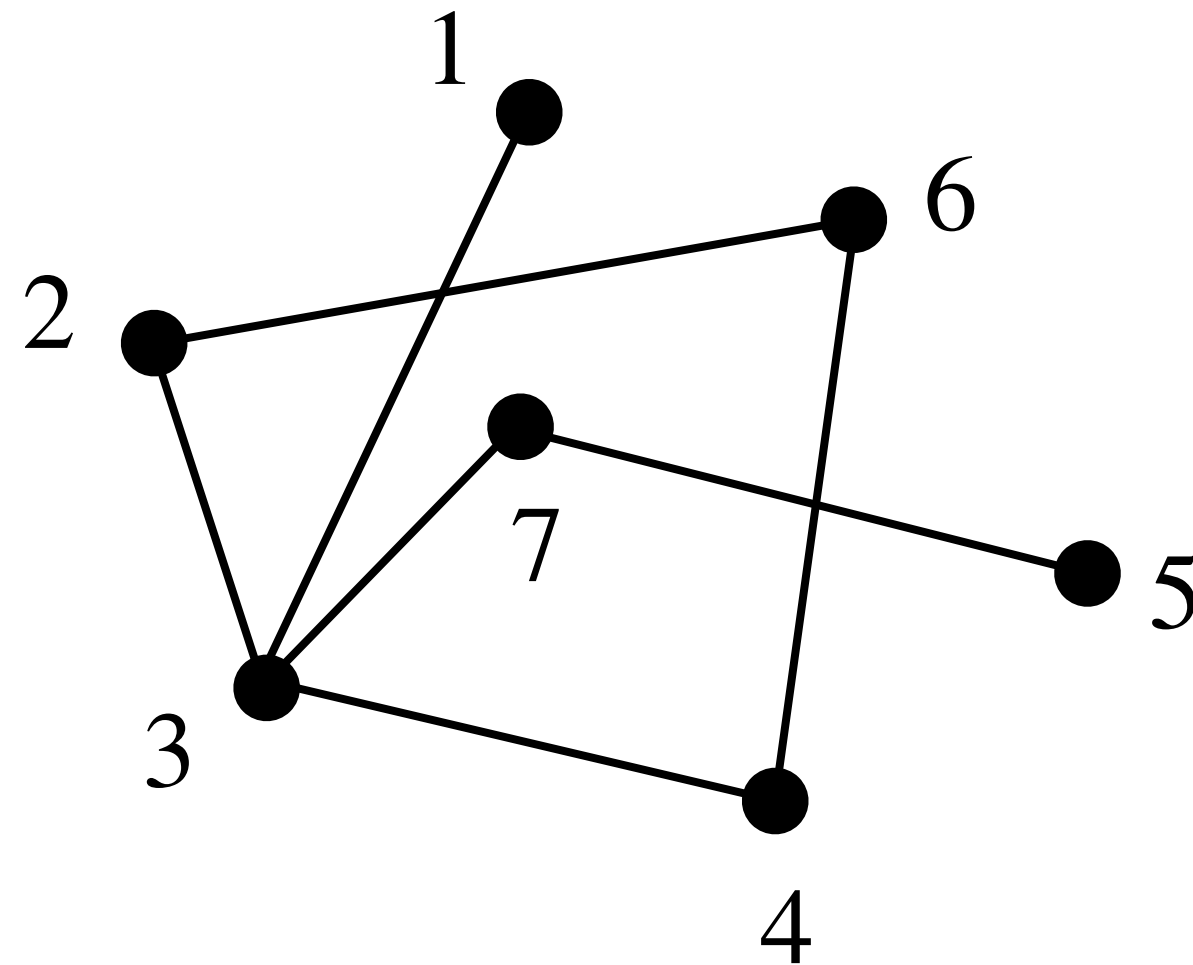
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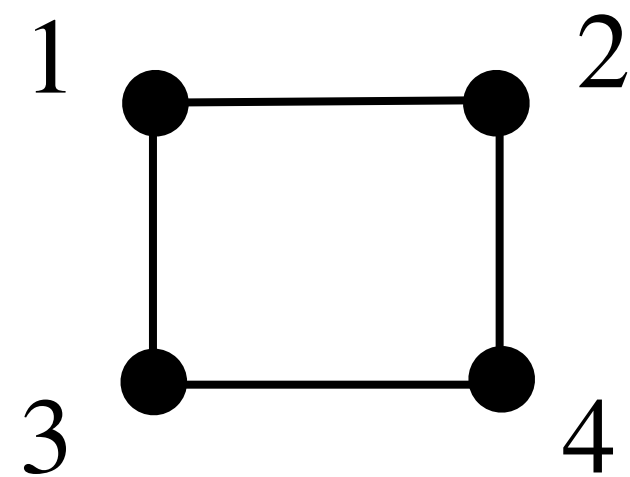


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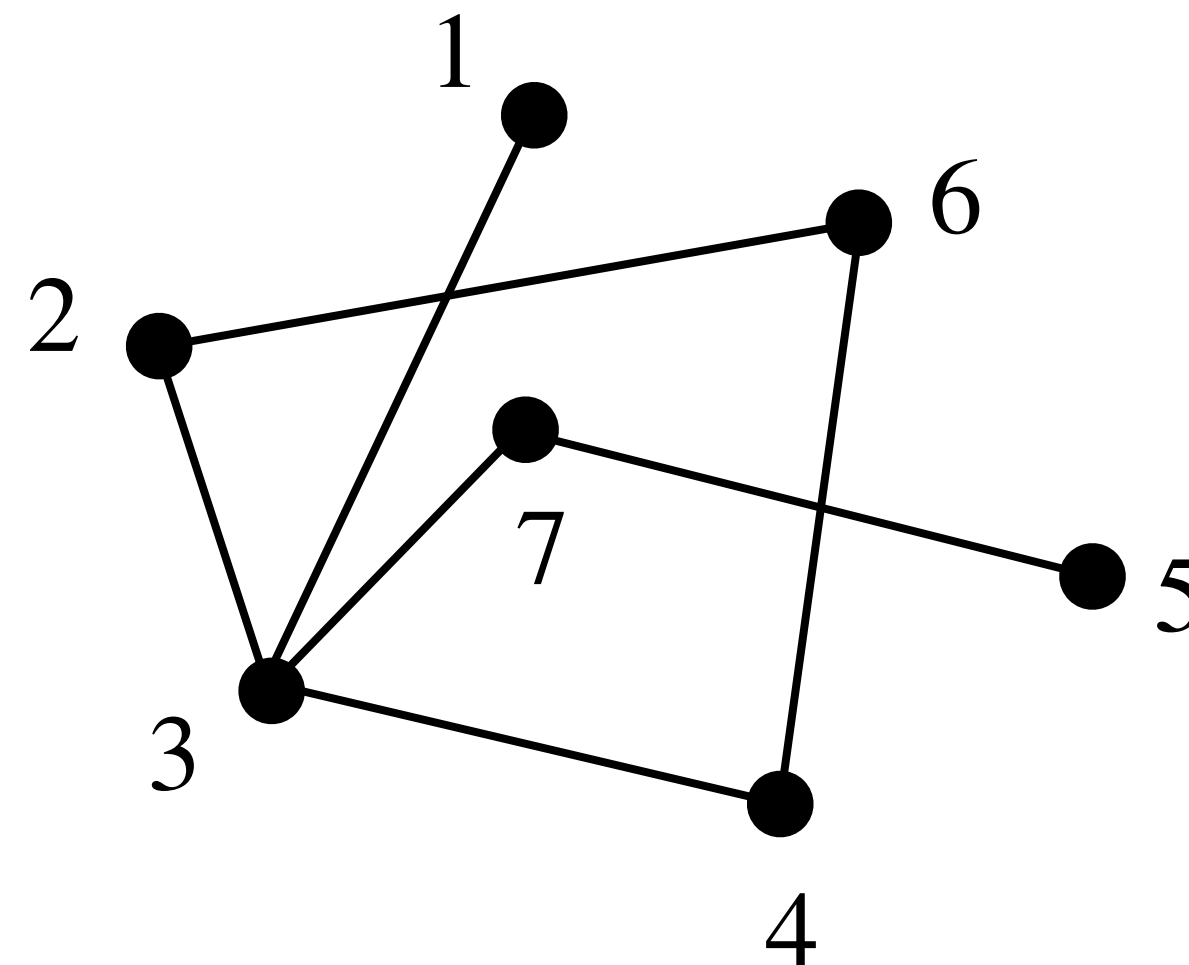
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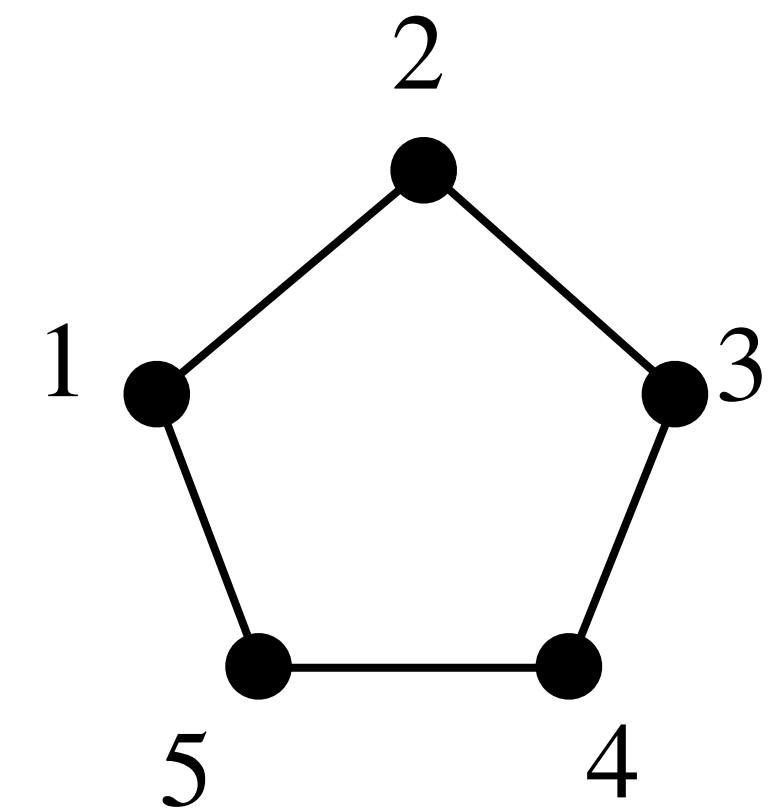
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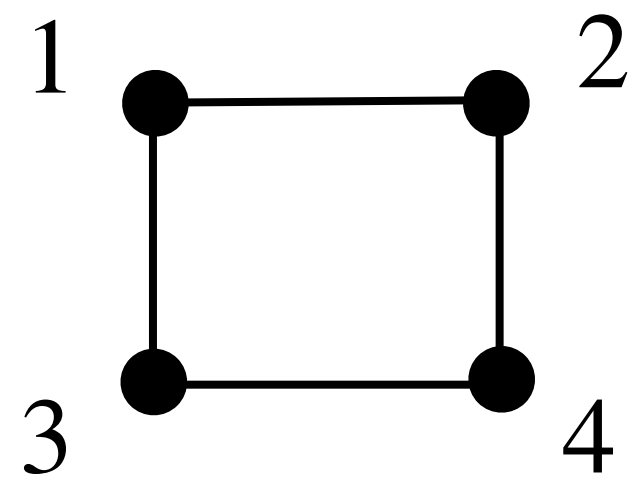
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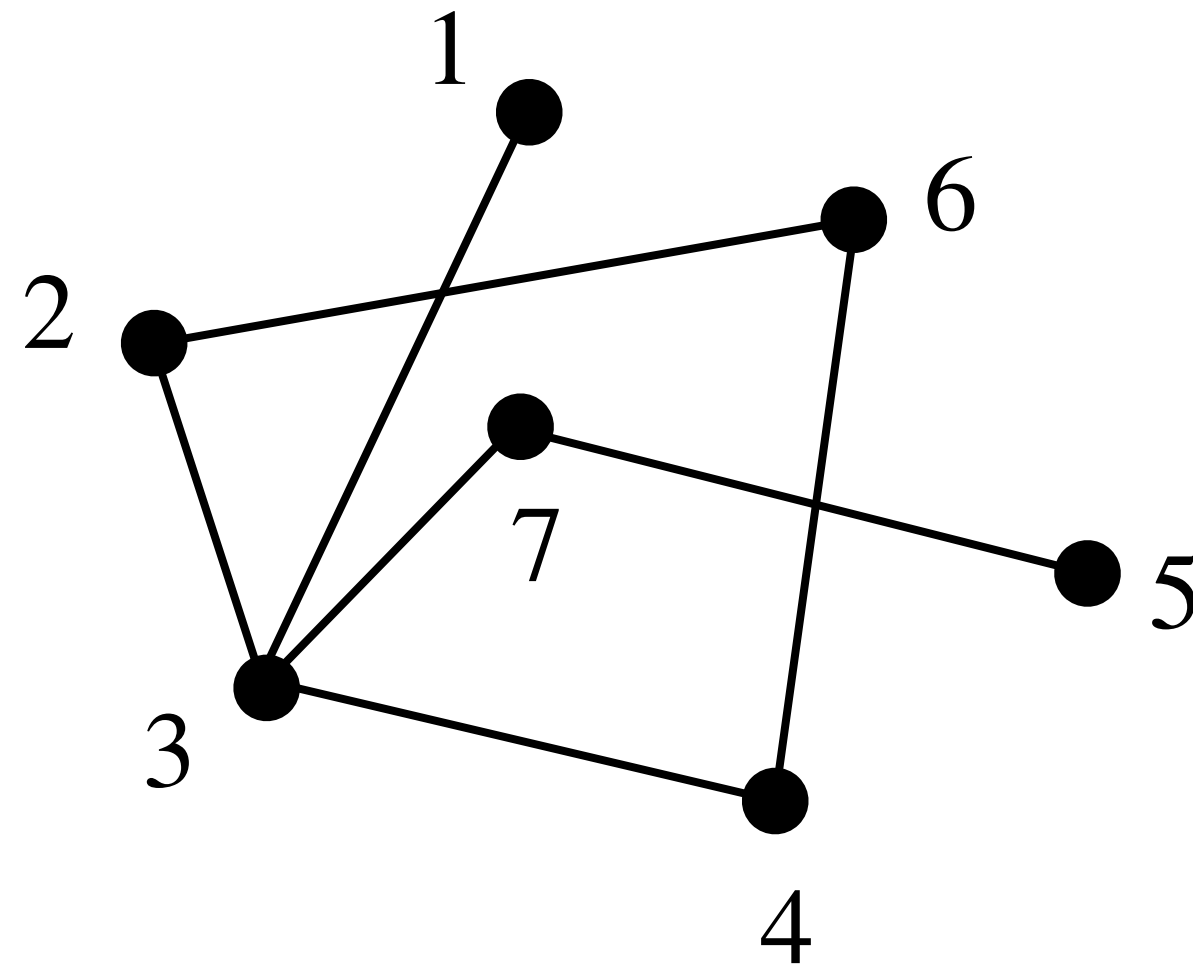
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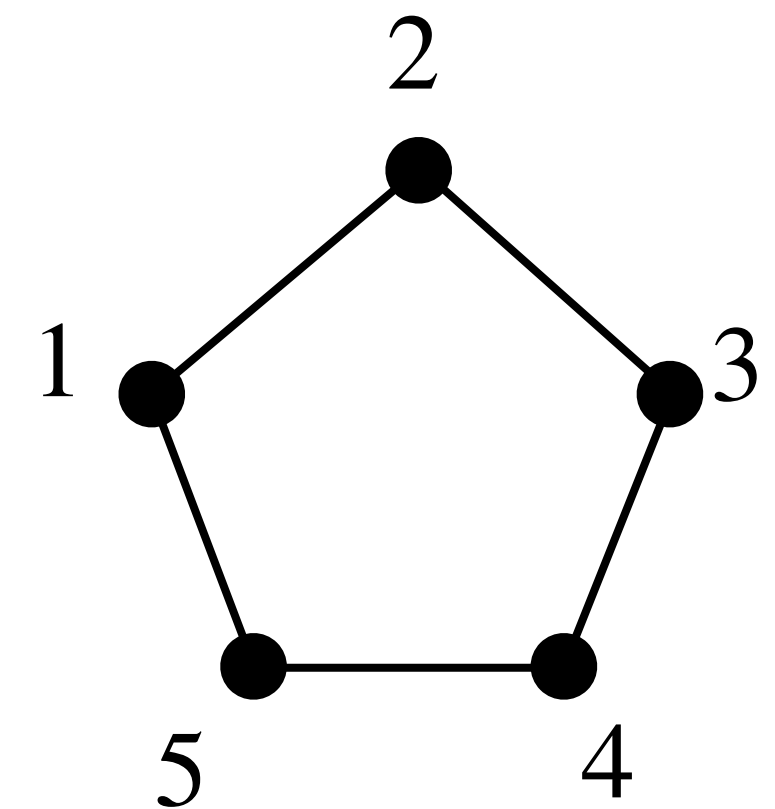
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Not bipartite

Maximum Bipartite Matching in Jobs

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Jobs	●	●	●	●	●

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Applicants

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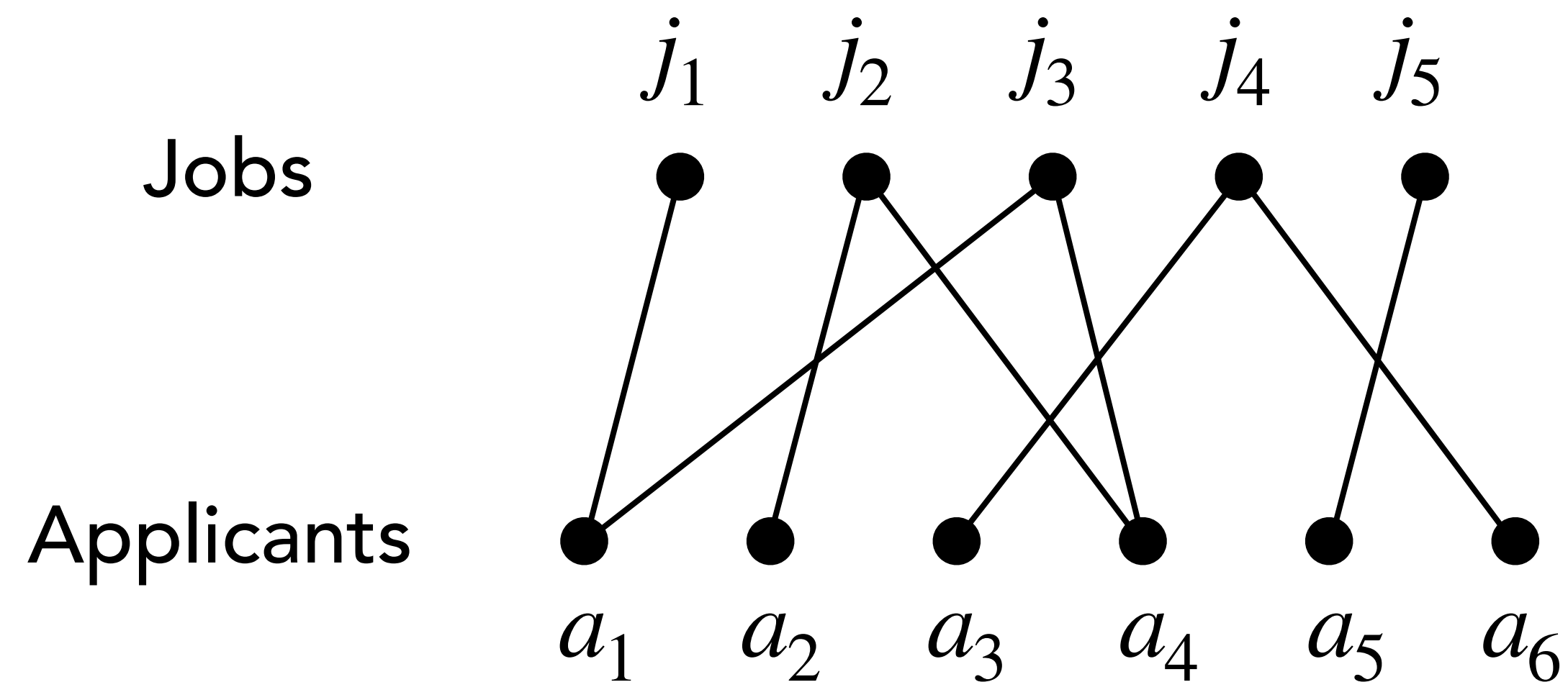
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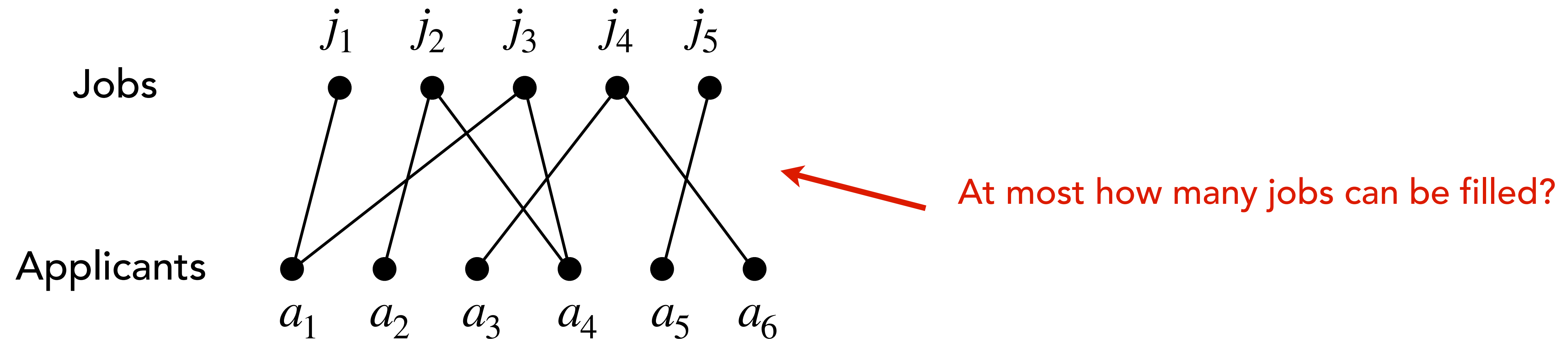
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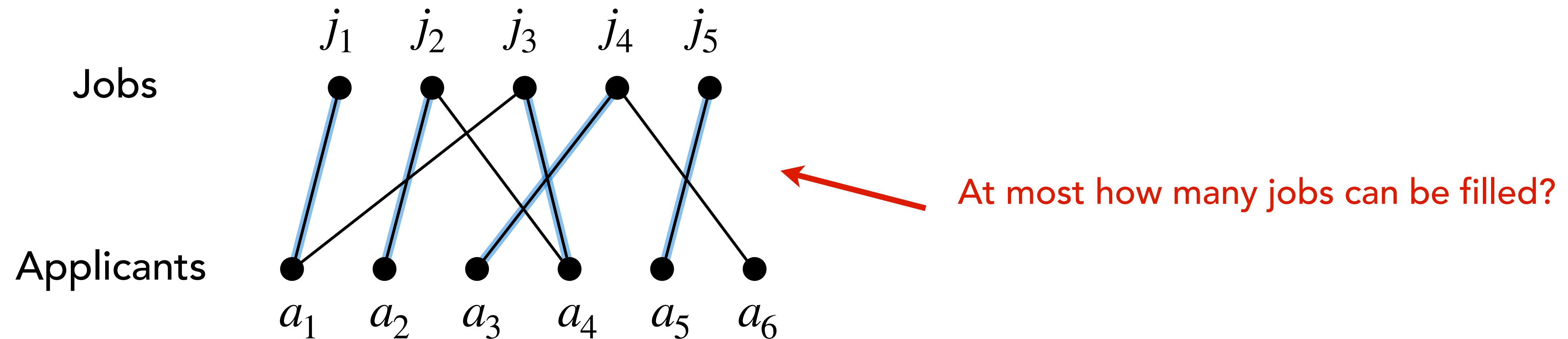
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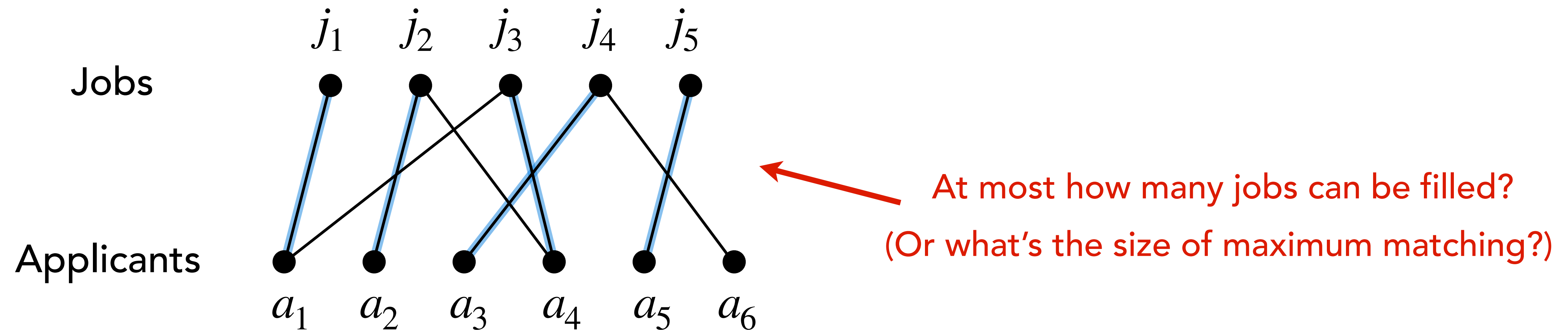
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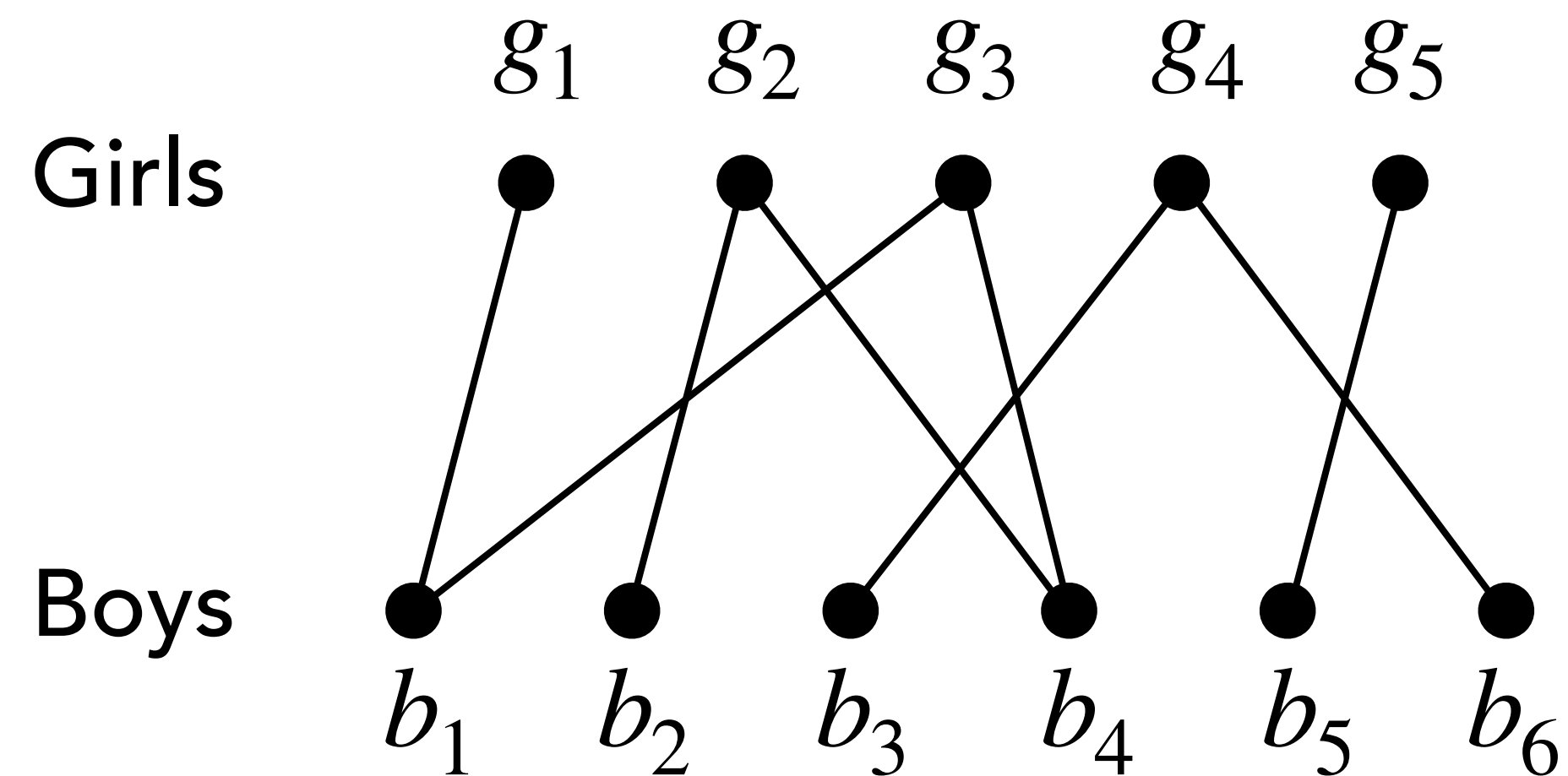
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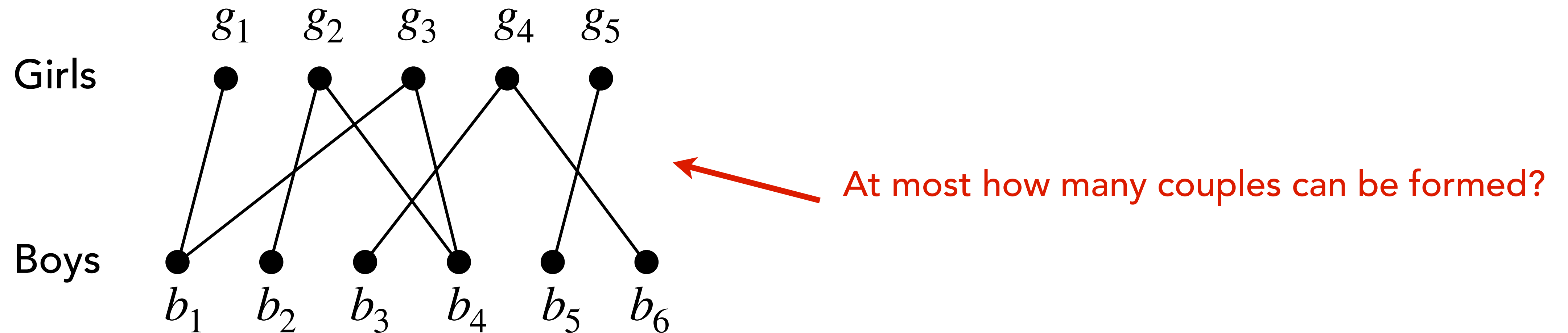
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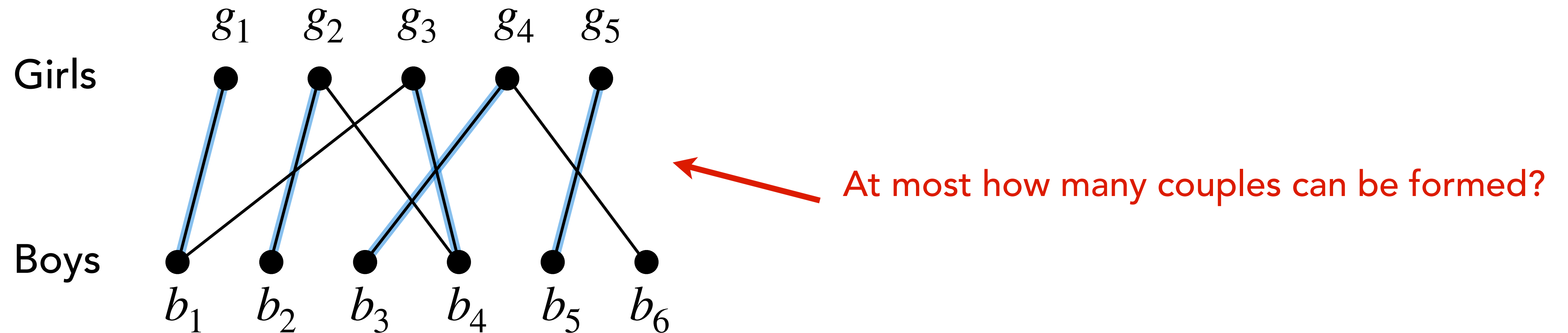
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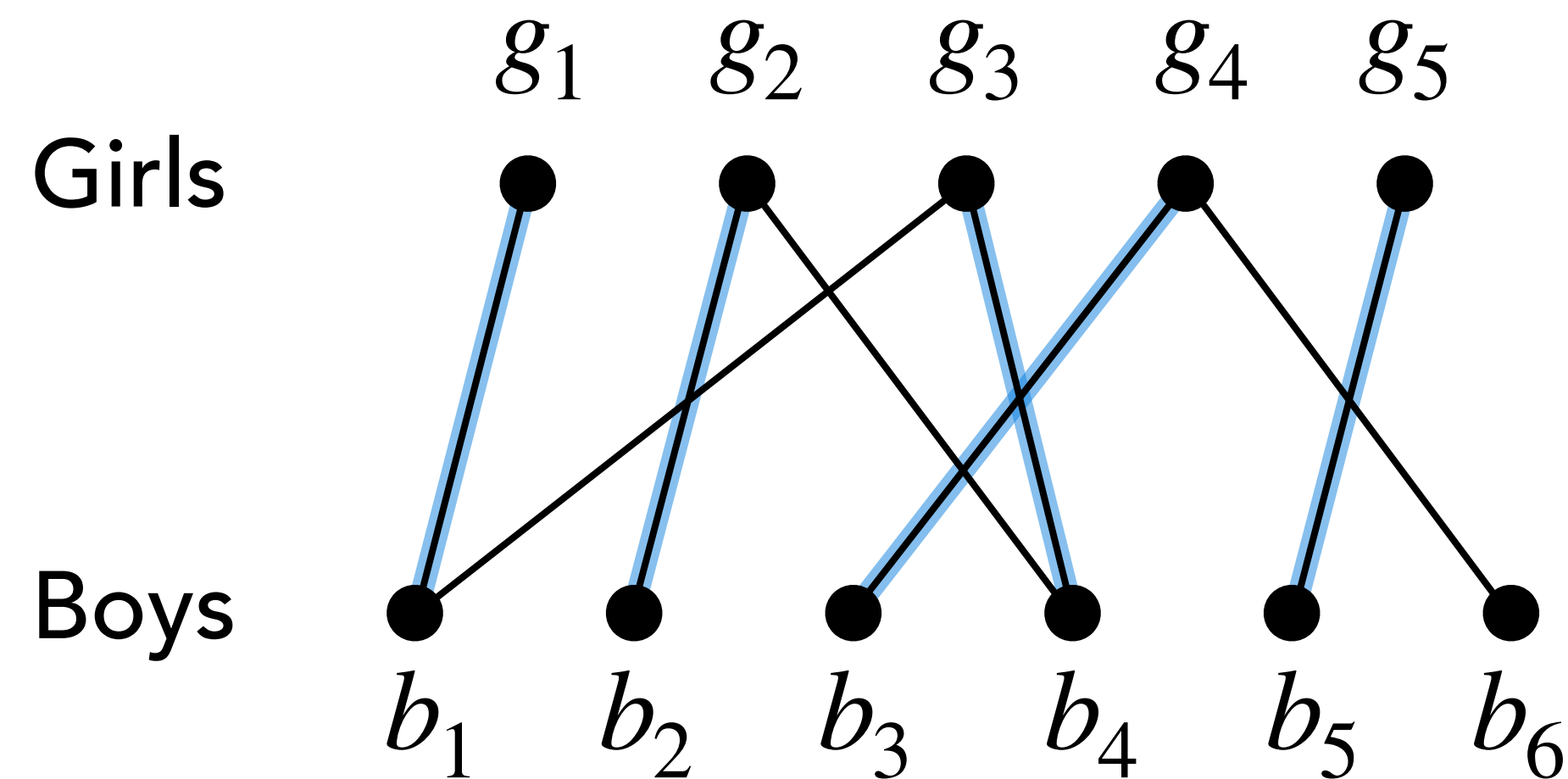
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At most how many couples can be formed?
(Or what's the size of maximum matching?)

Maximum Bipartite Matching in Factories

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Suppose there are 5 machine and 6 tasks.

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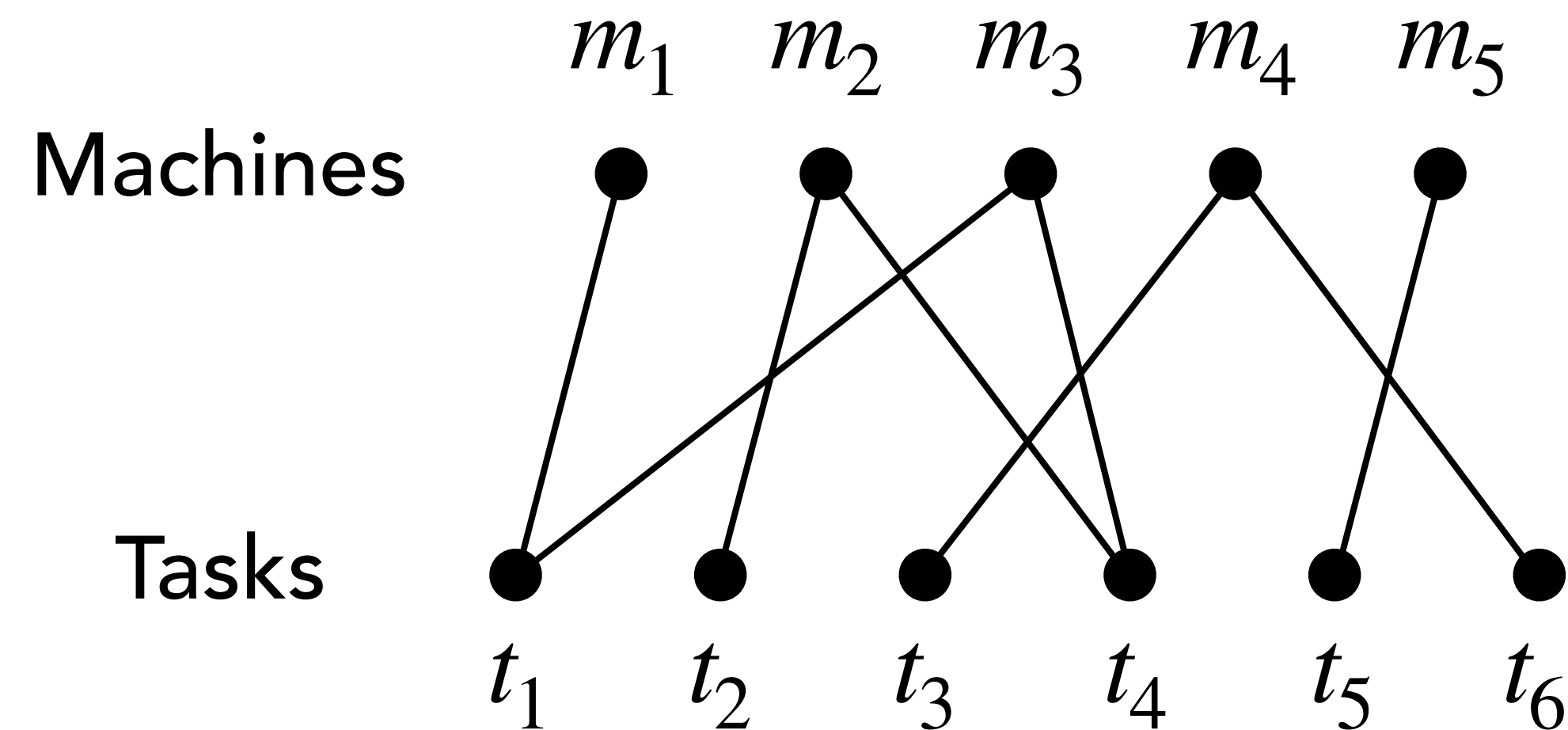
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Suppose there are 5 machine and 6 tasks. Each machine can do one task at a time and one task can be done by at most one machine.

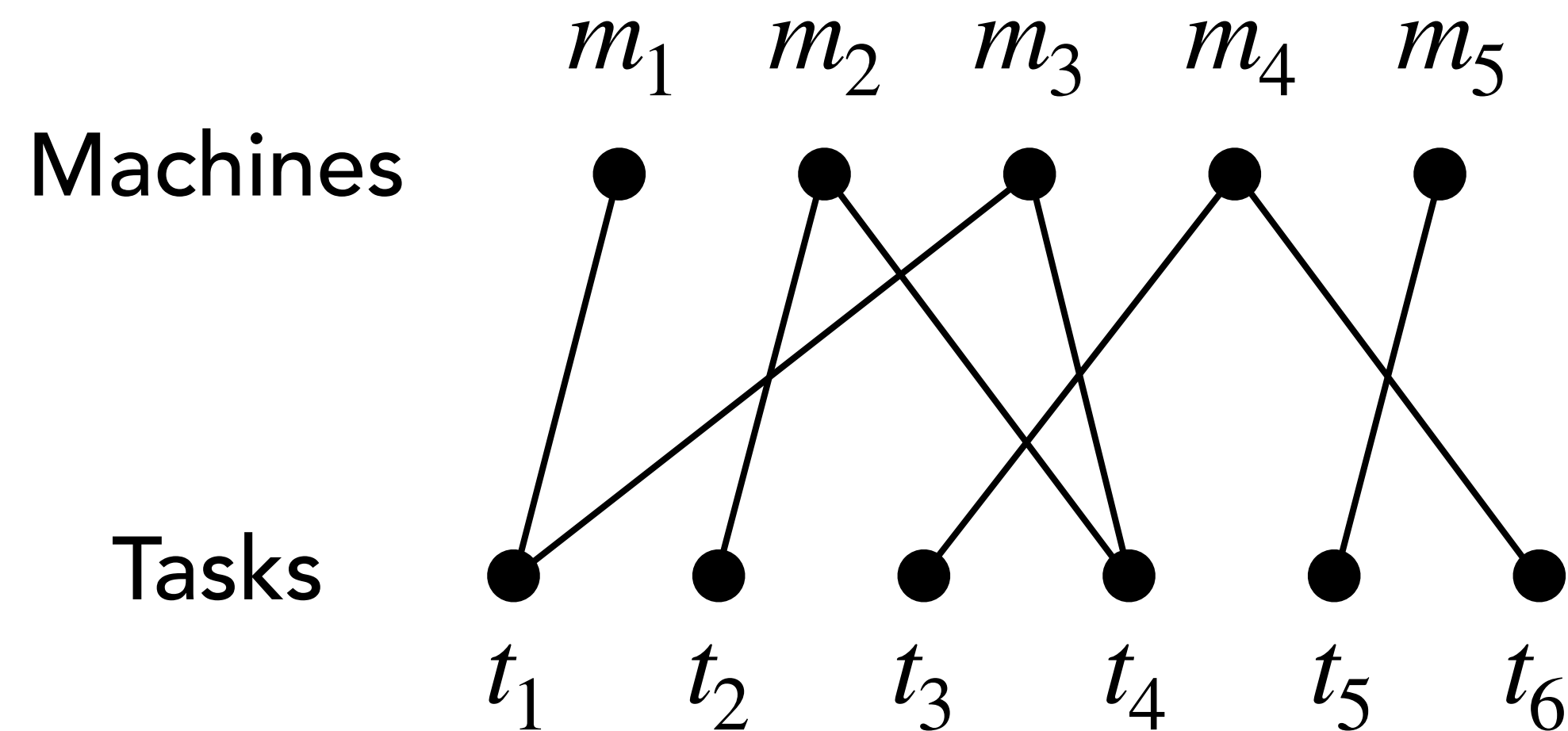
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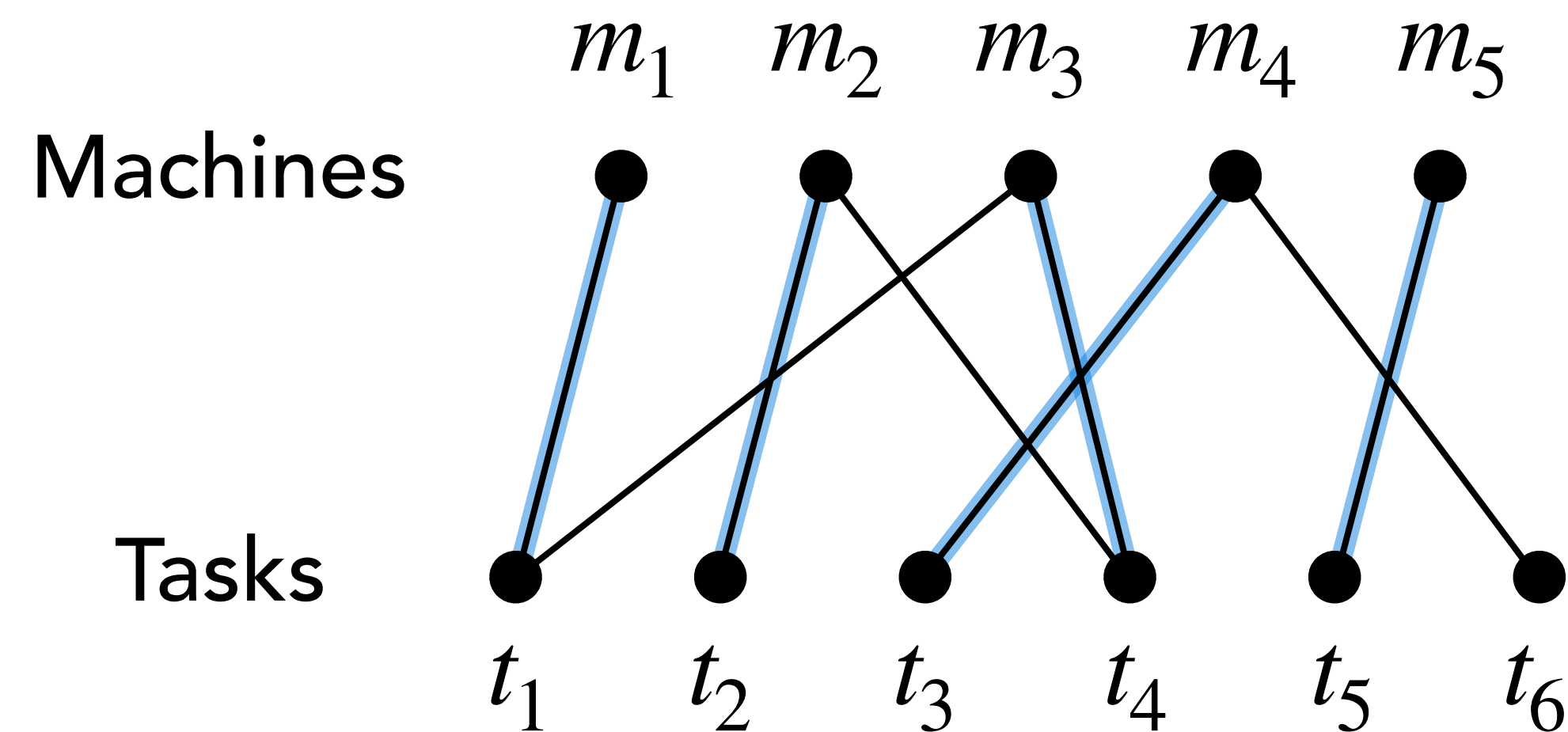
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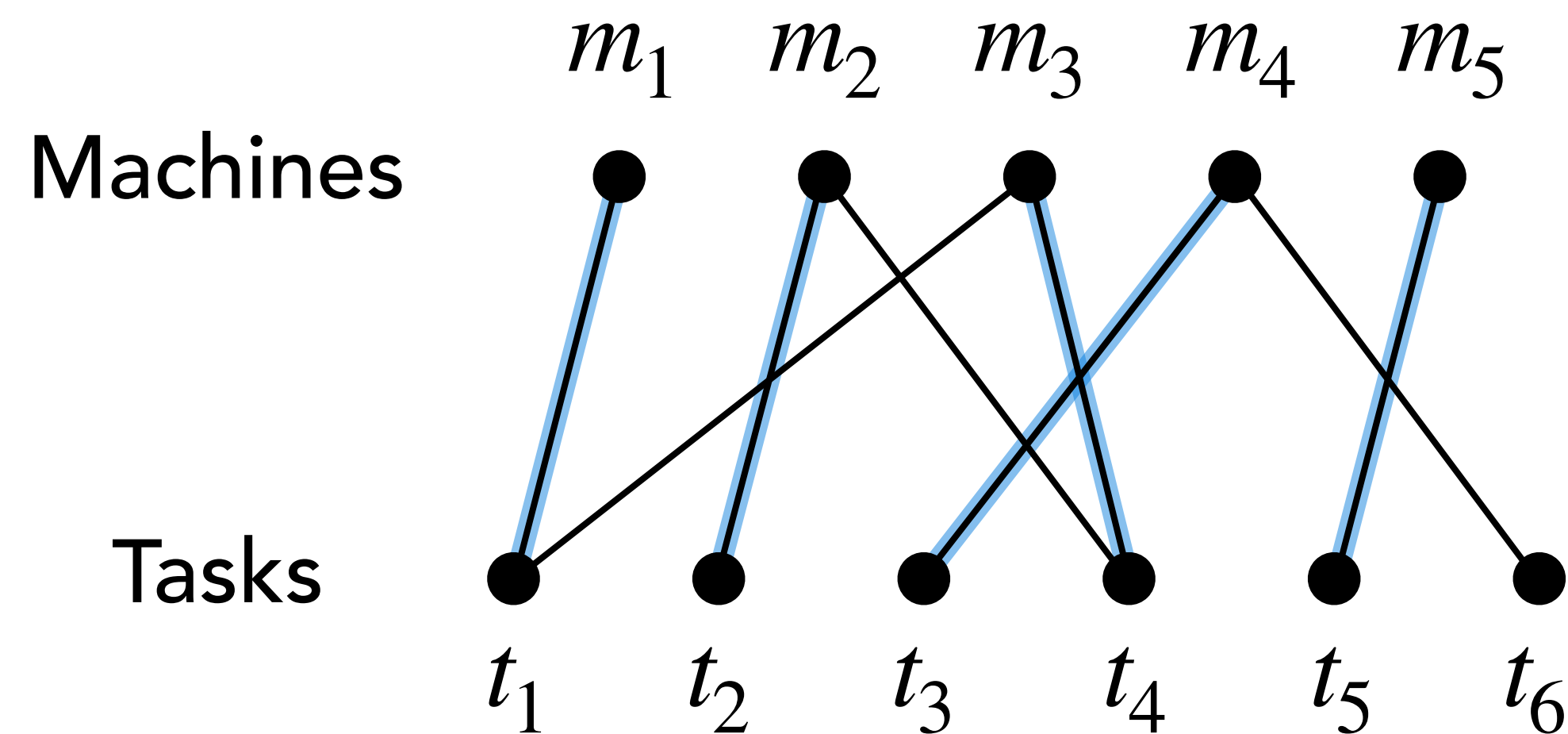
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Input: A bipartite graph G .

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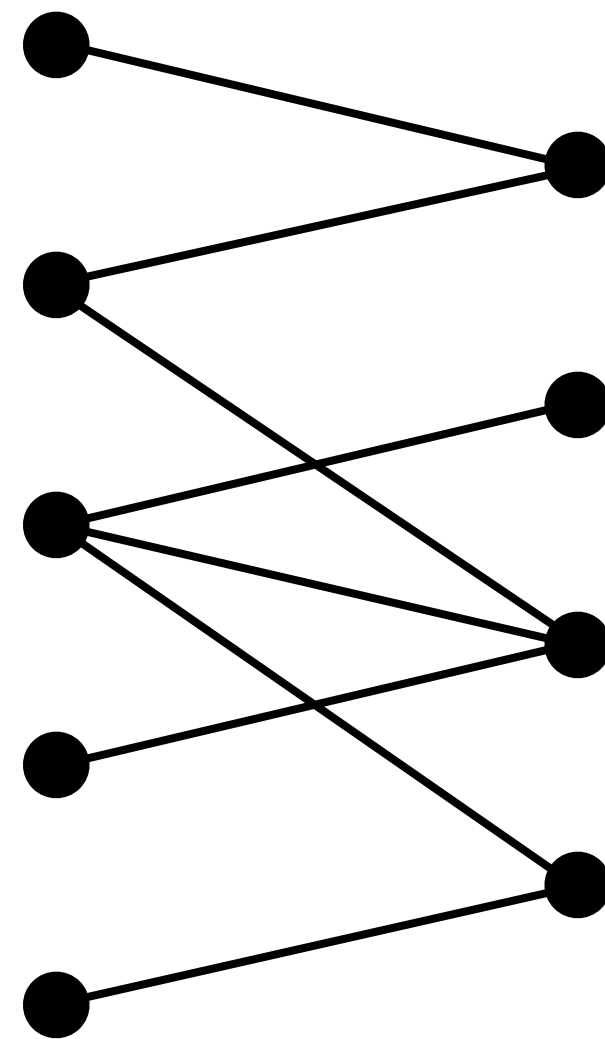


← Has a surprising connection to *Max-flow* problem.

Bipartite Matching to Flow

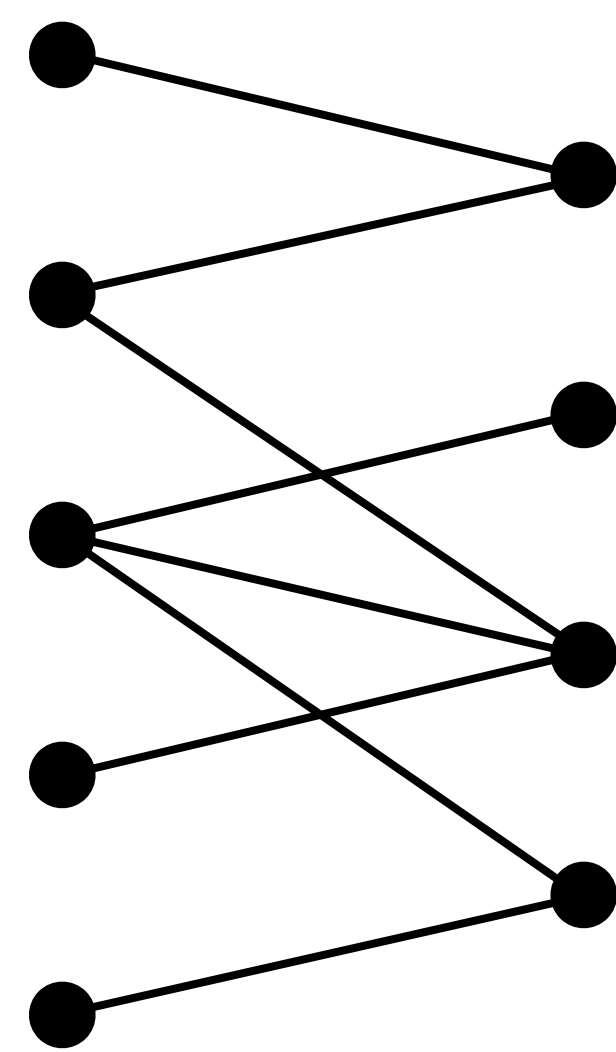
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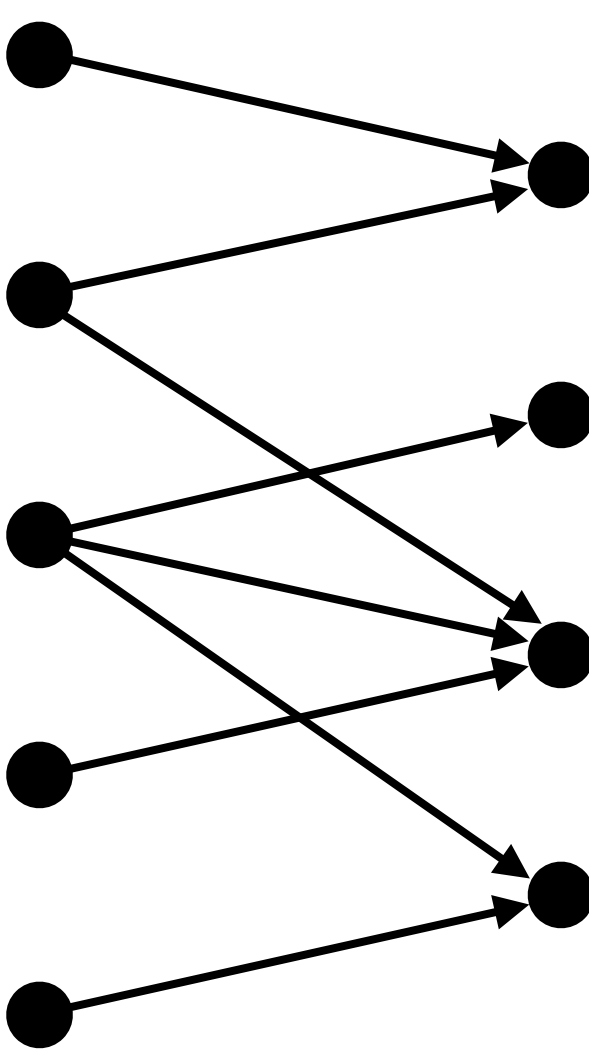


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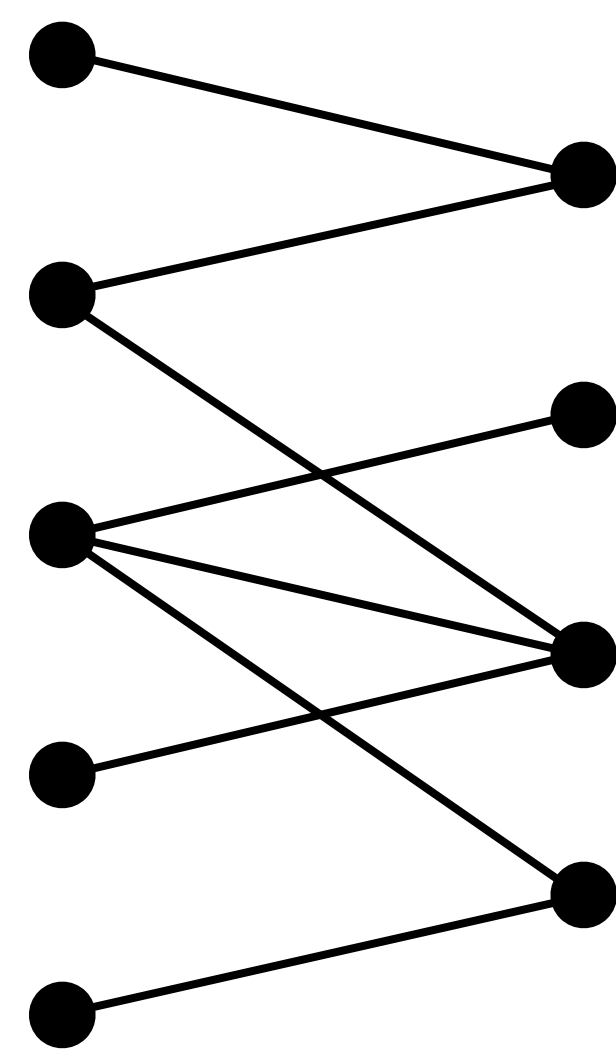


$G' = (V', E')$

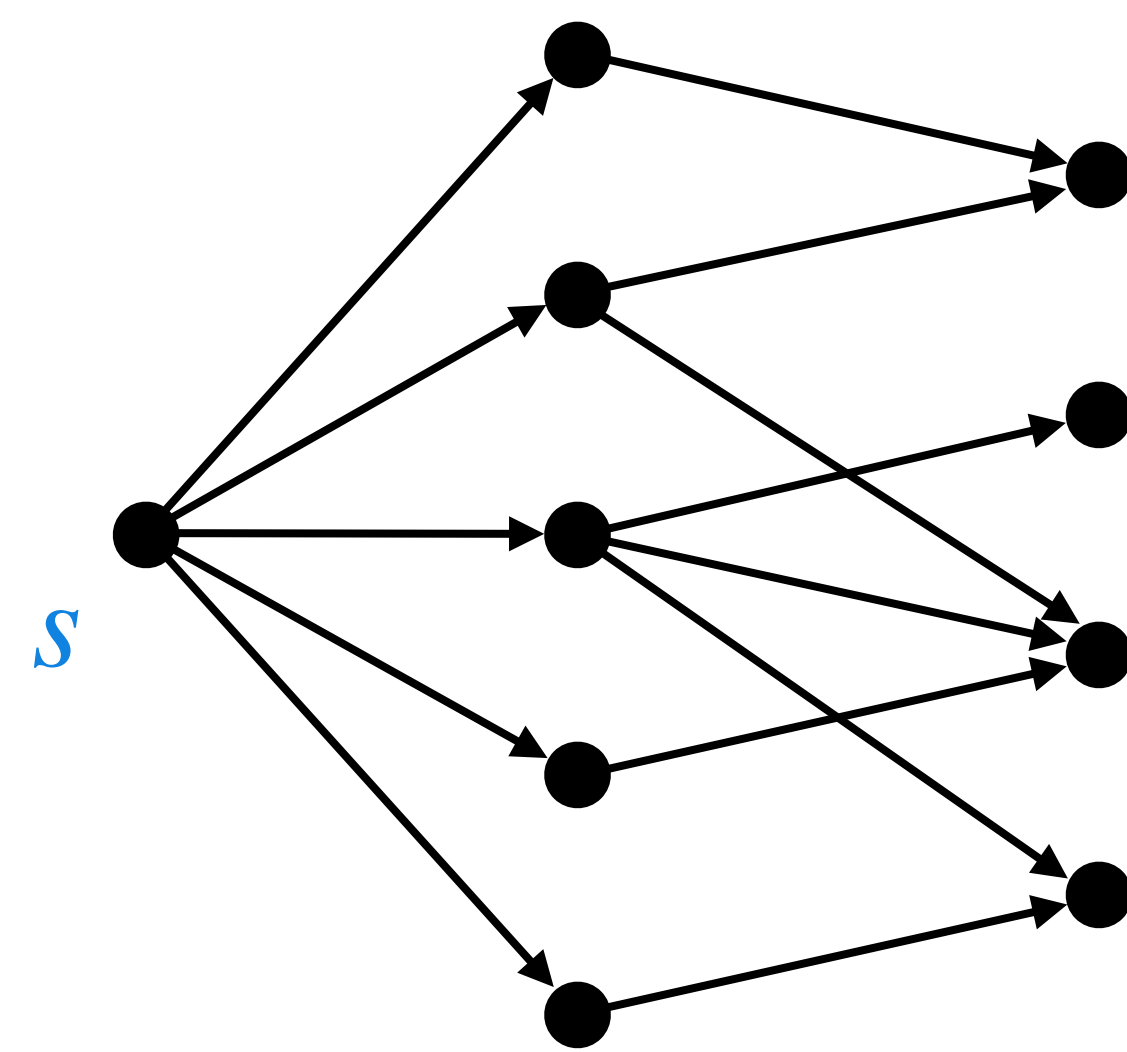


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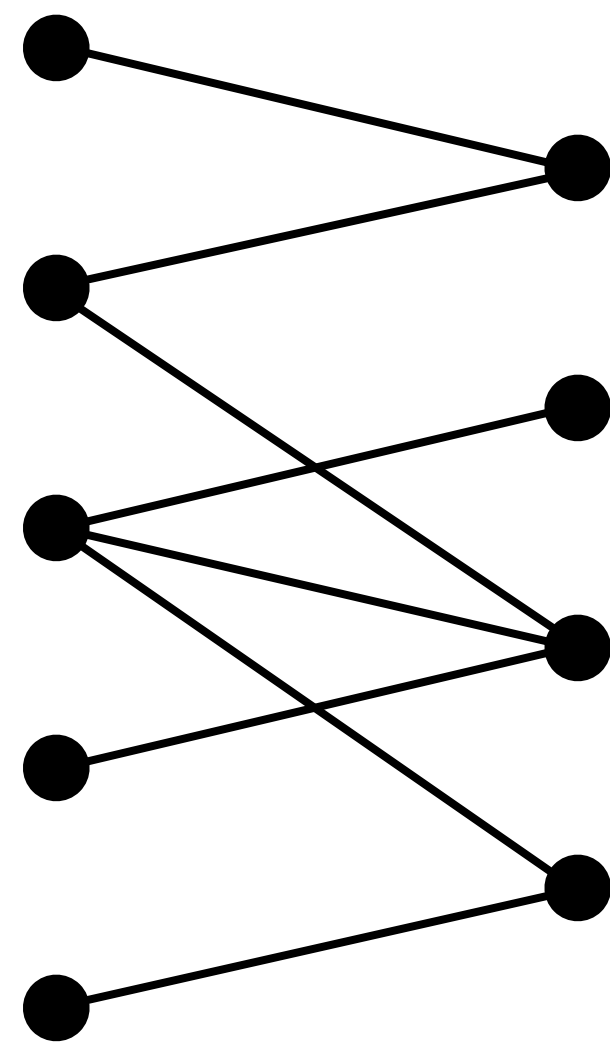


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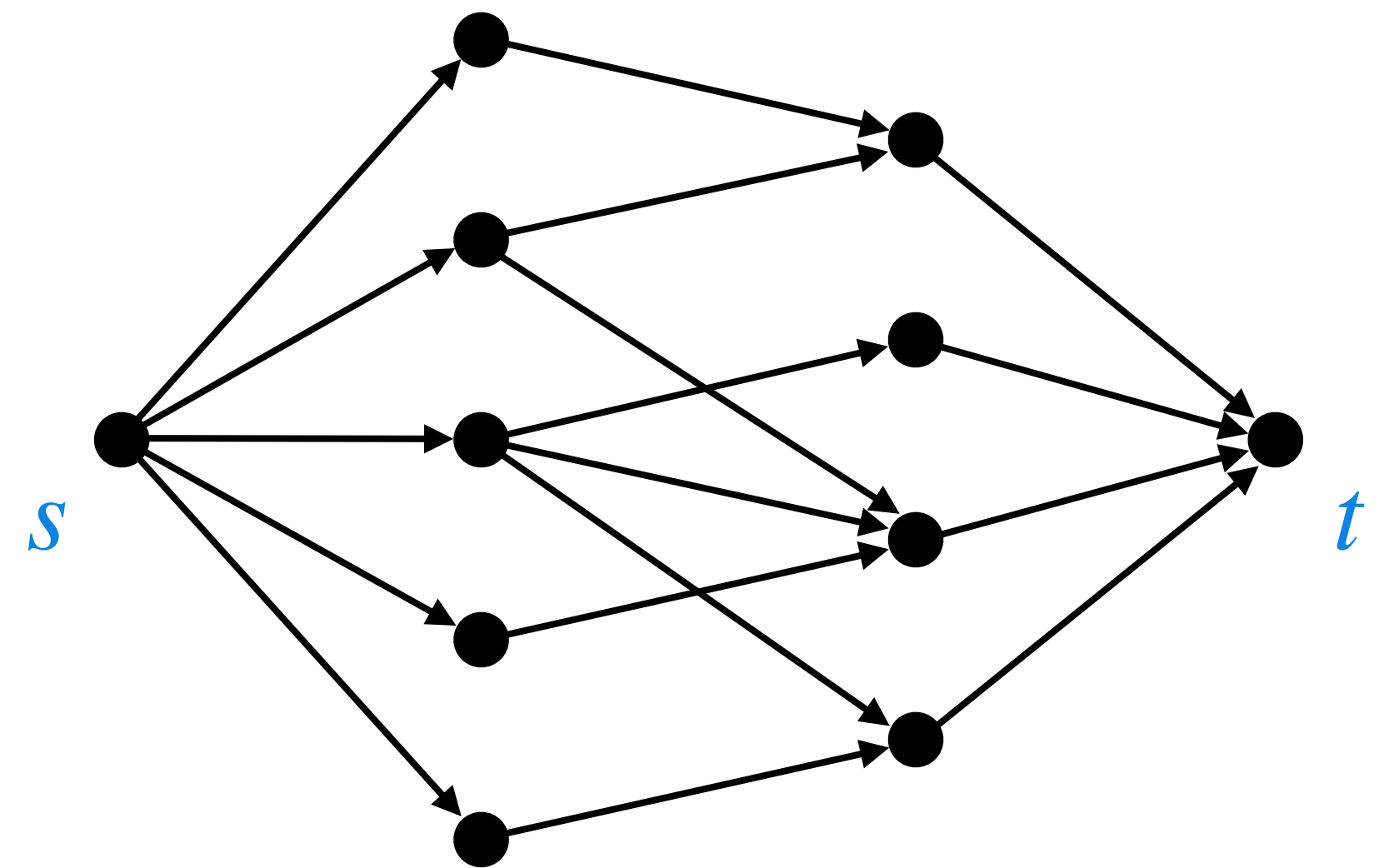


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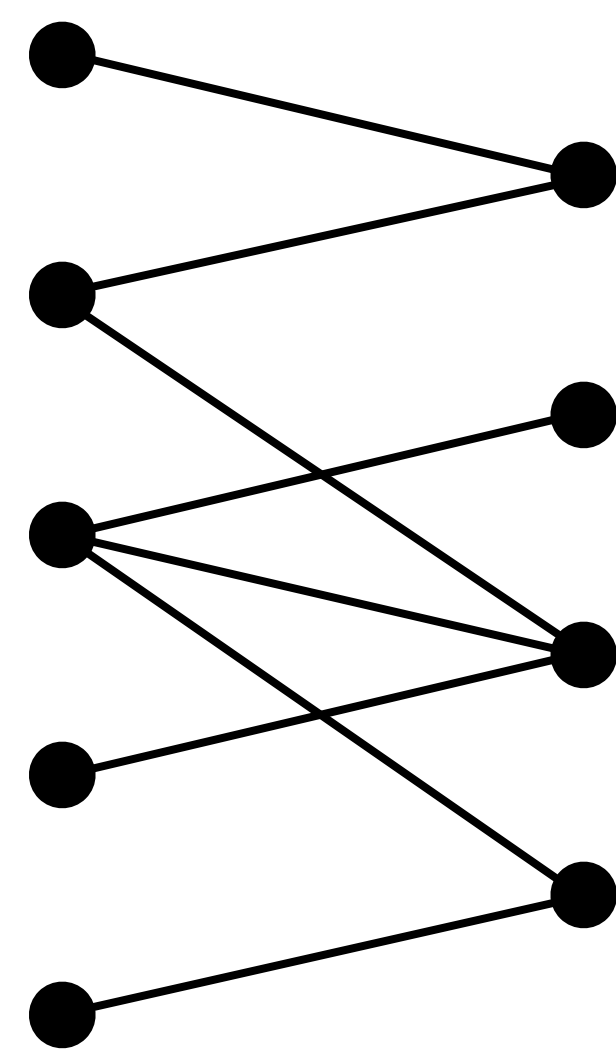


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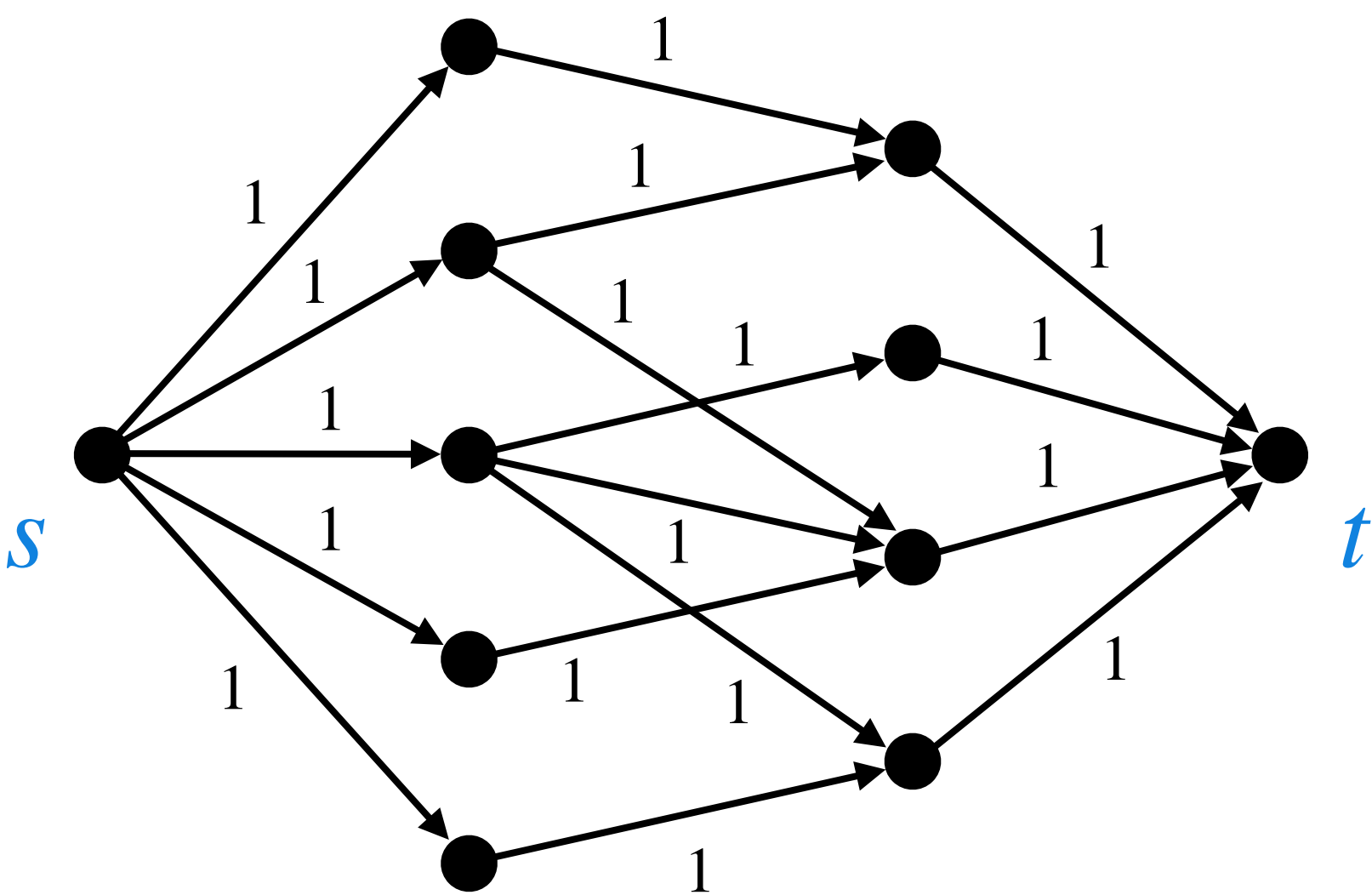


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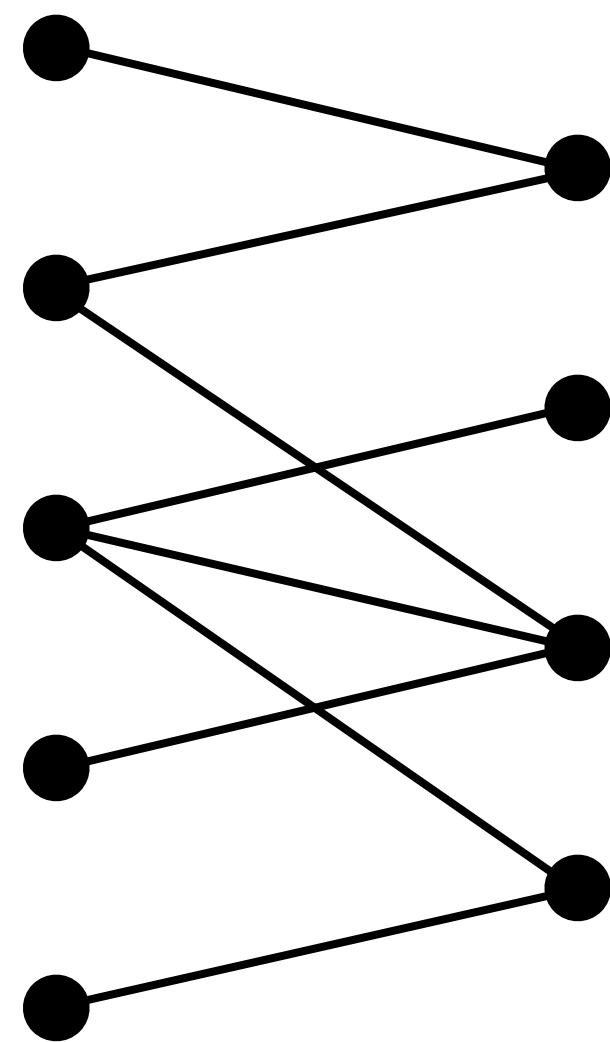
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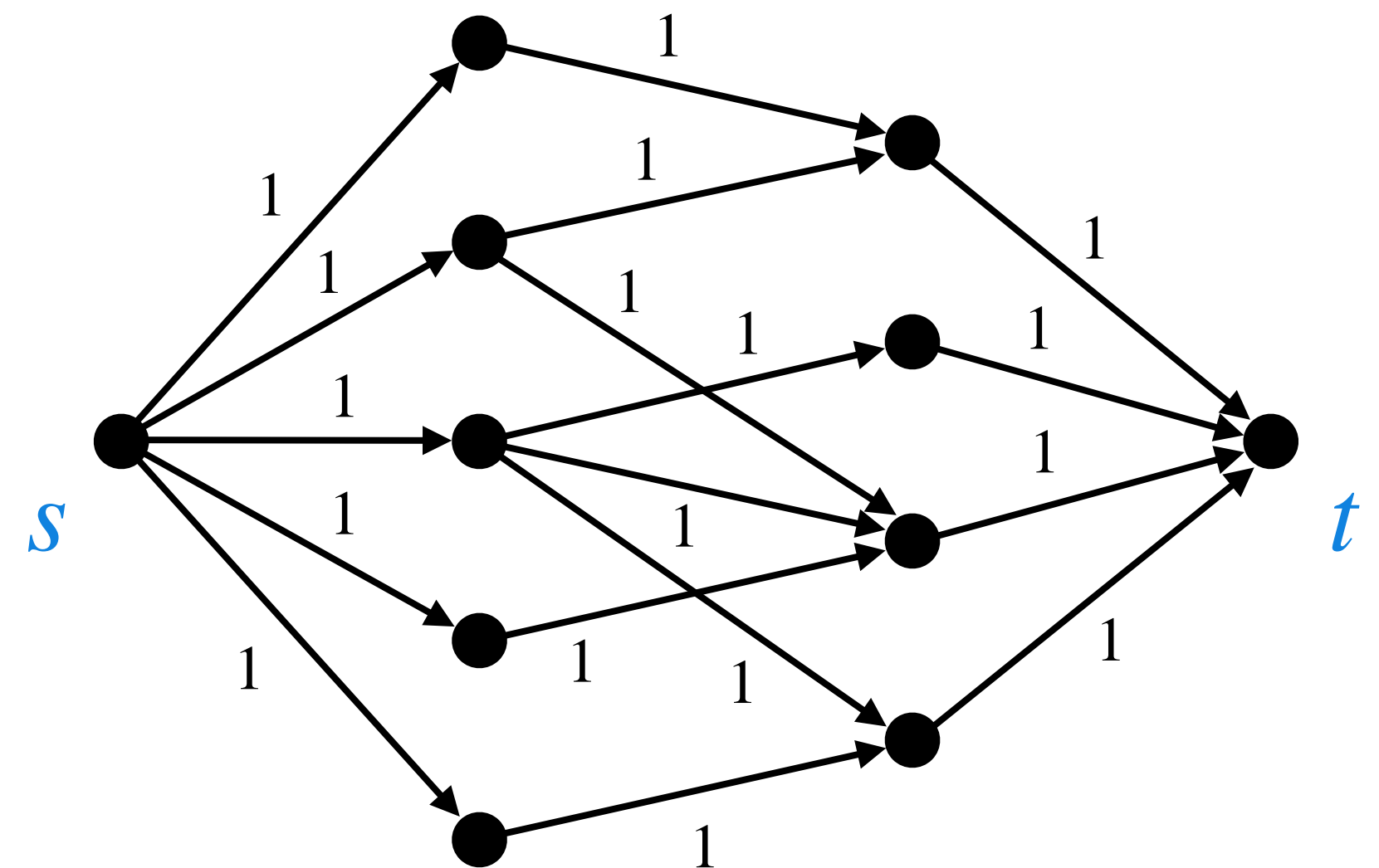
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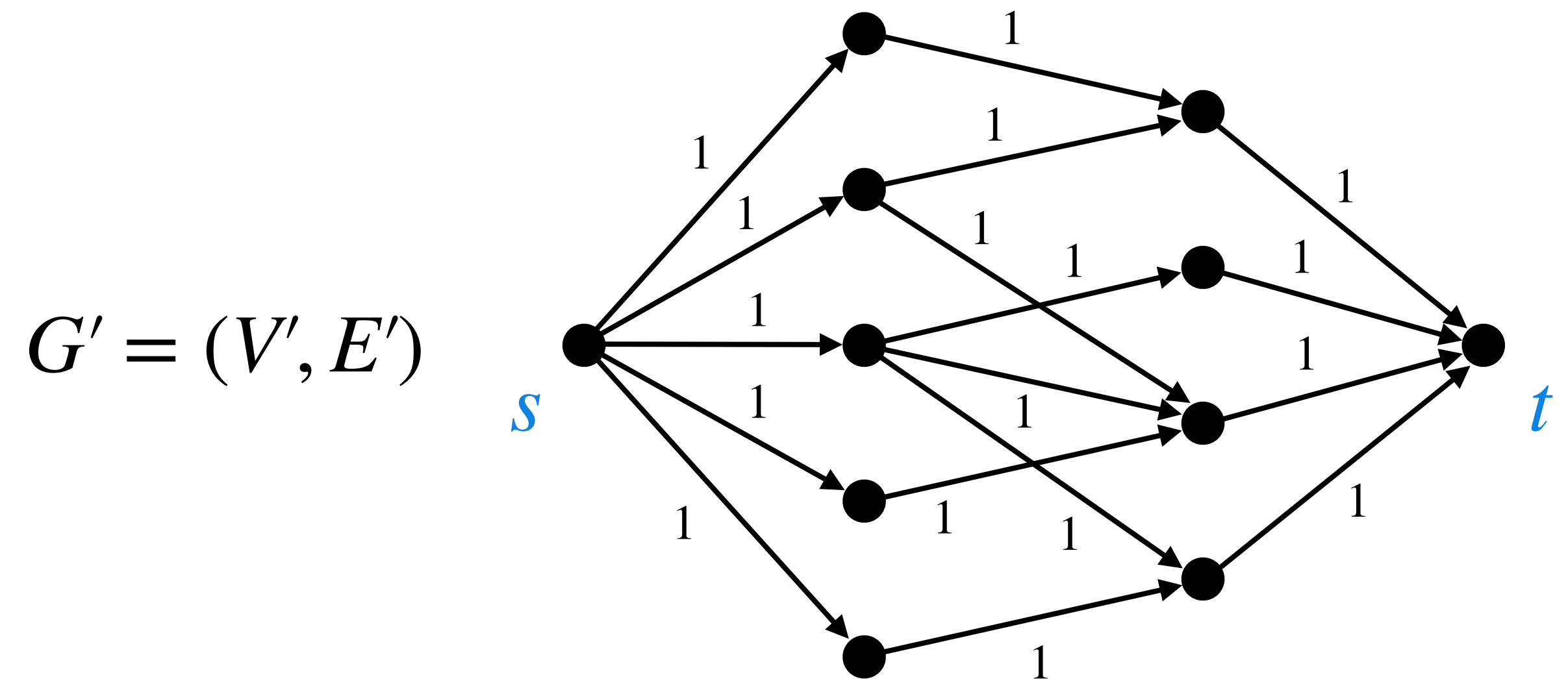
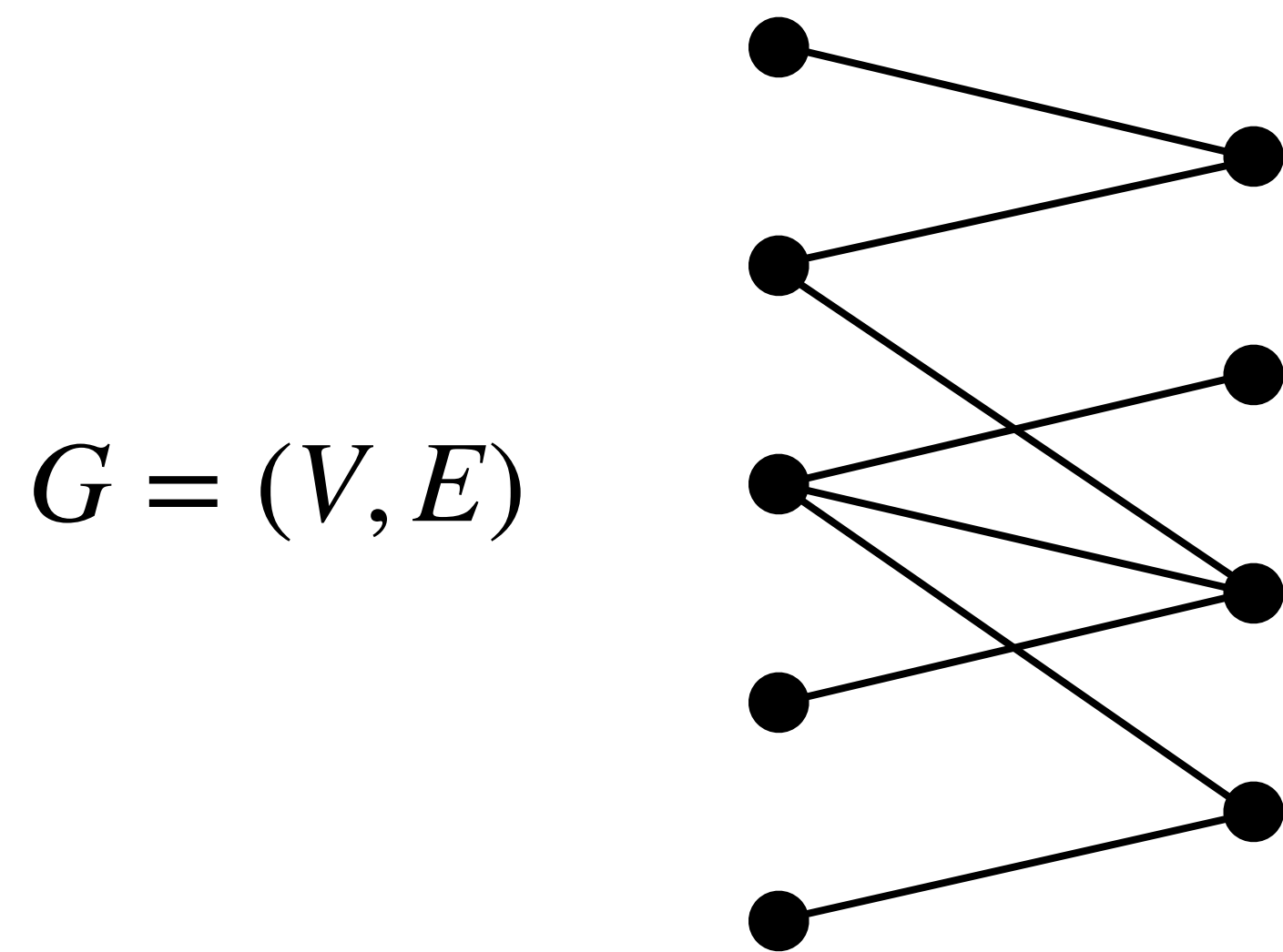


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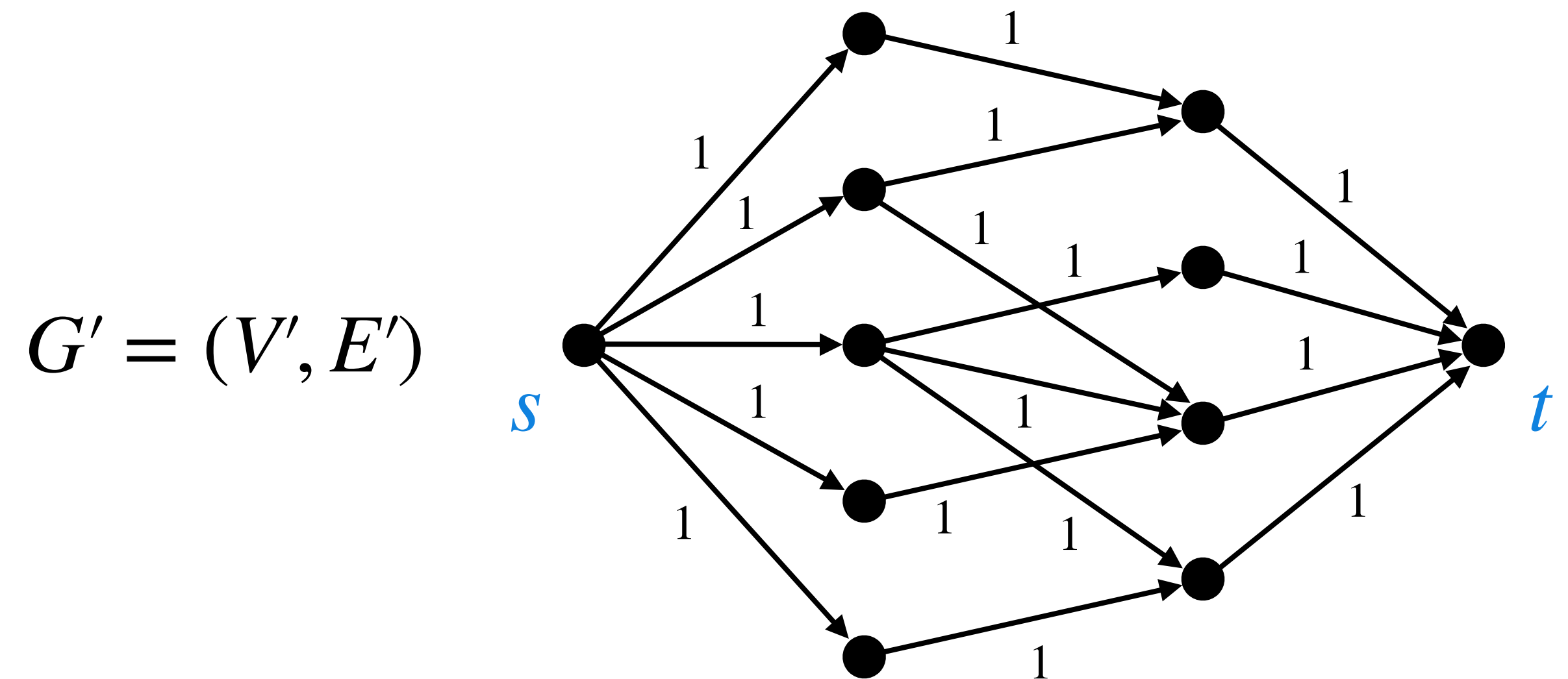
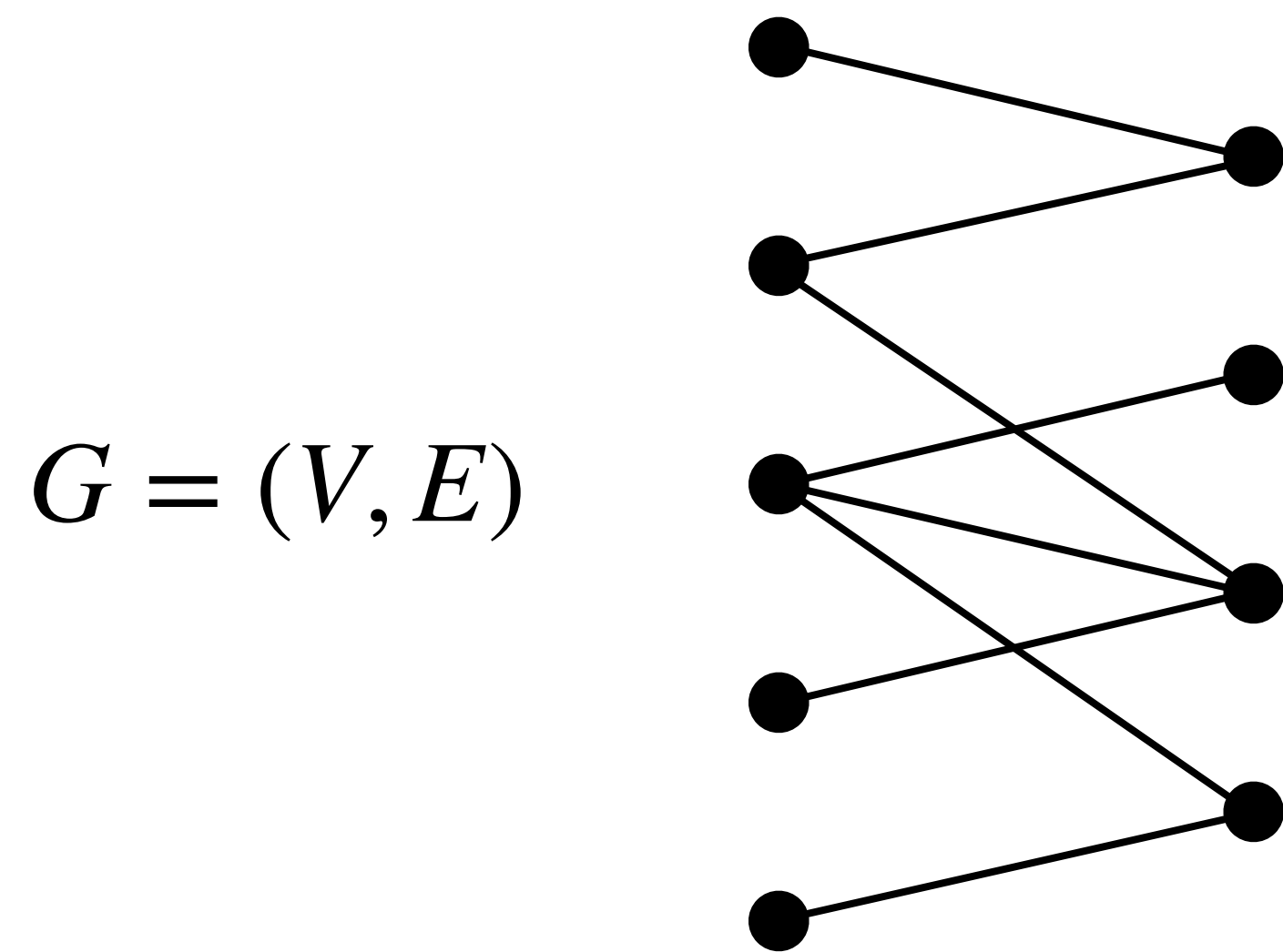
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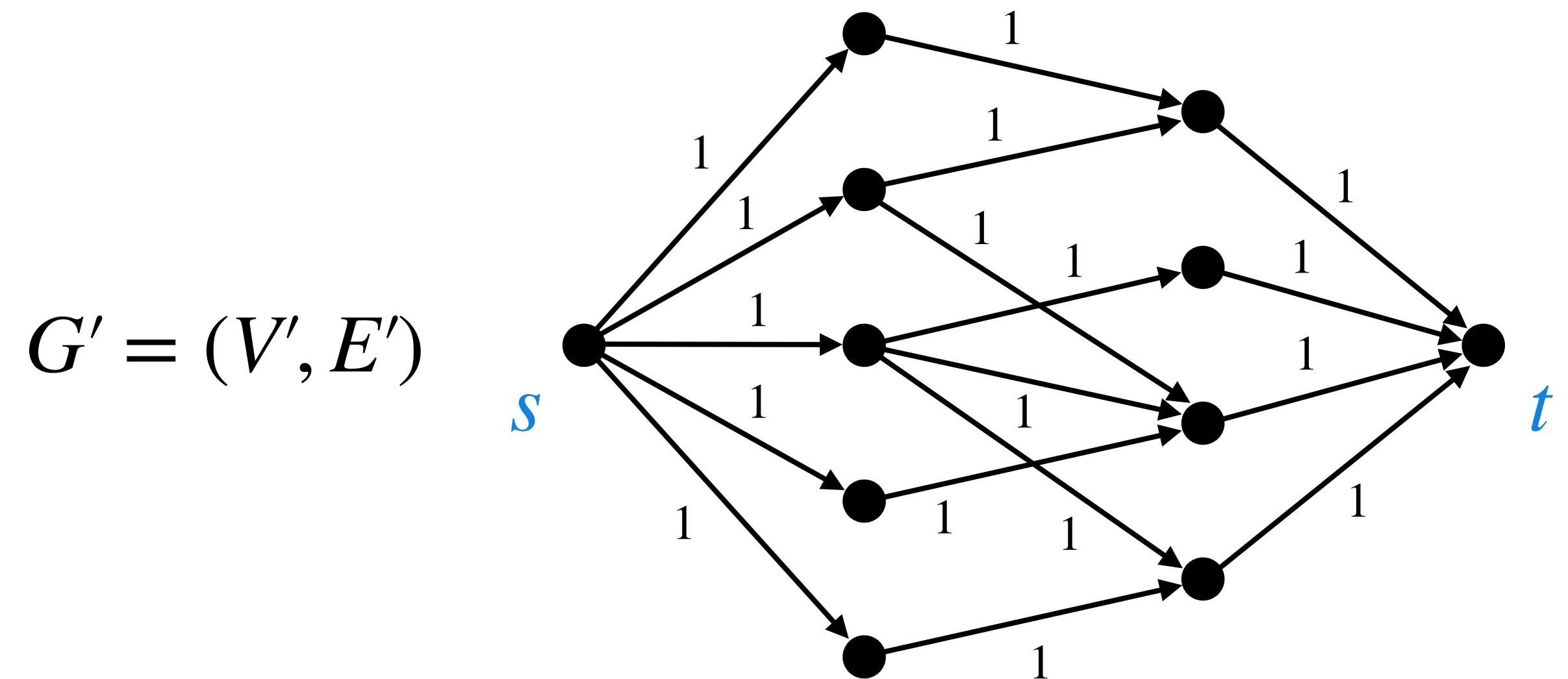
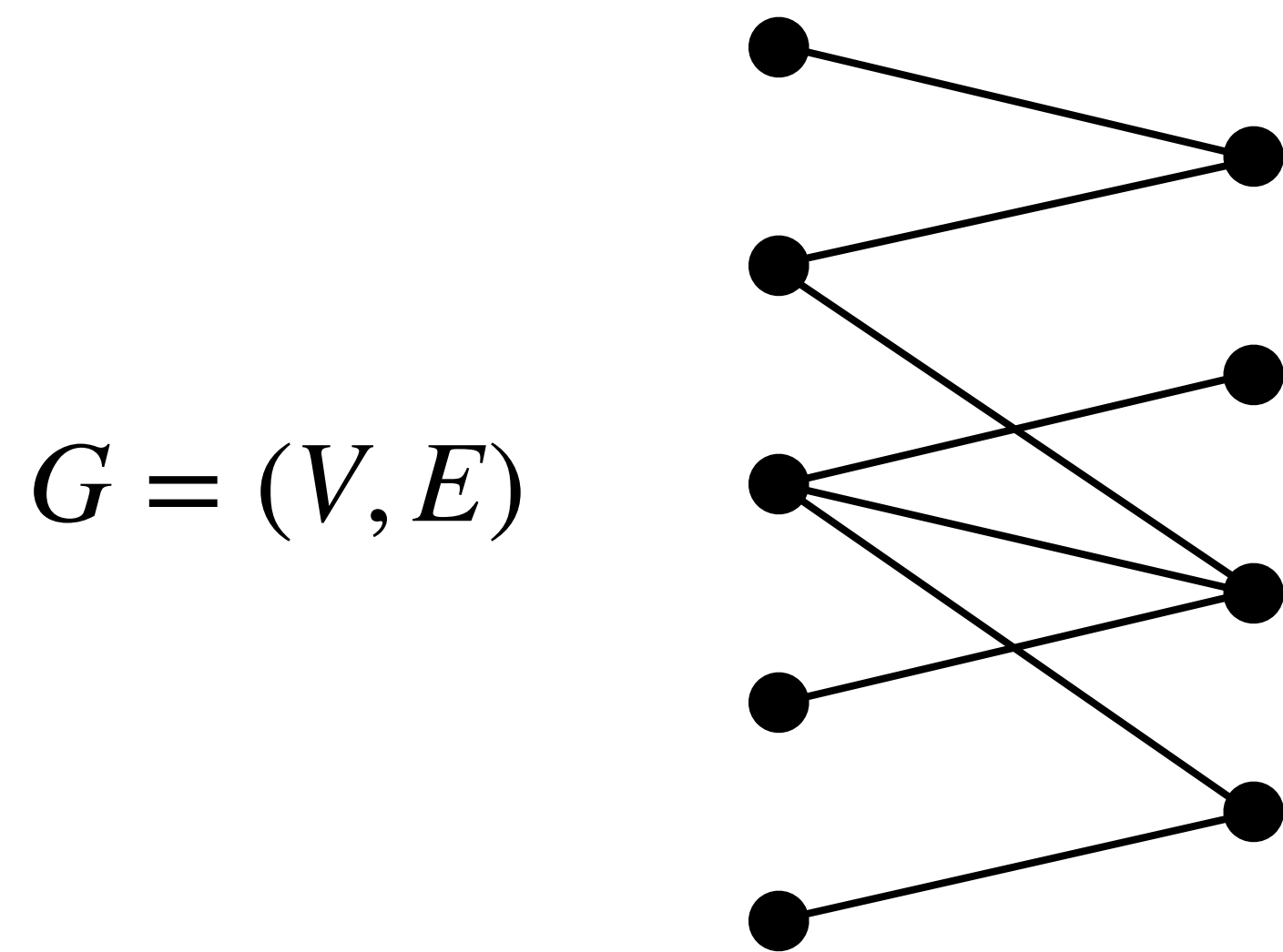
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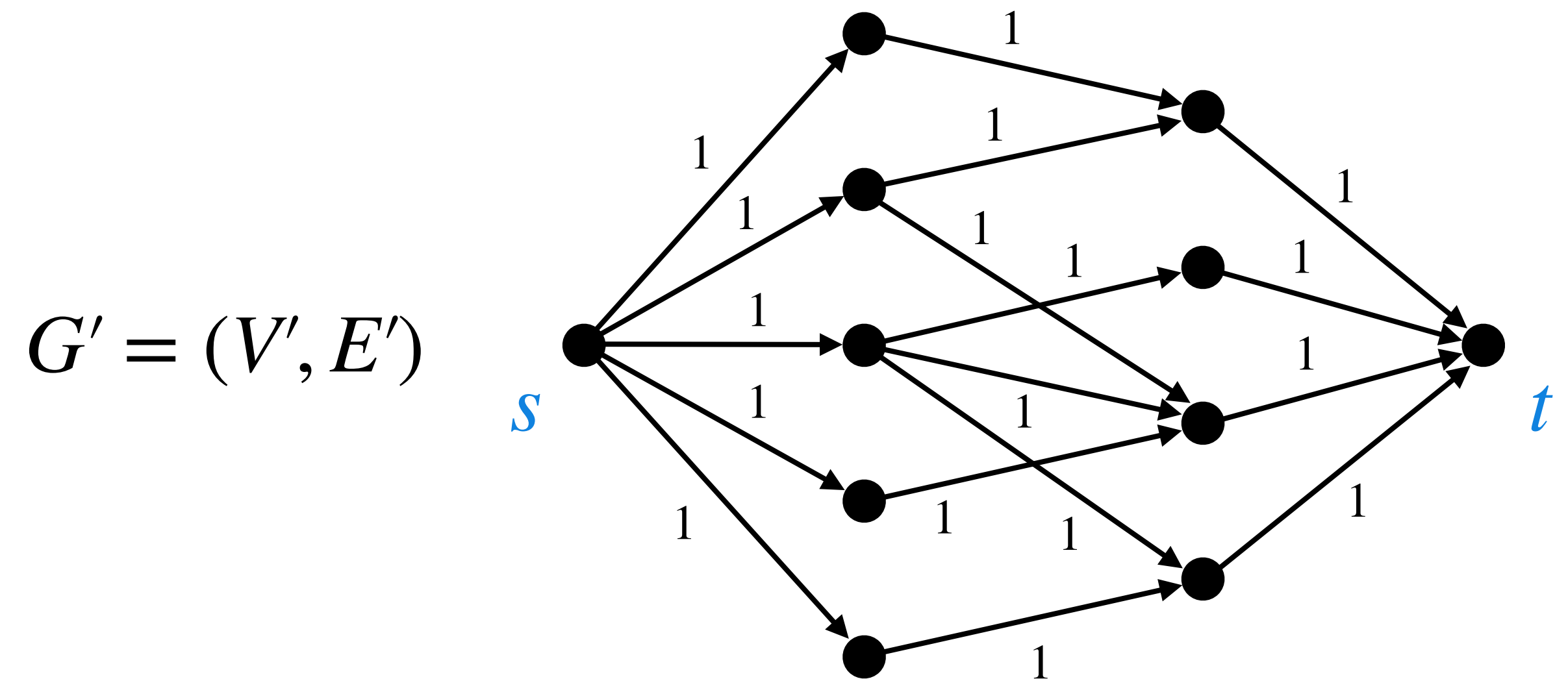
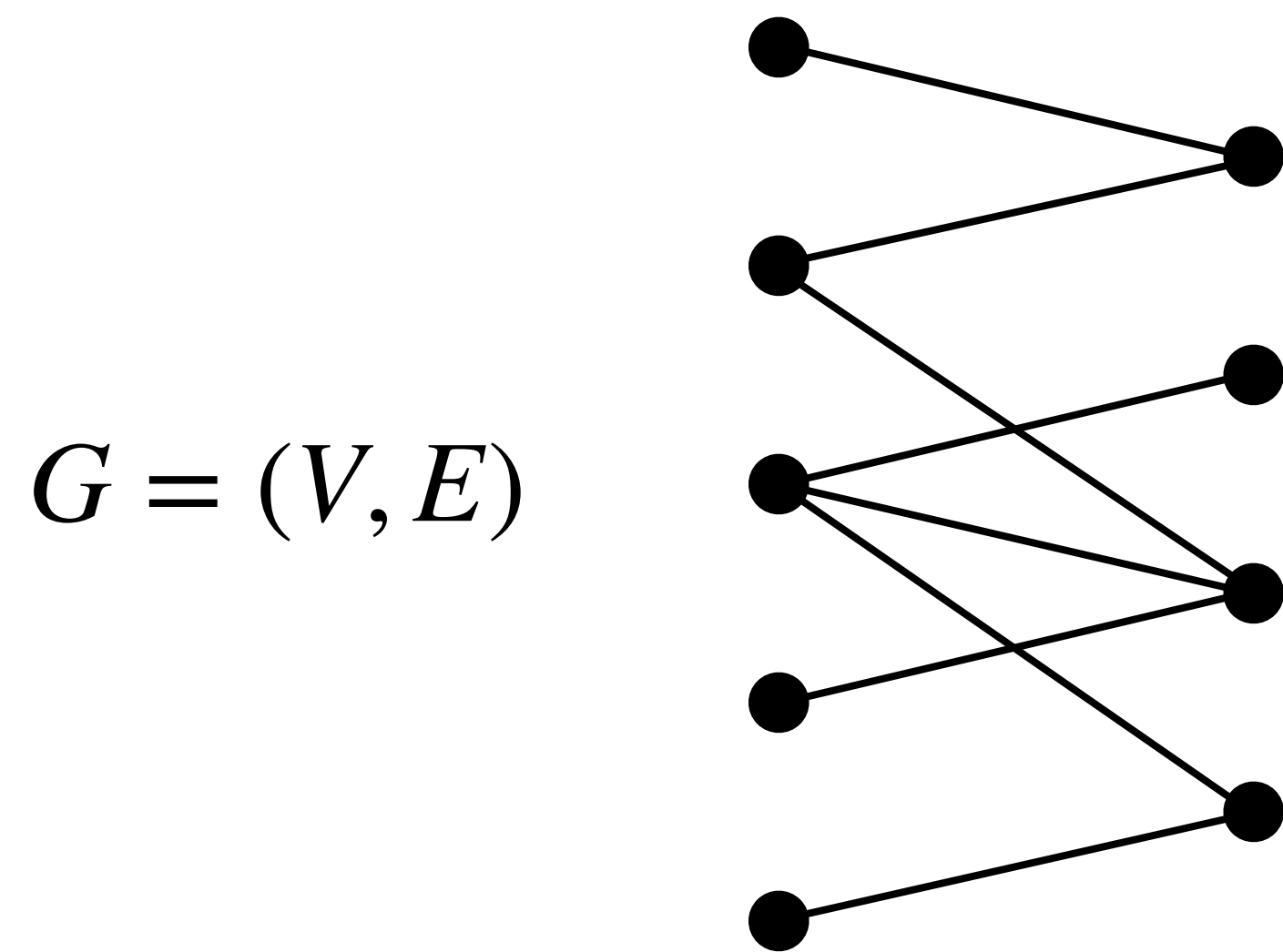
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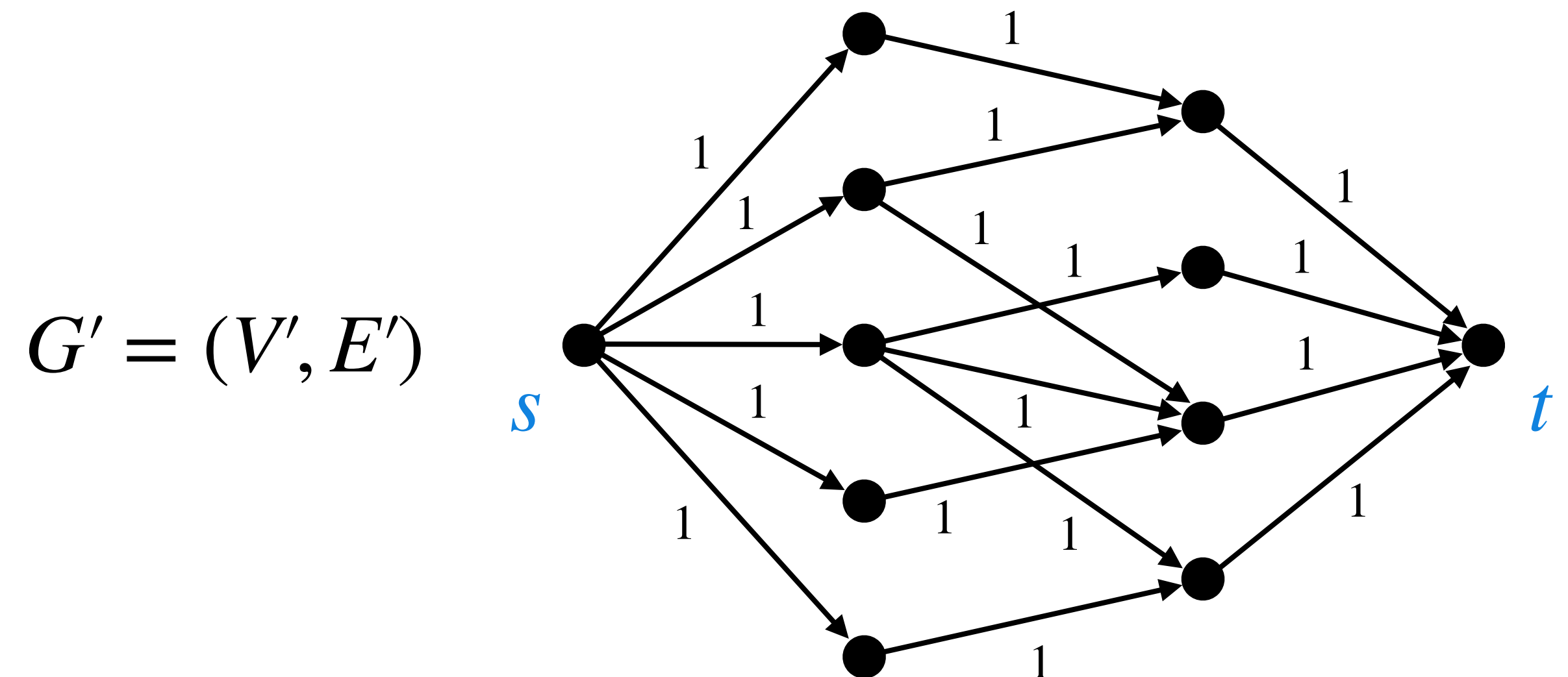
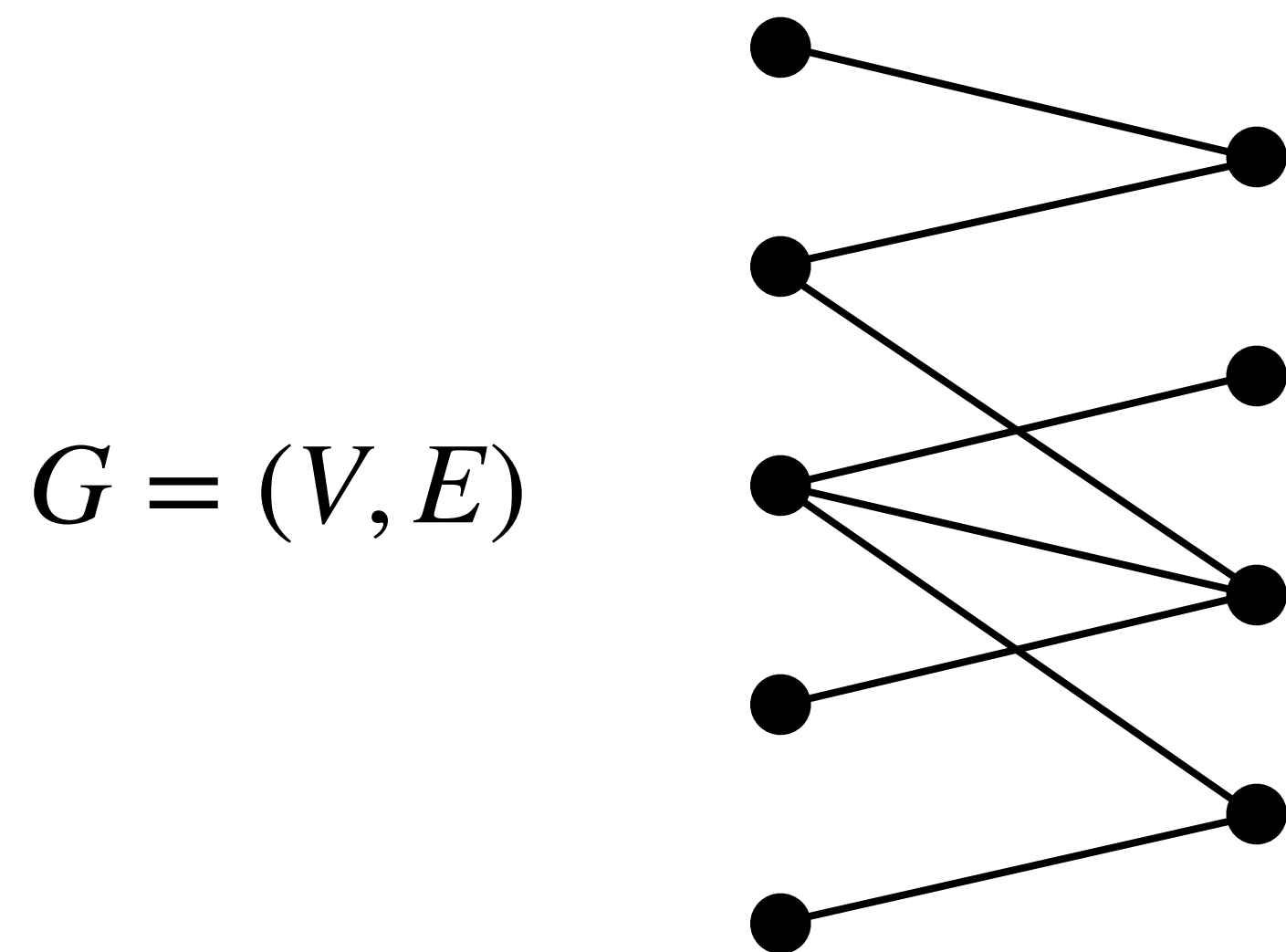
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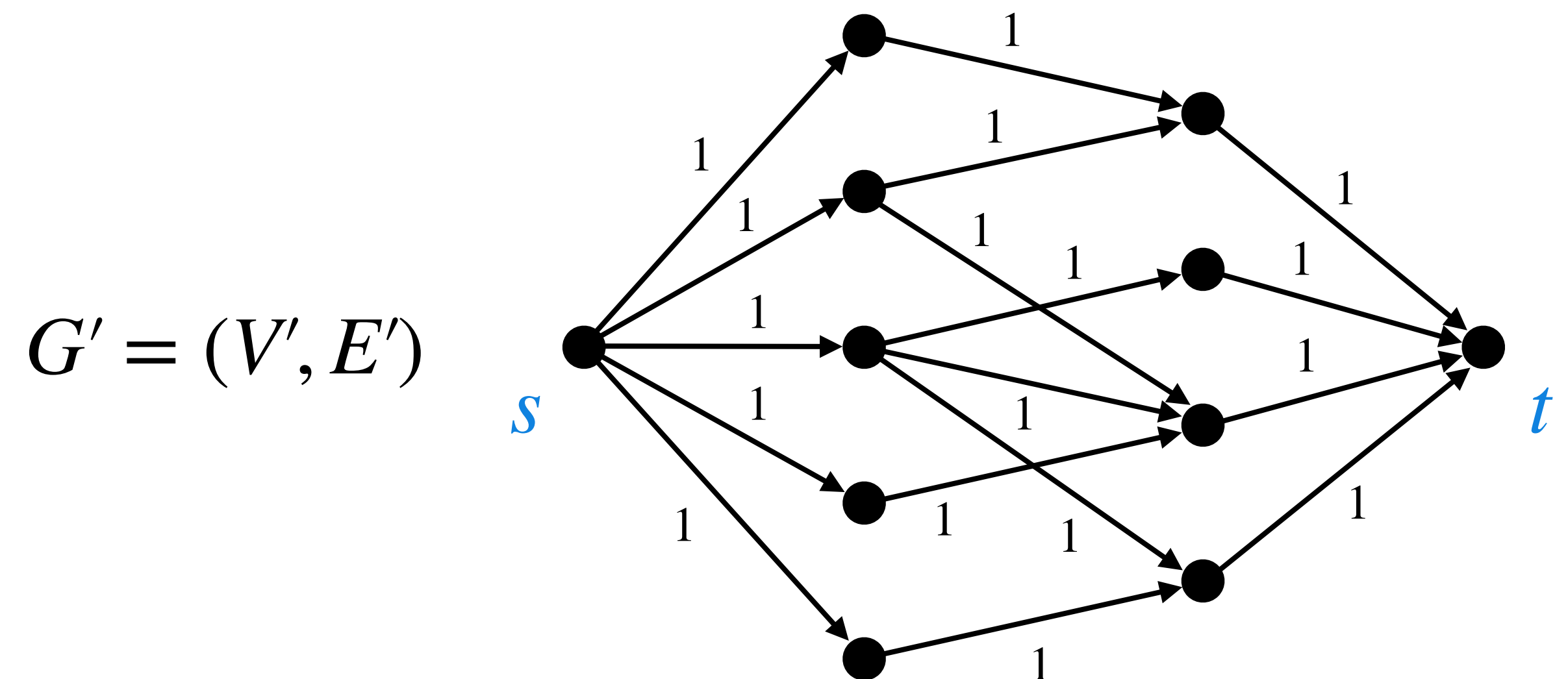
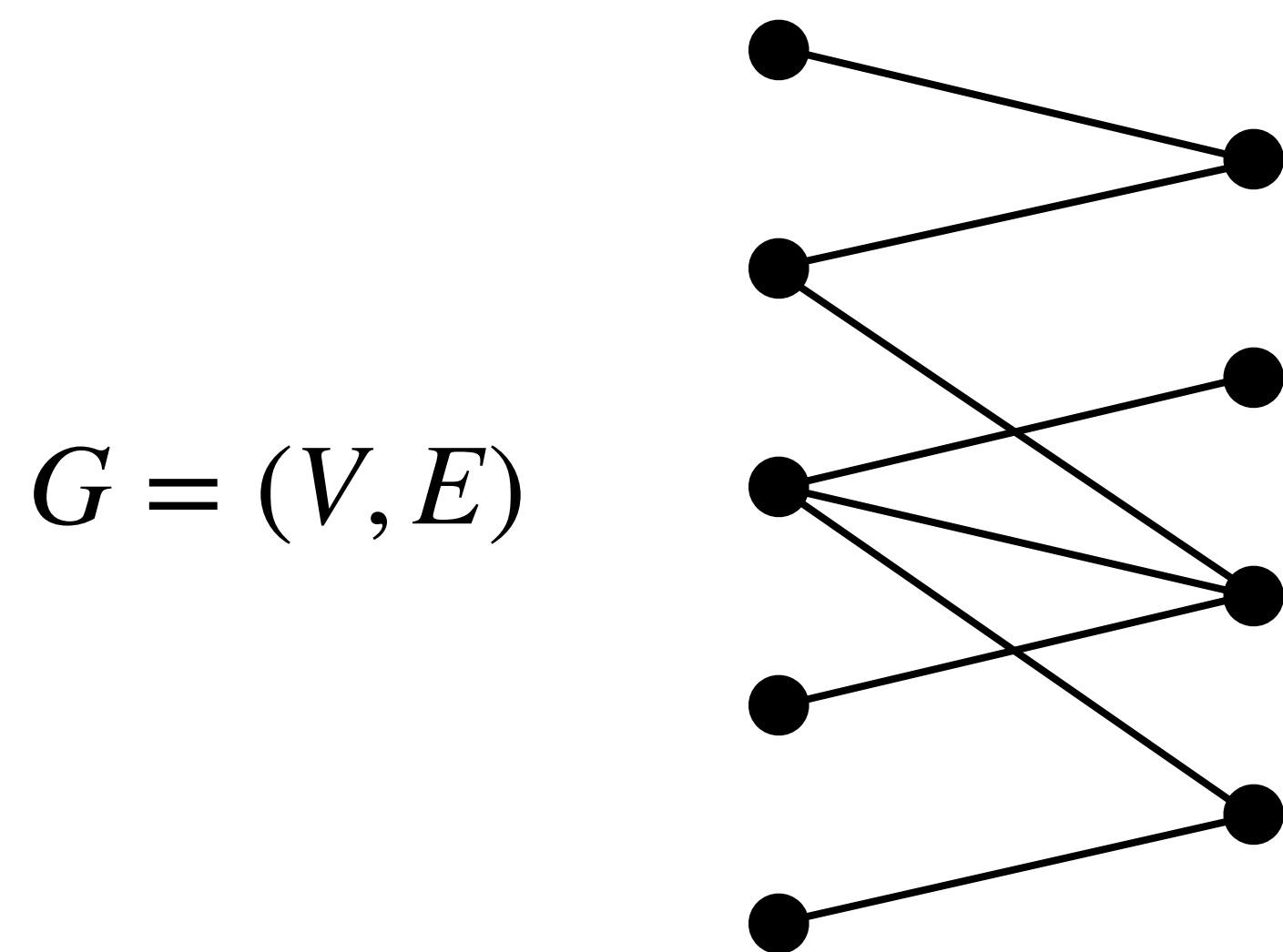
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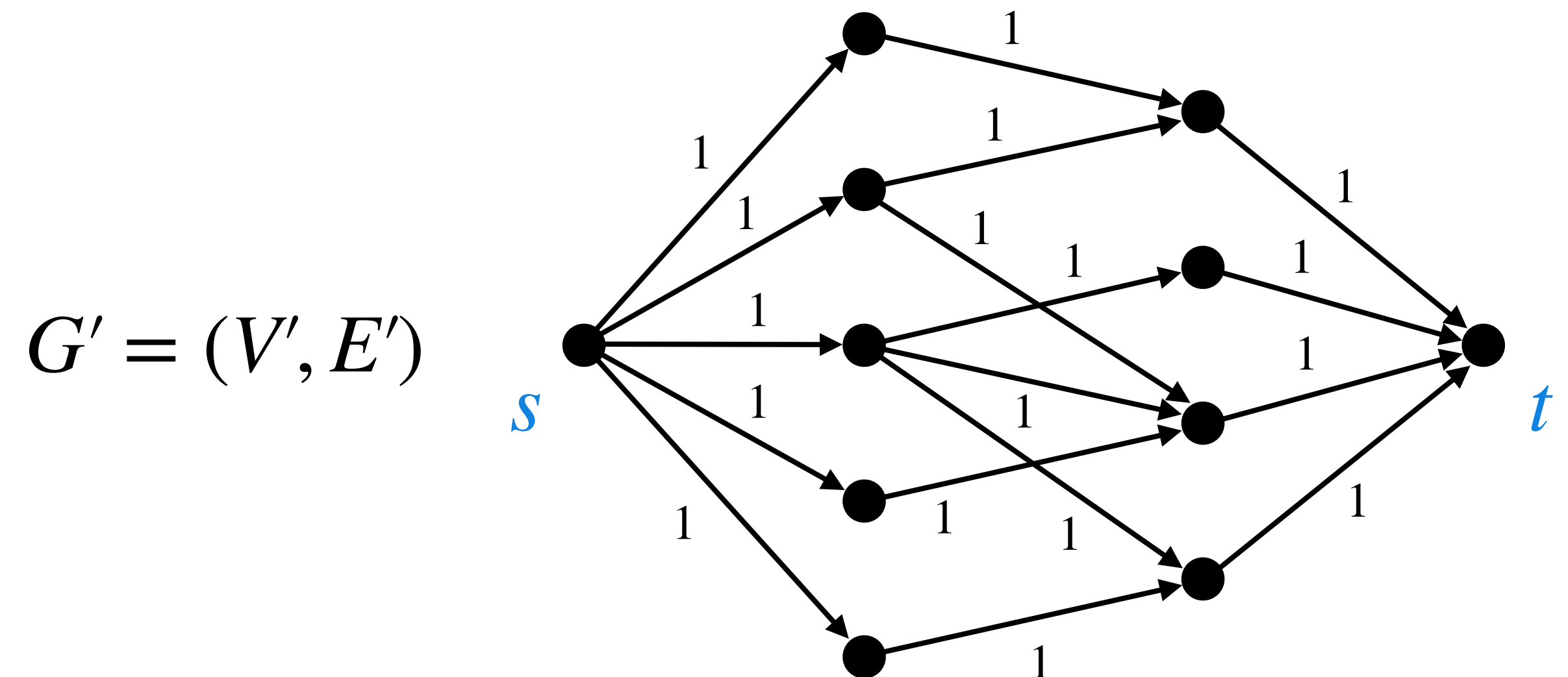
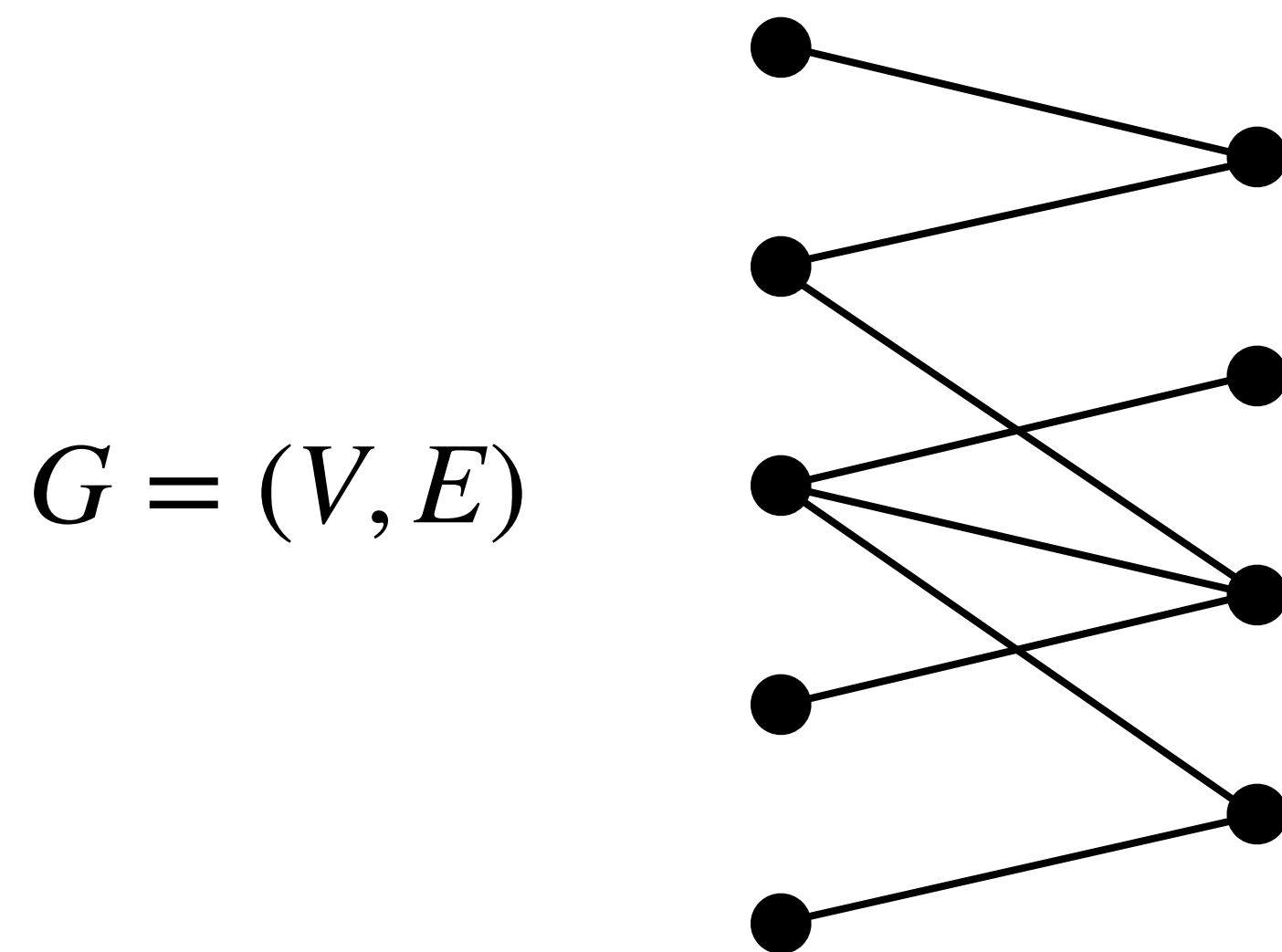
- $V' = V \cup \{s, t\}$, where s is the source and t is the sink.
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Bipartite Matching to Flow

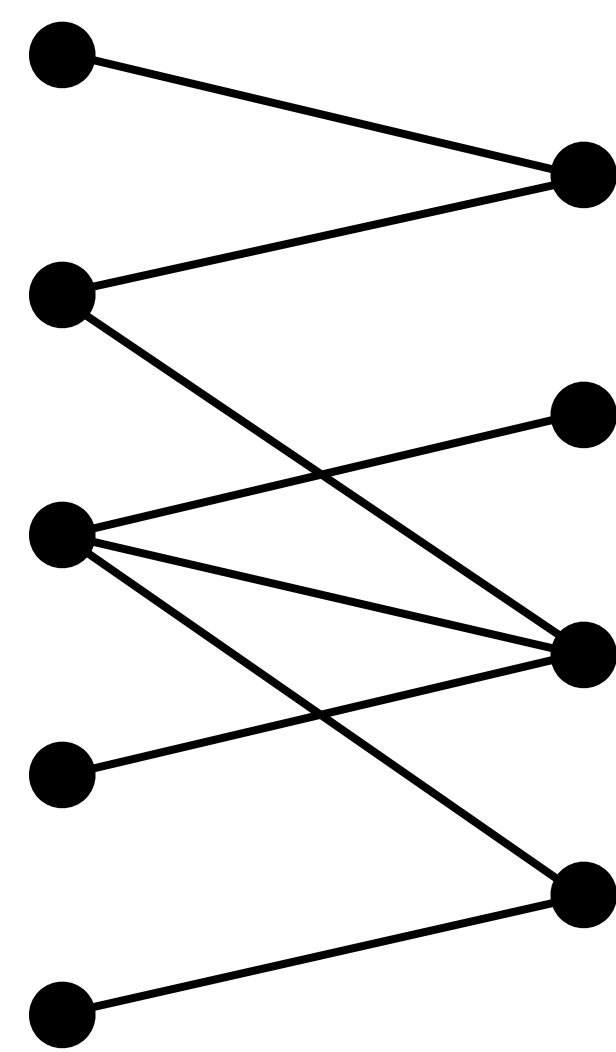
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- Every edge has capacity **one**.

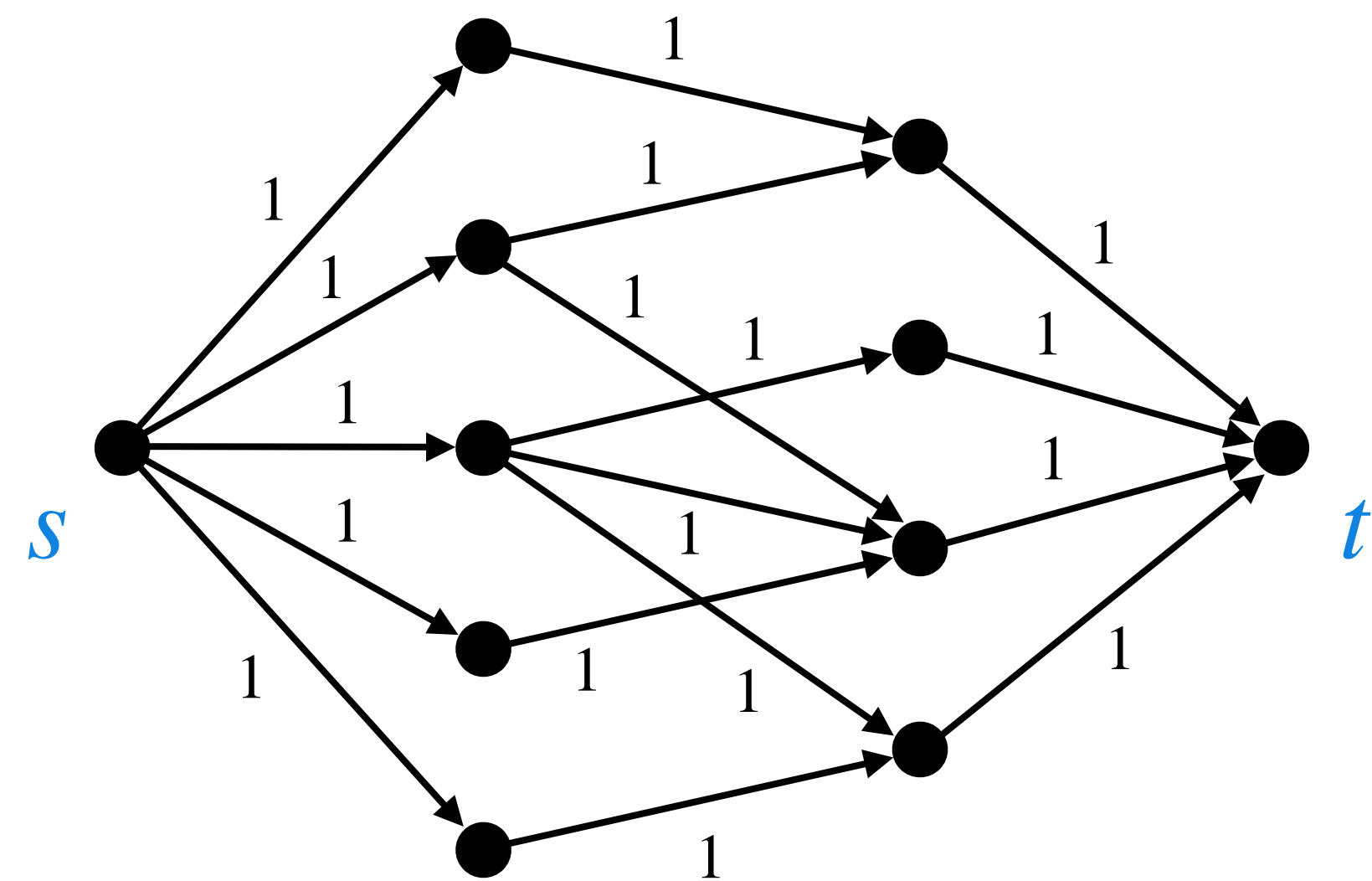


Bipartite Matching to Flow

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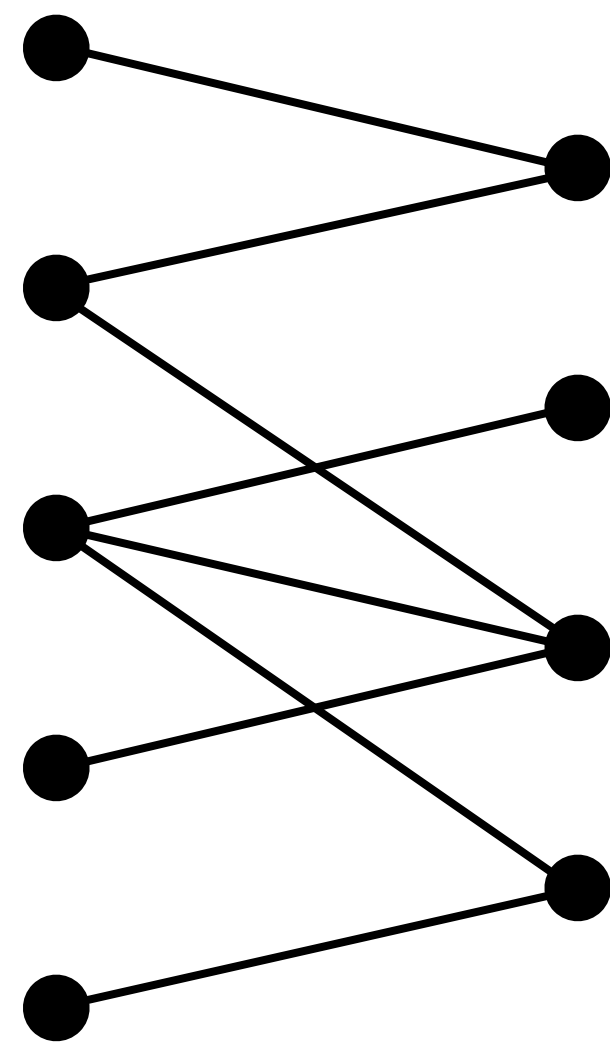
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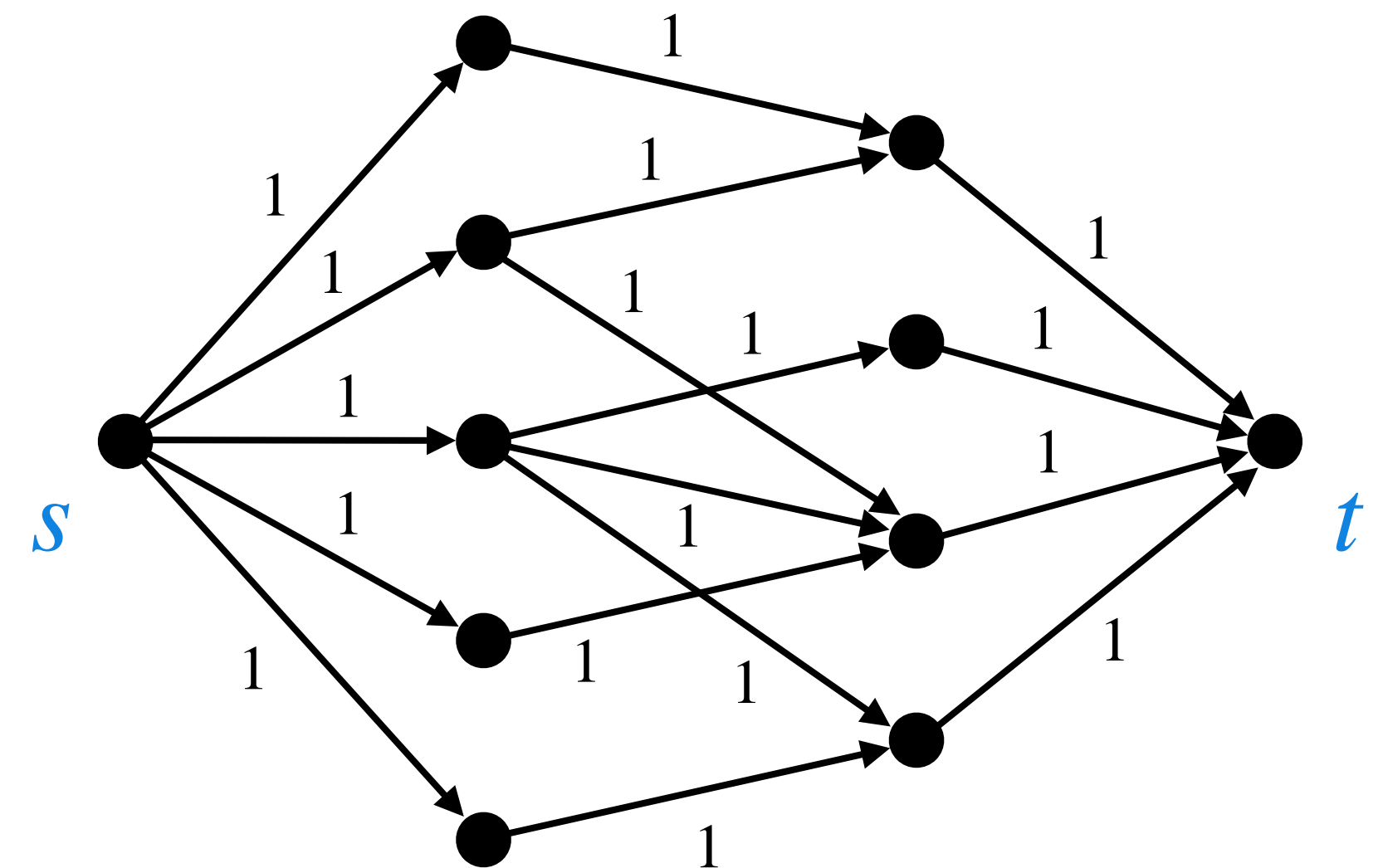
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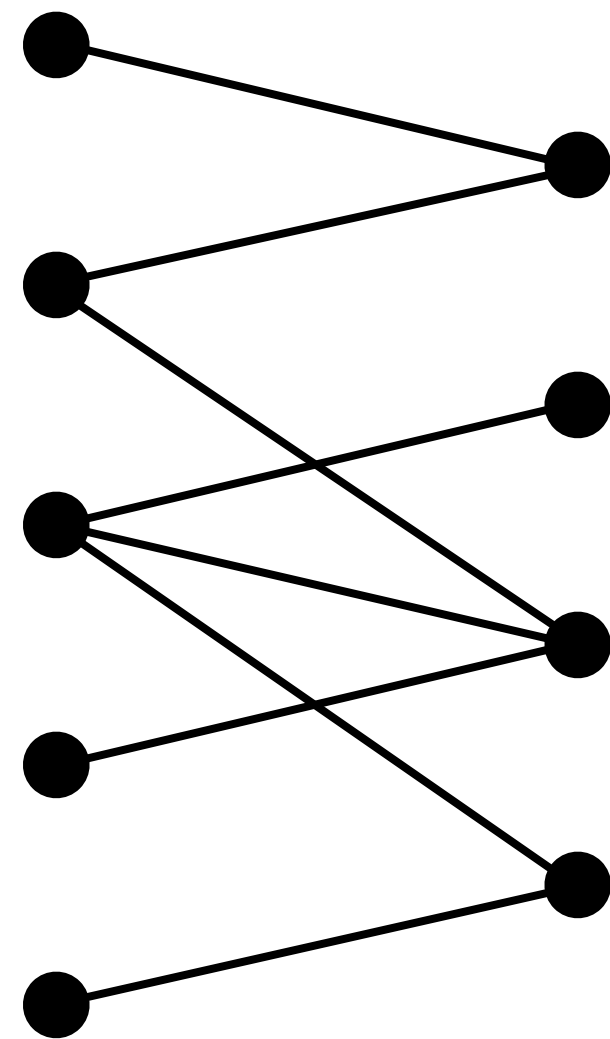


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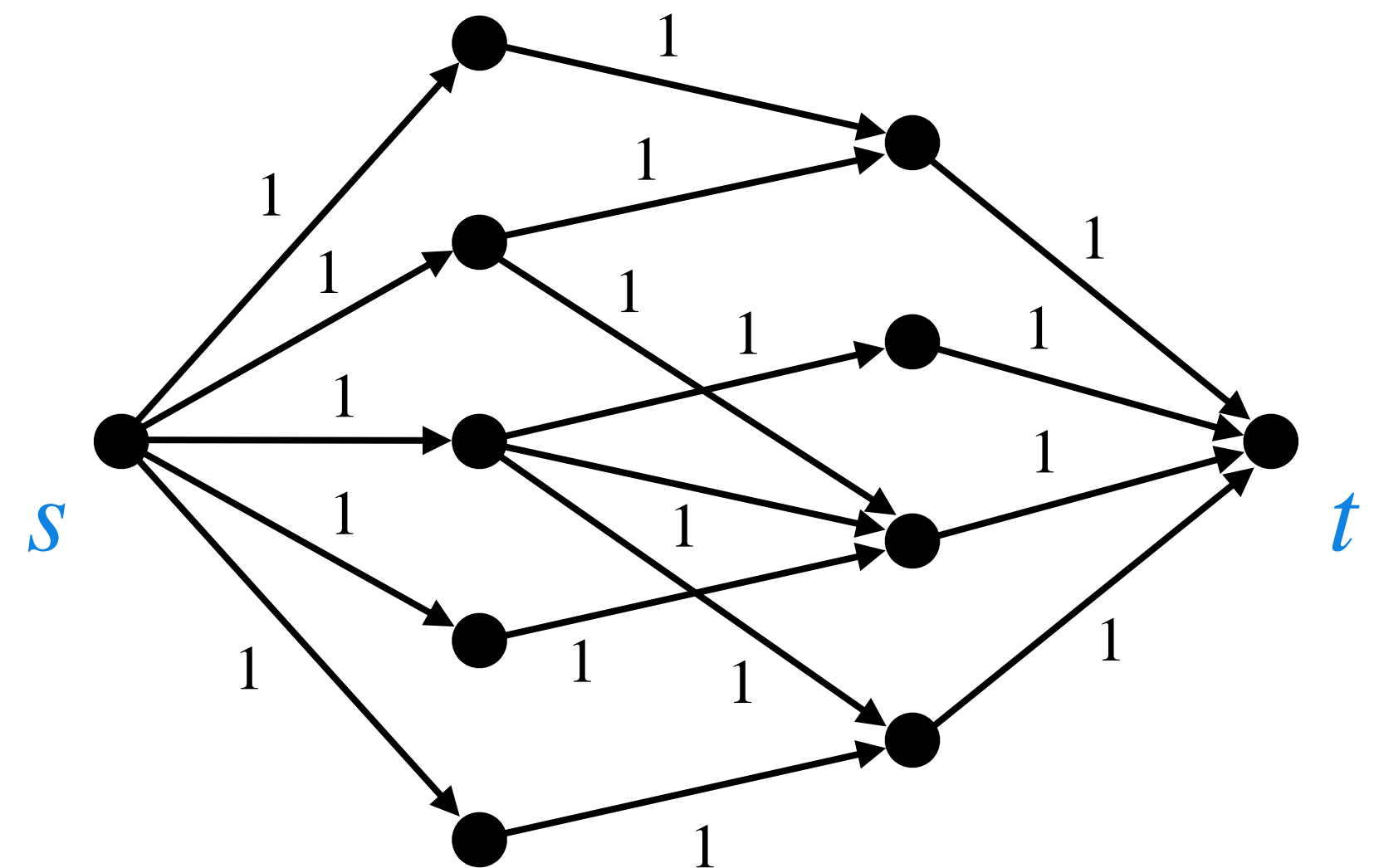
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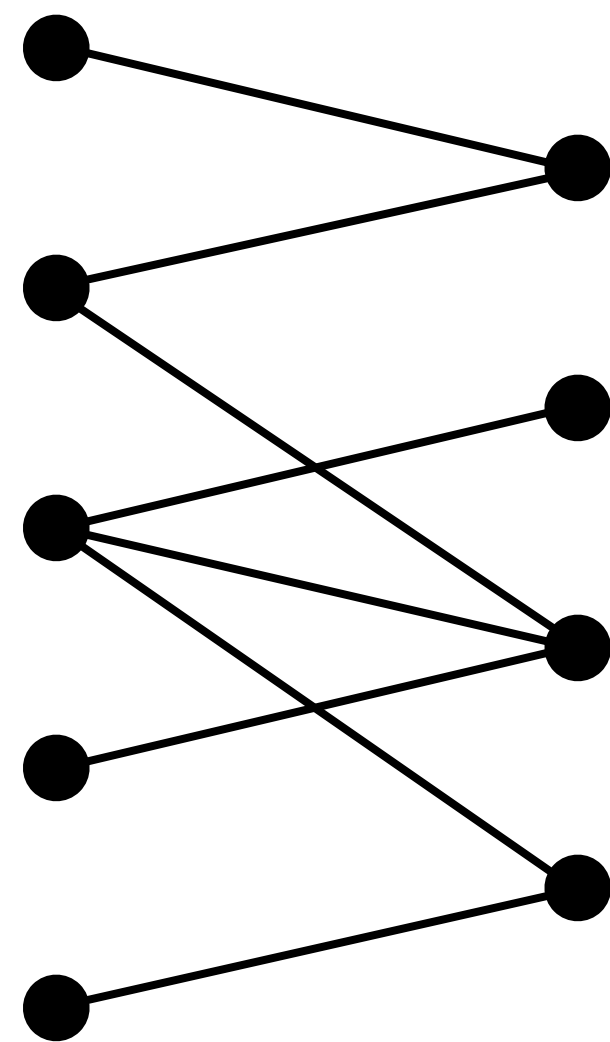
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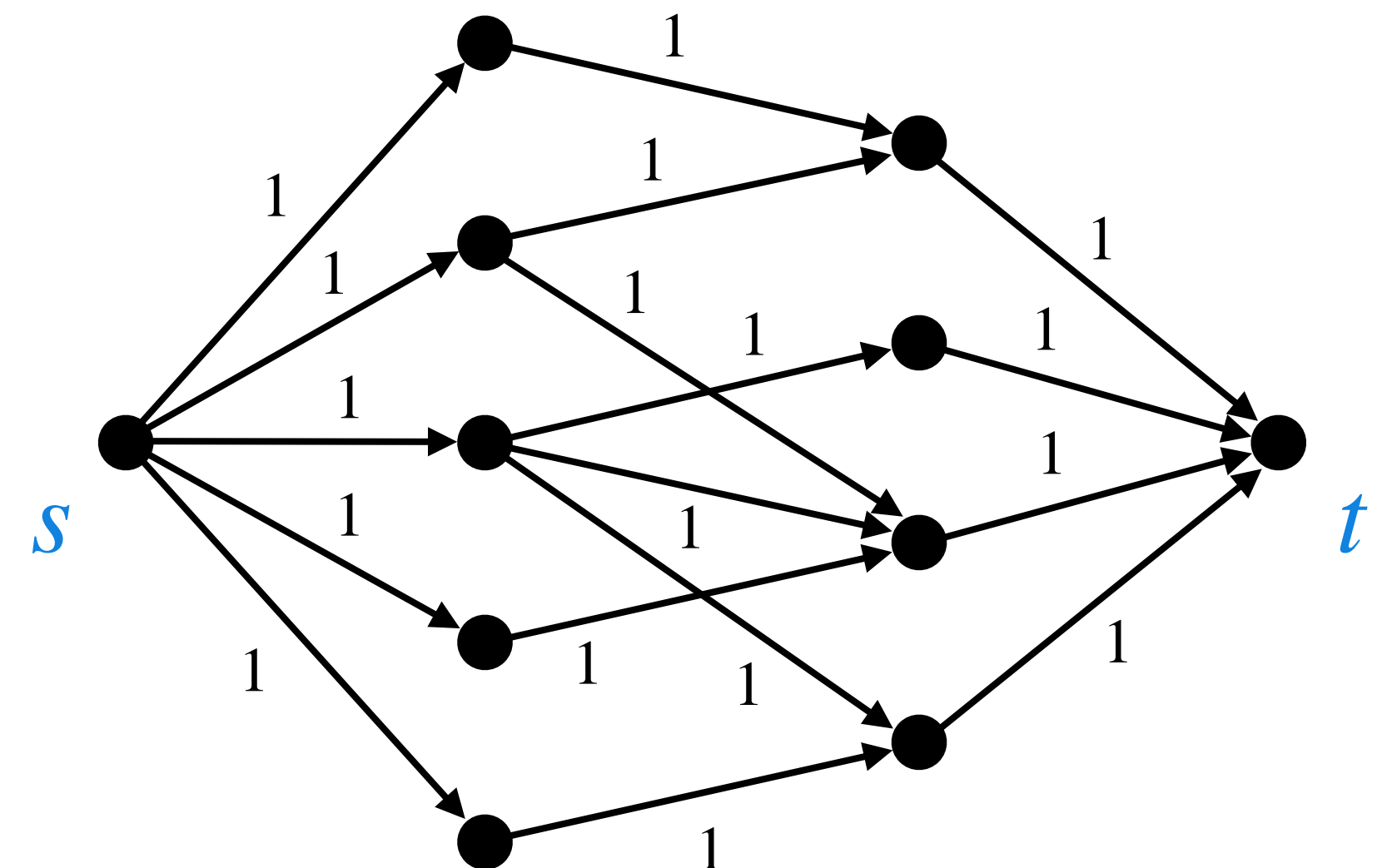
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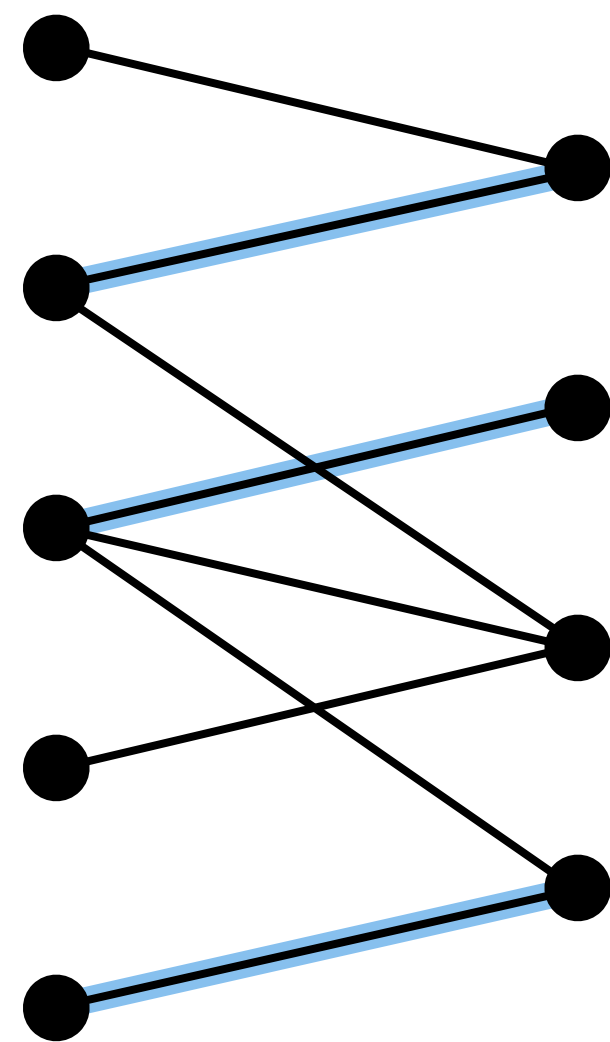
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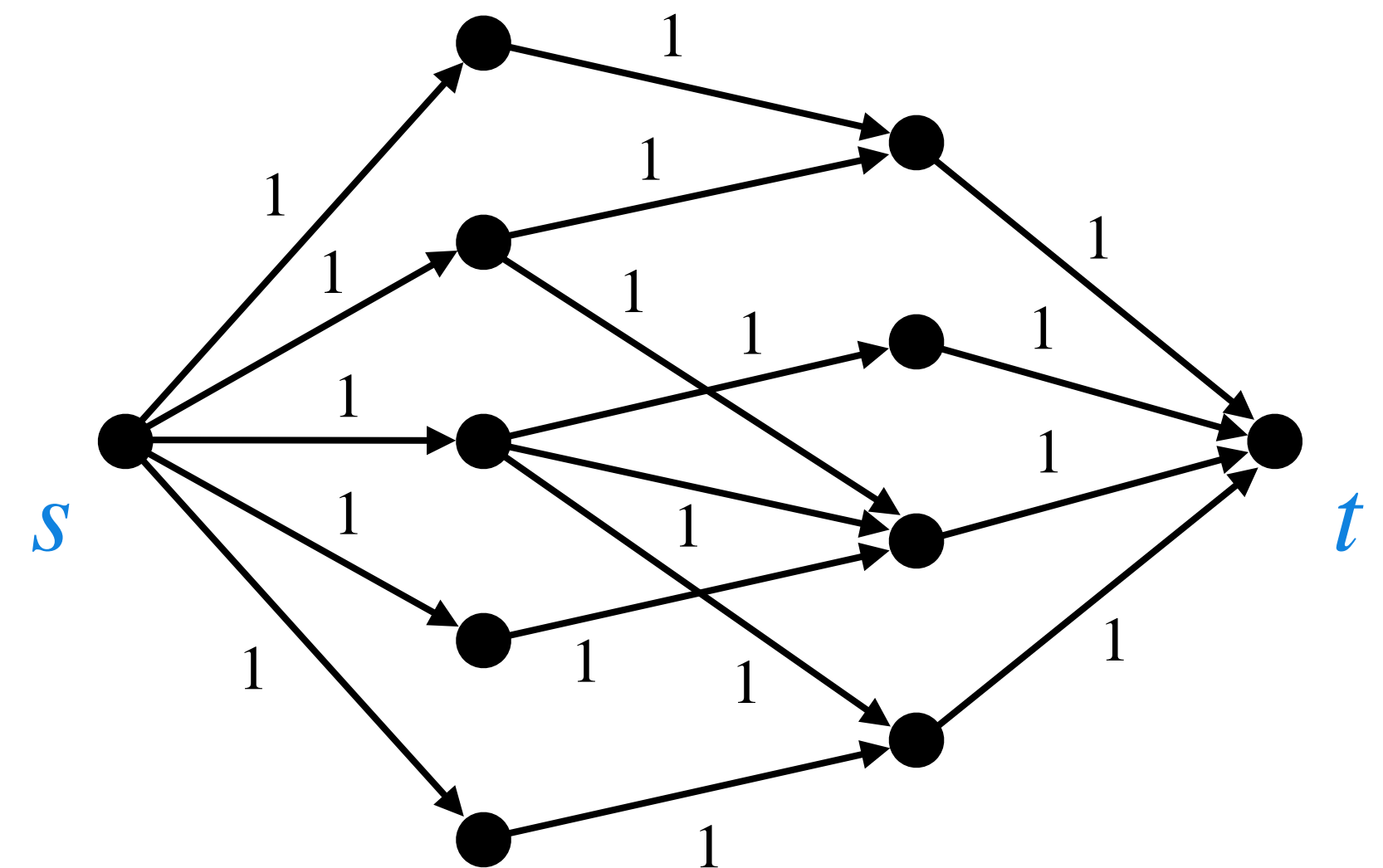
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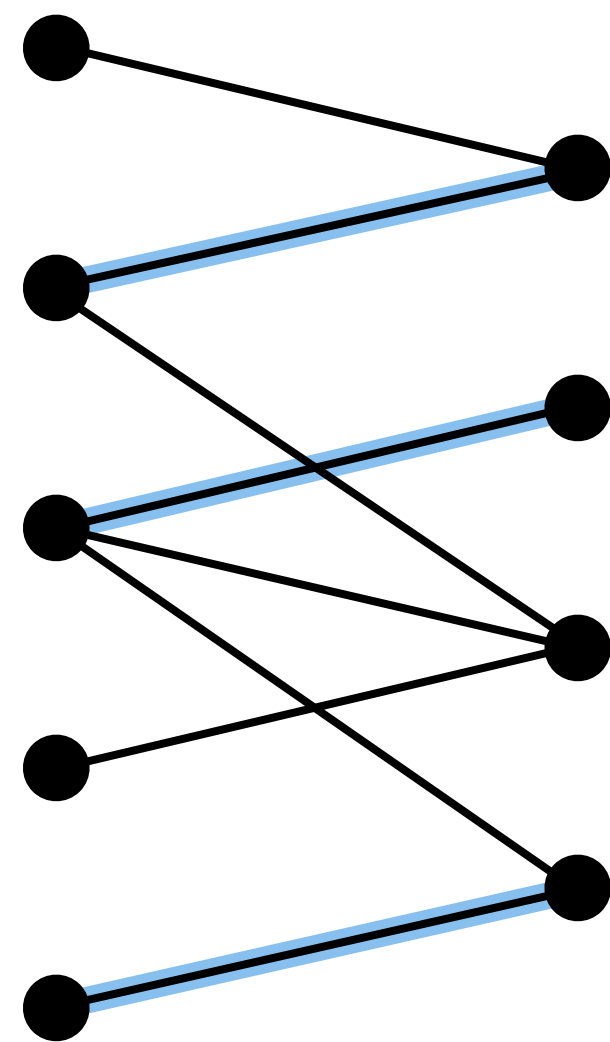
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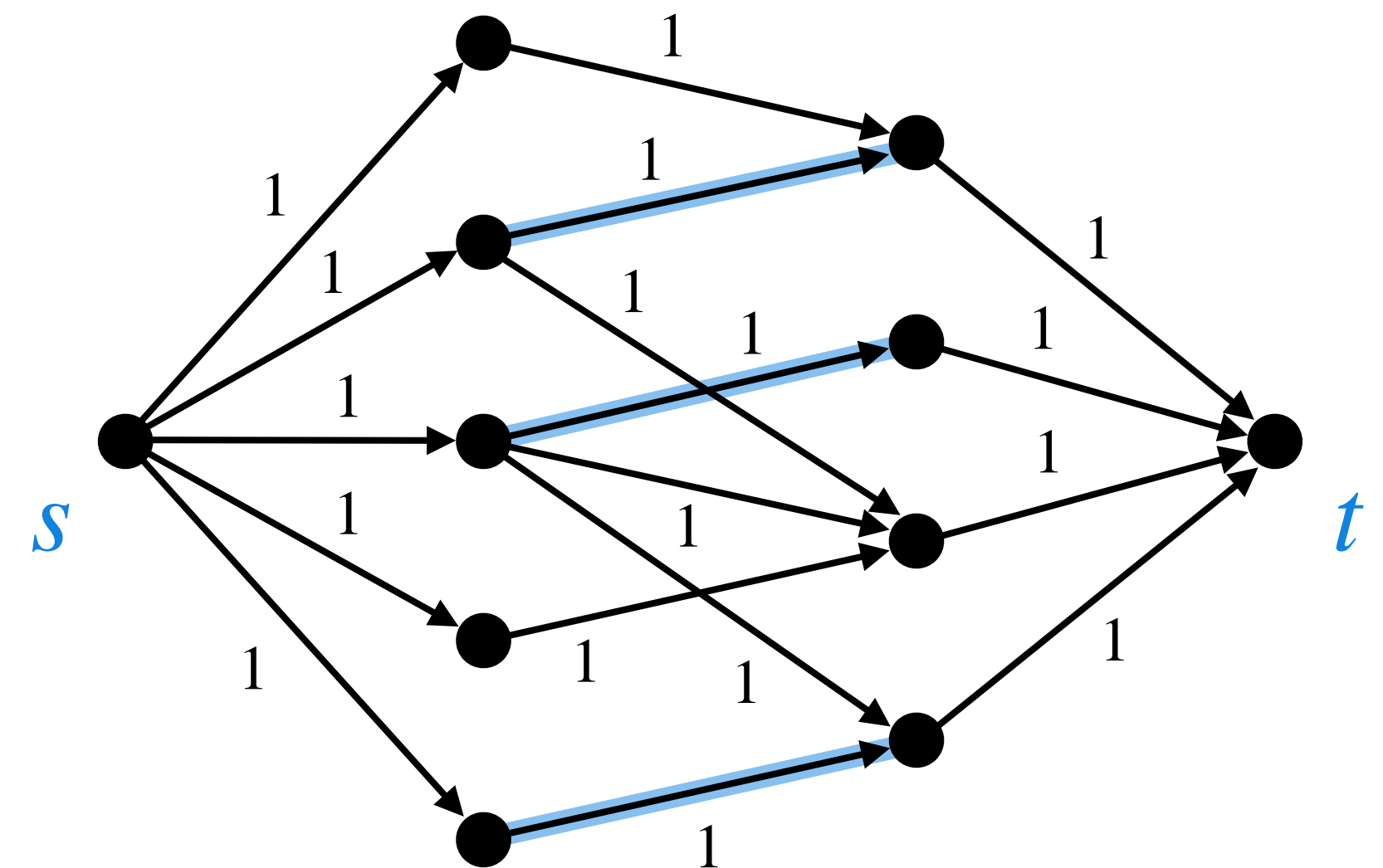
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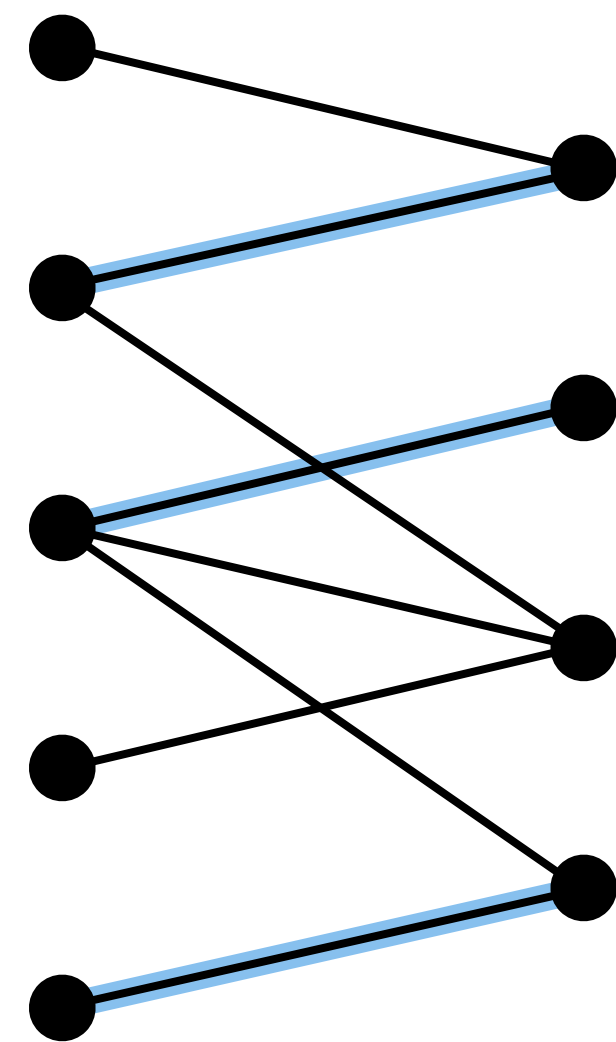
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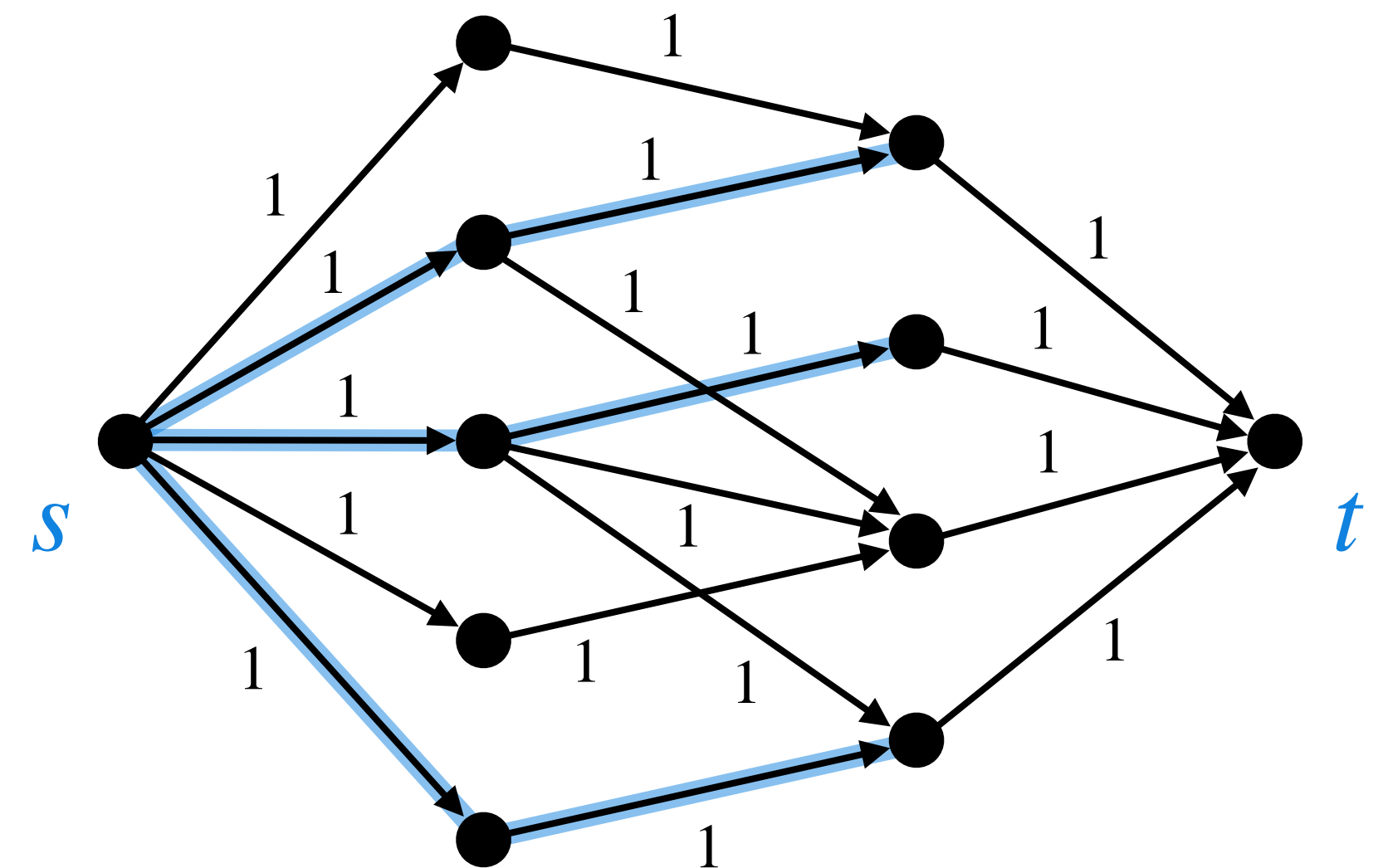
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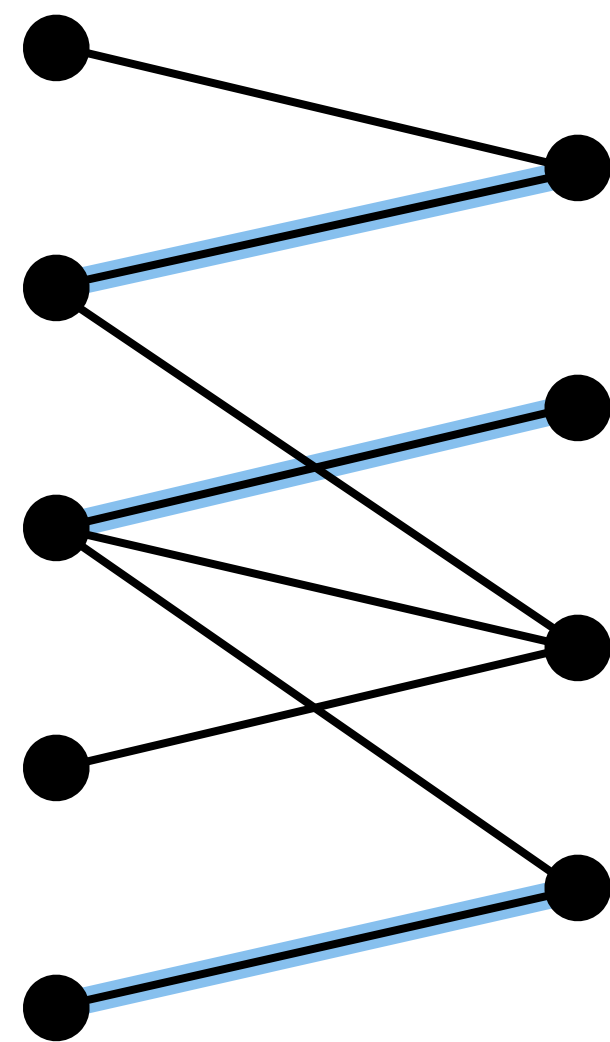
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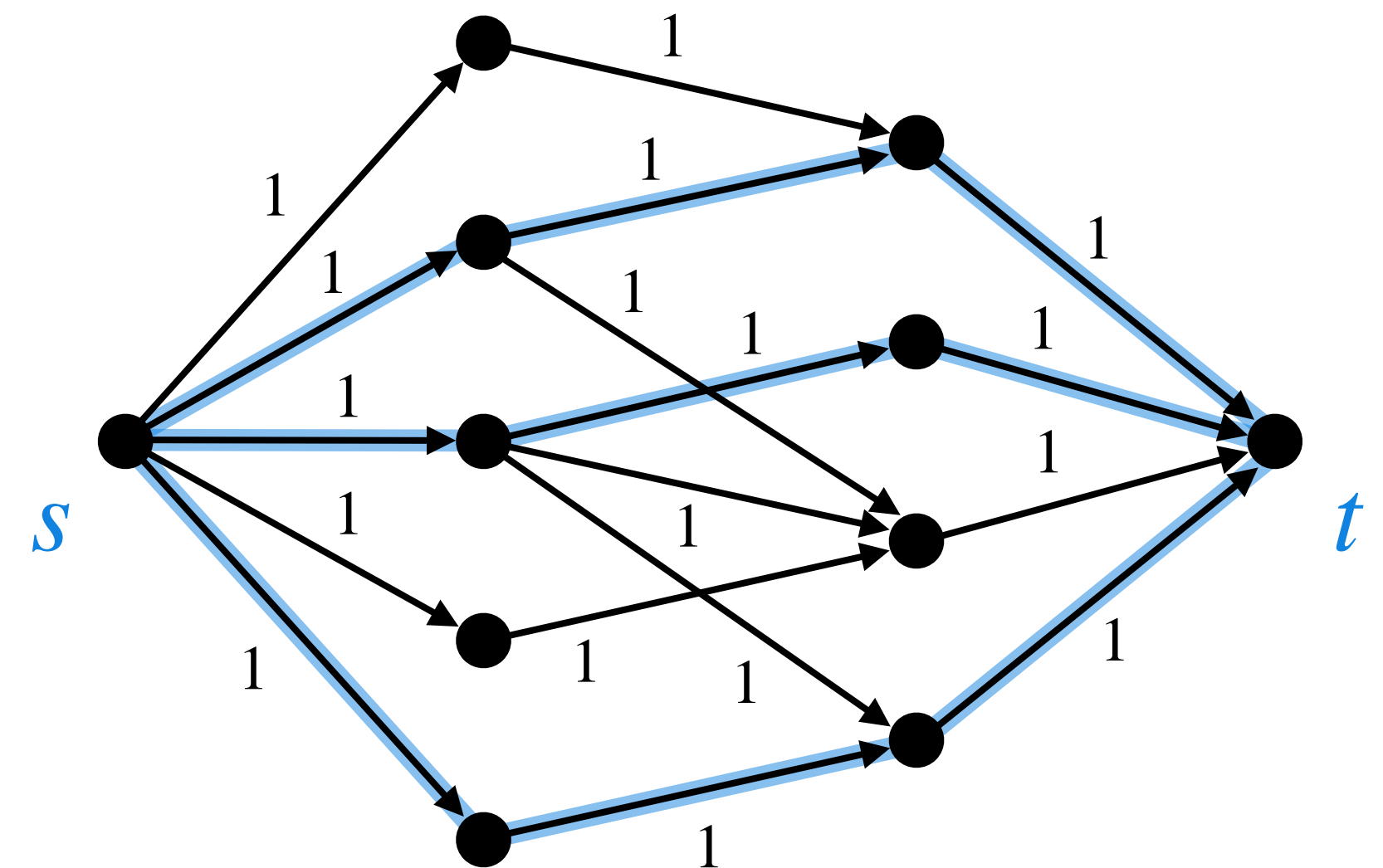
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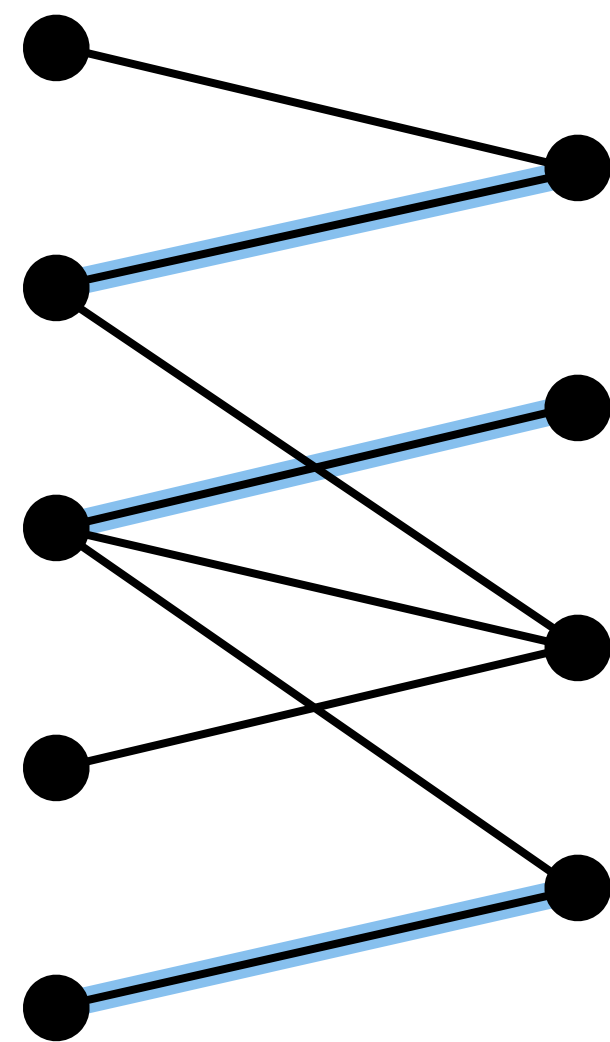


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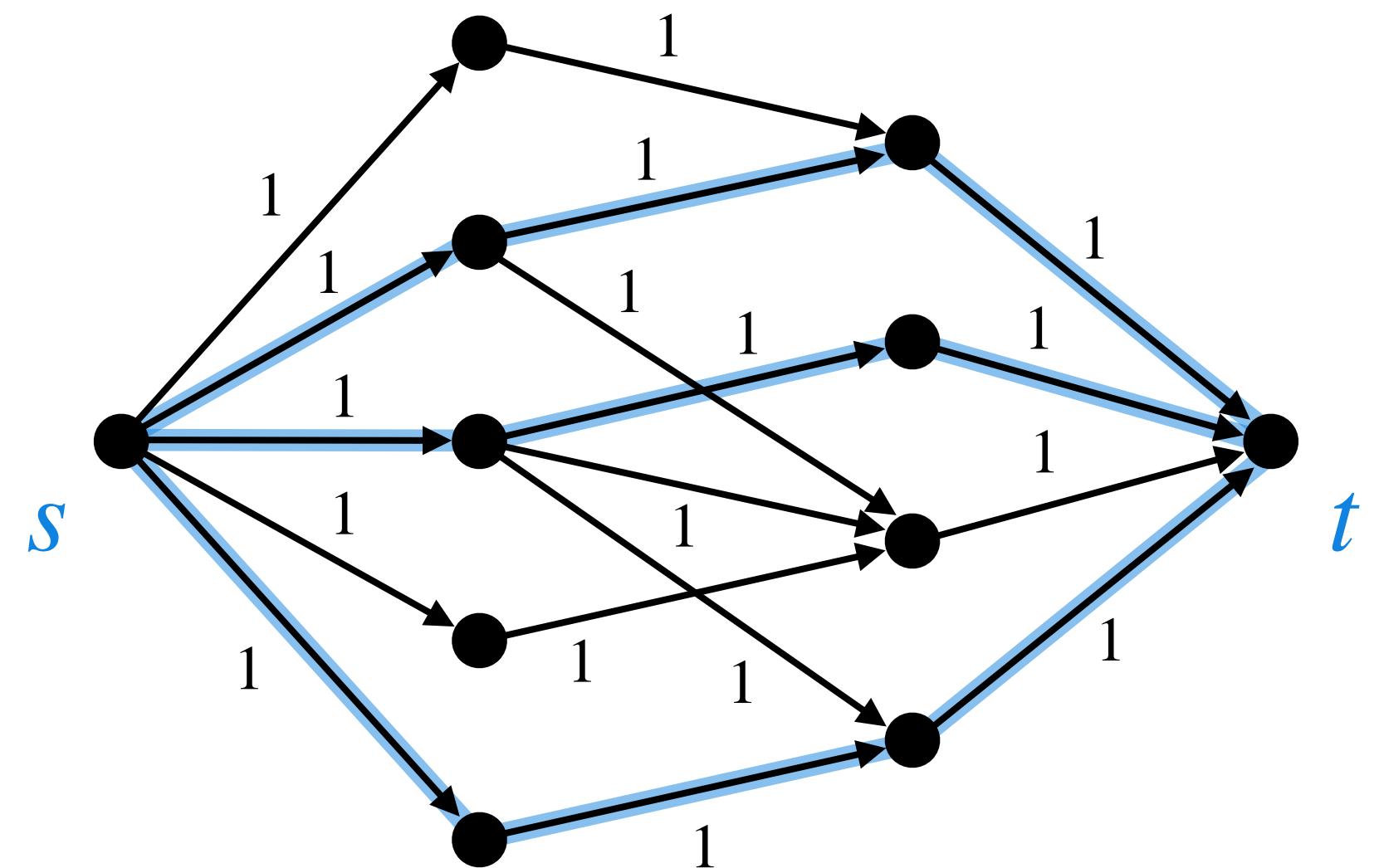
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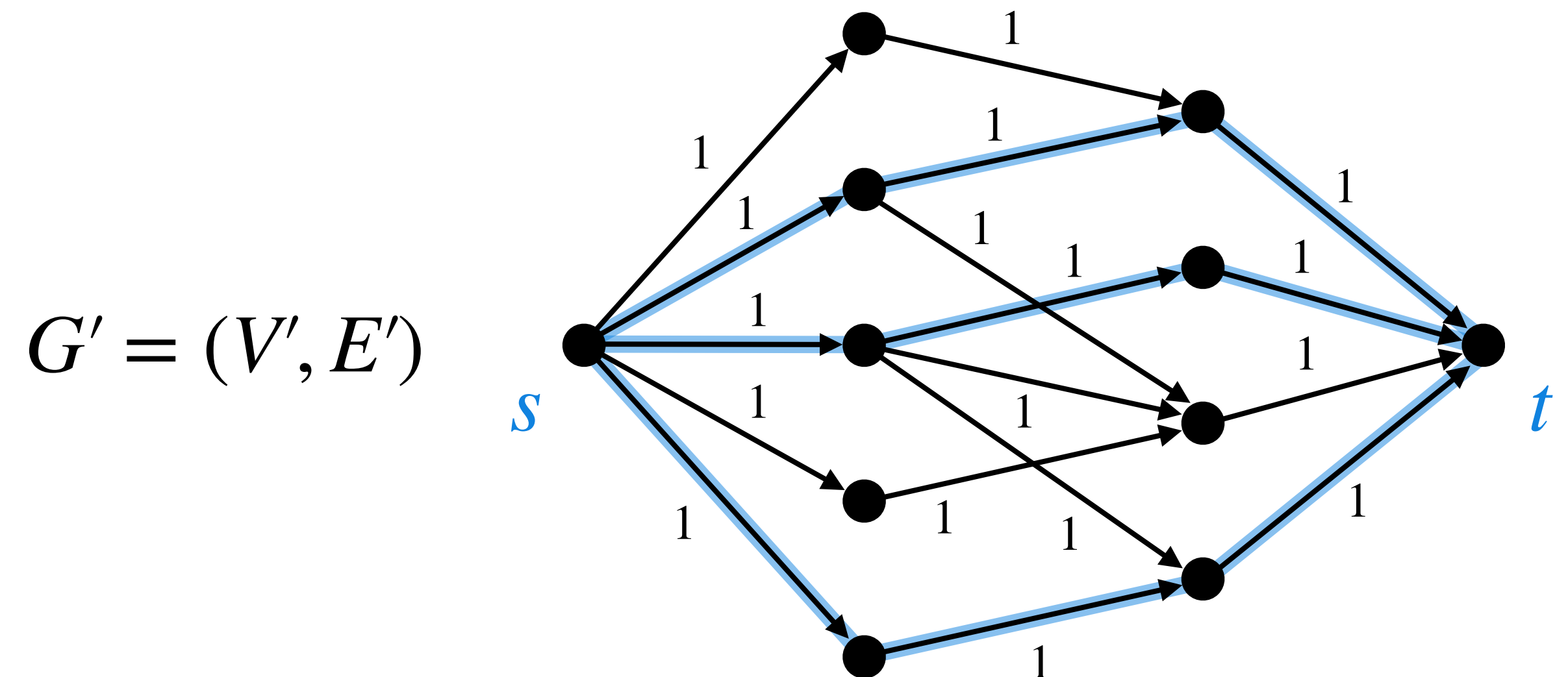
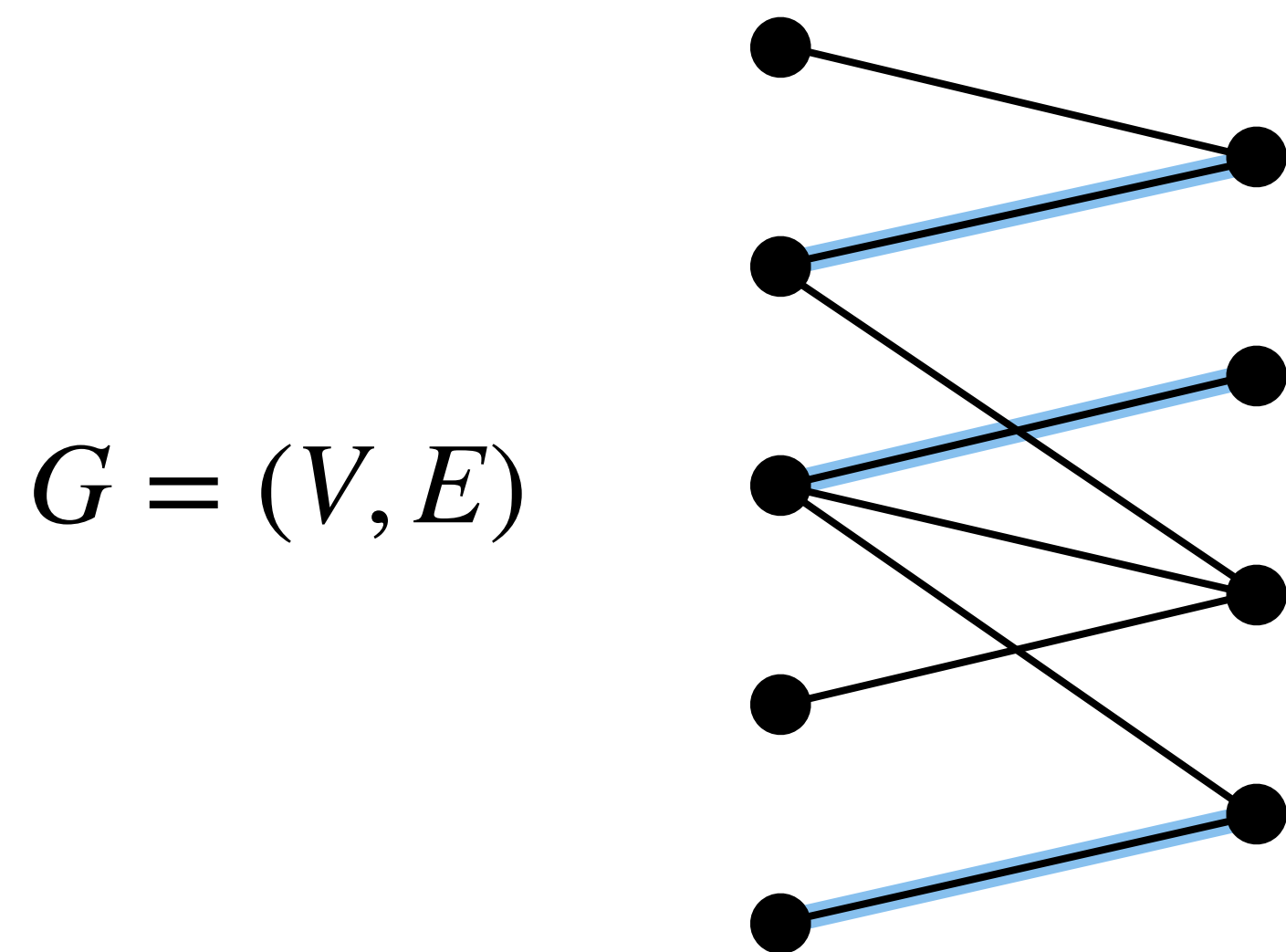
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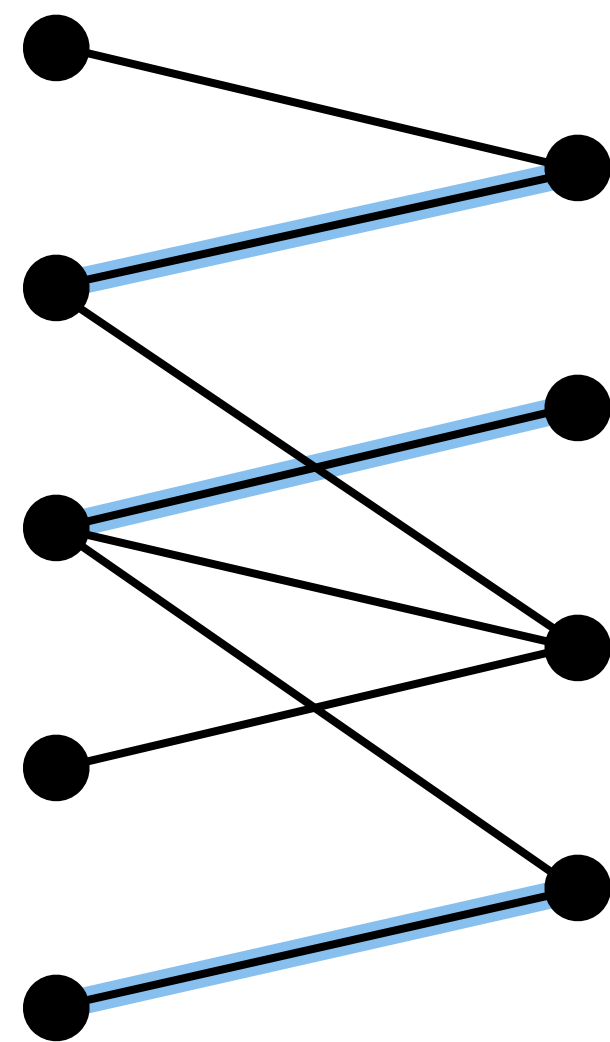
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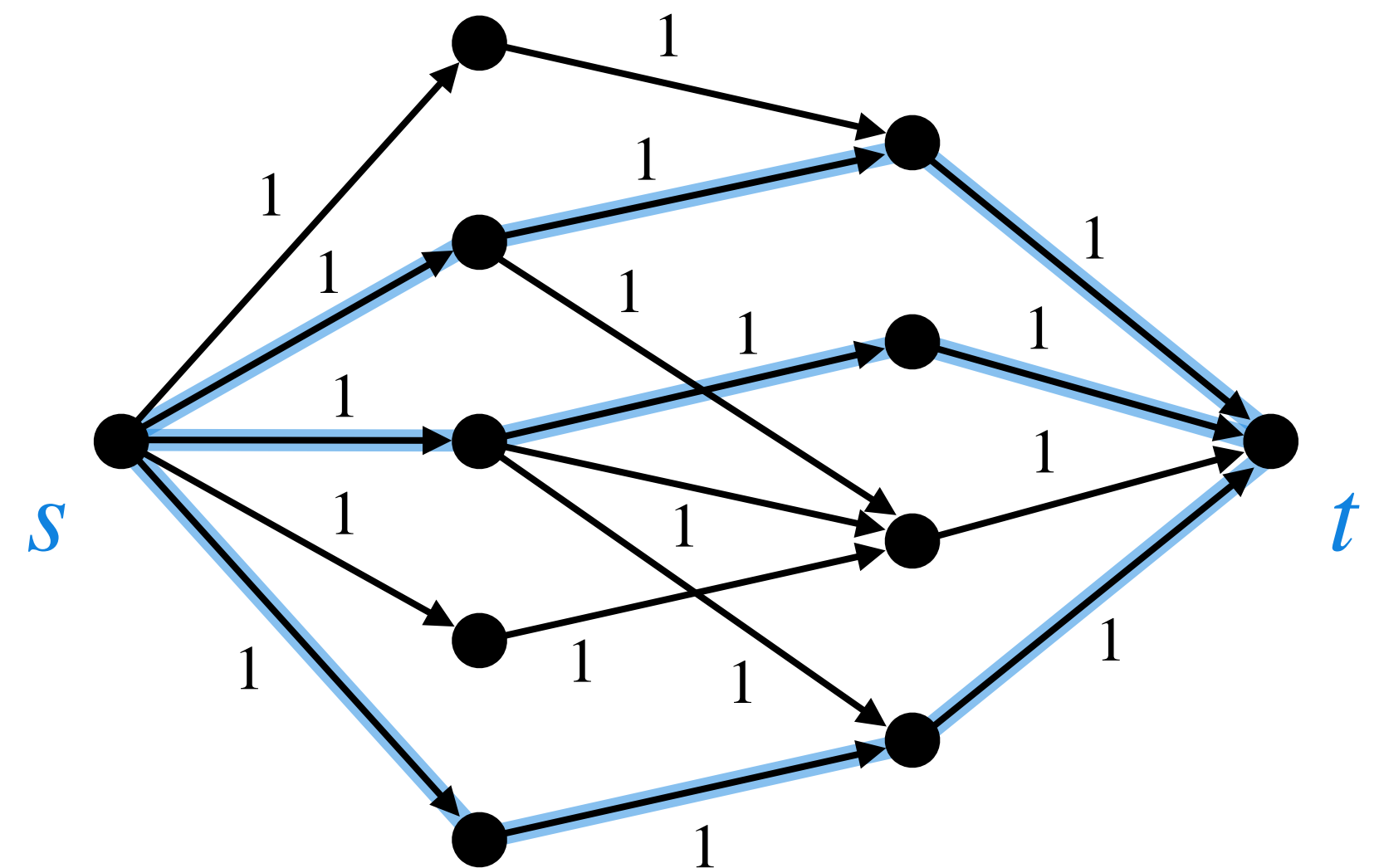
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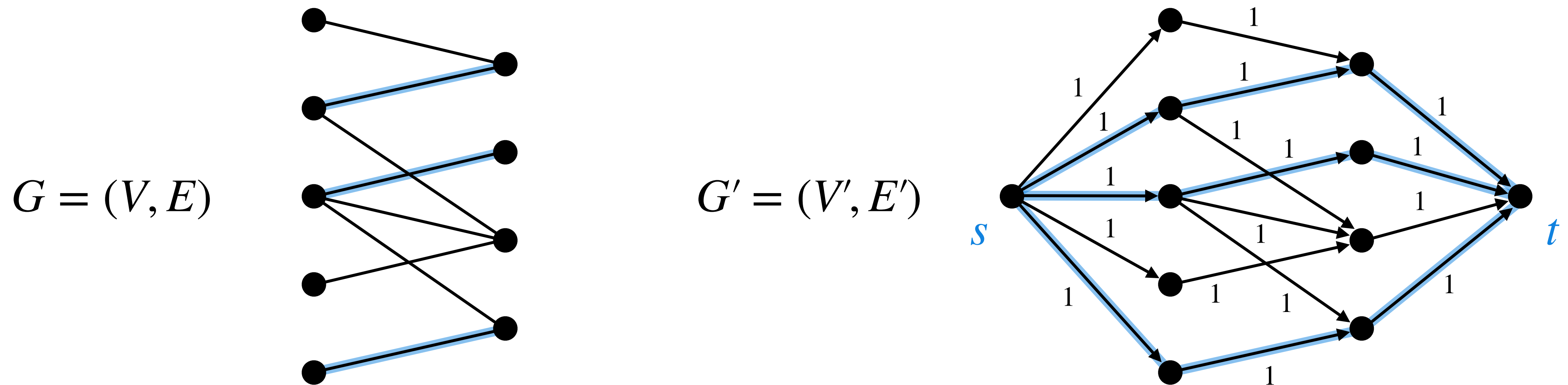


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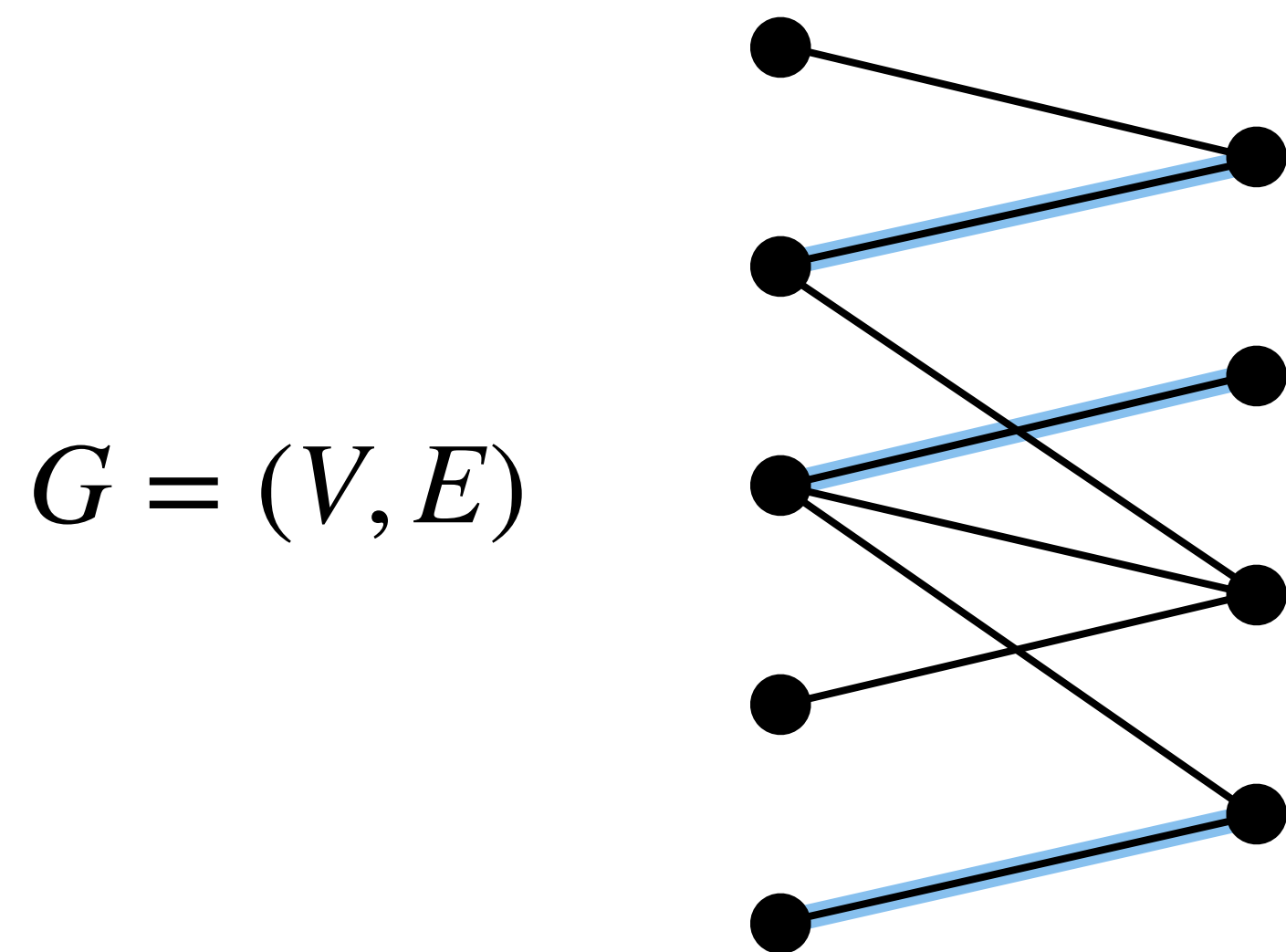


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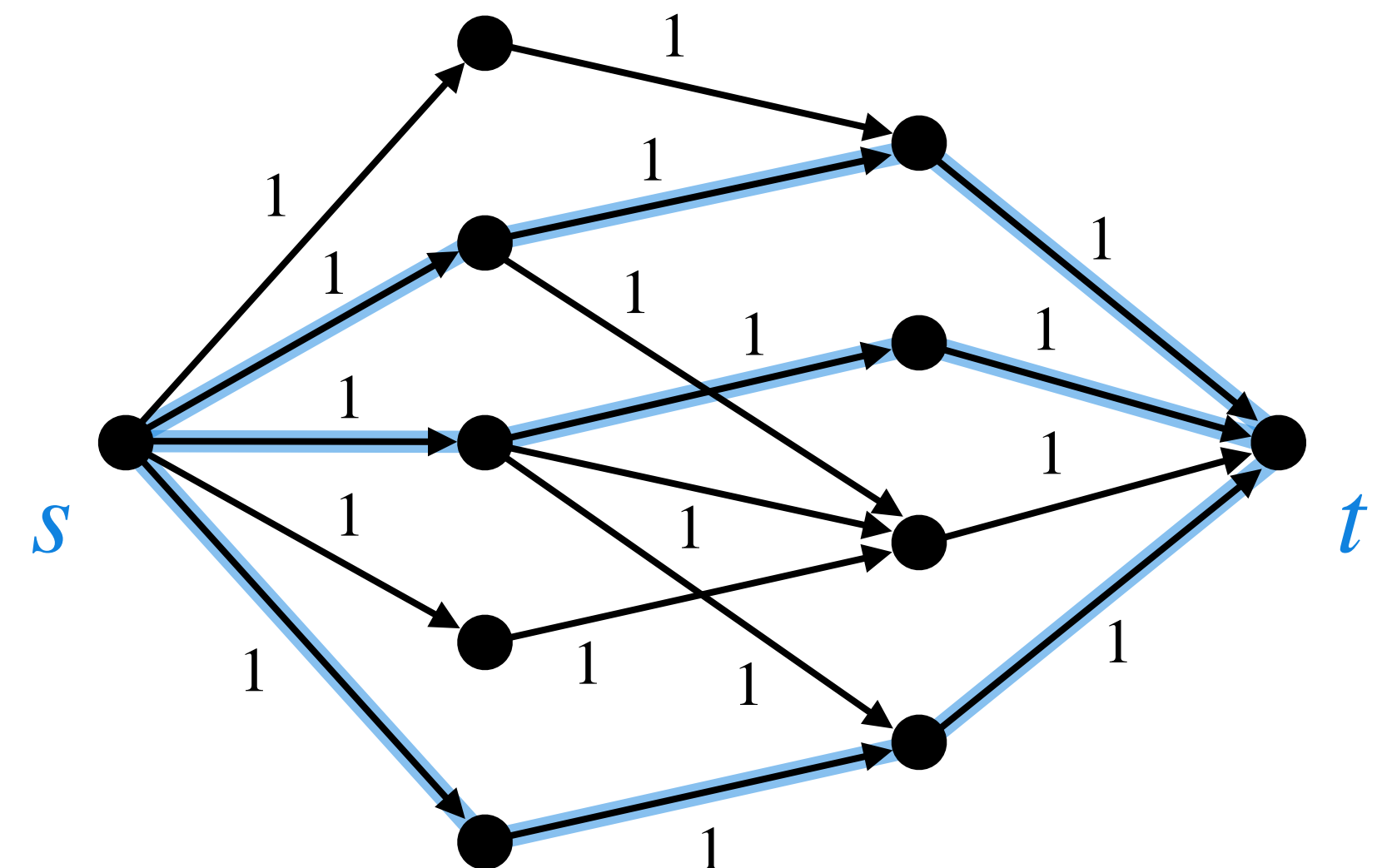
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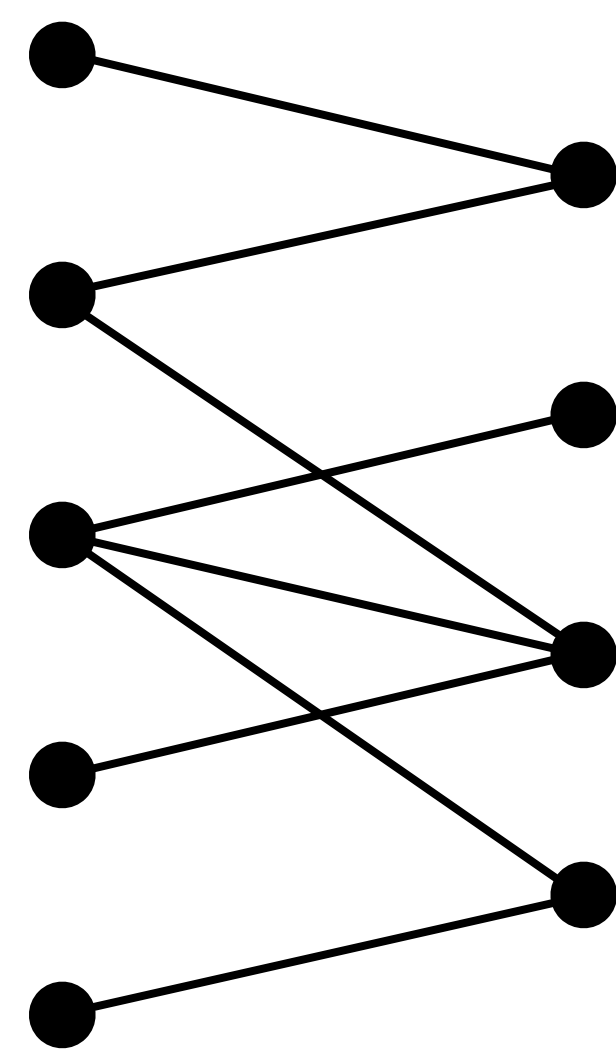


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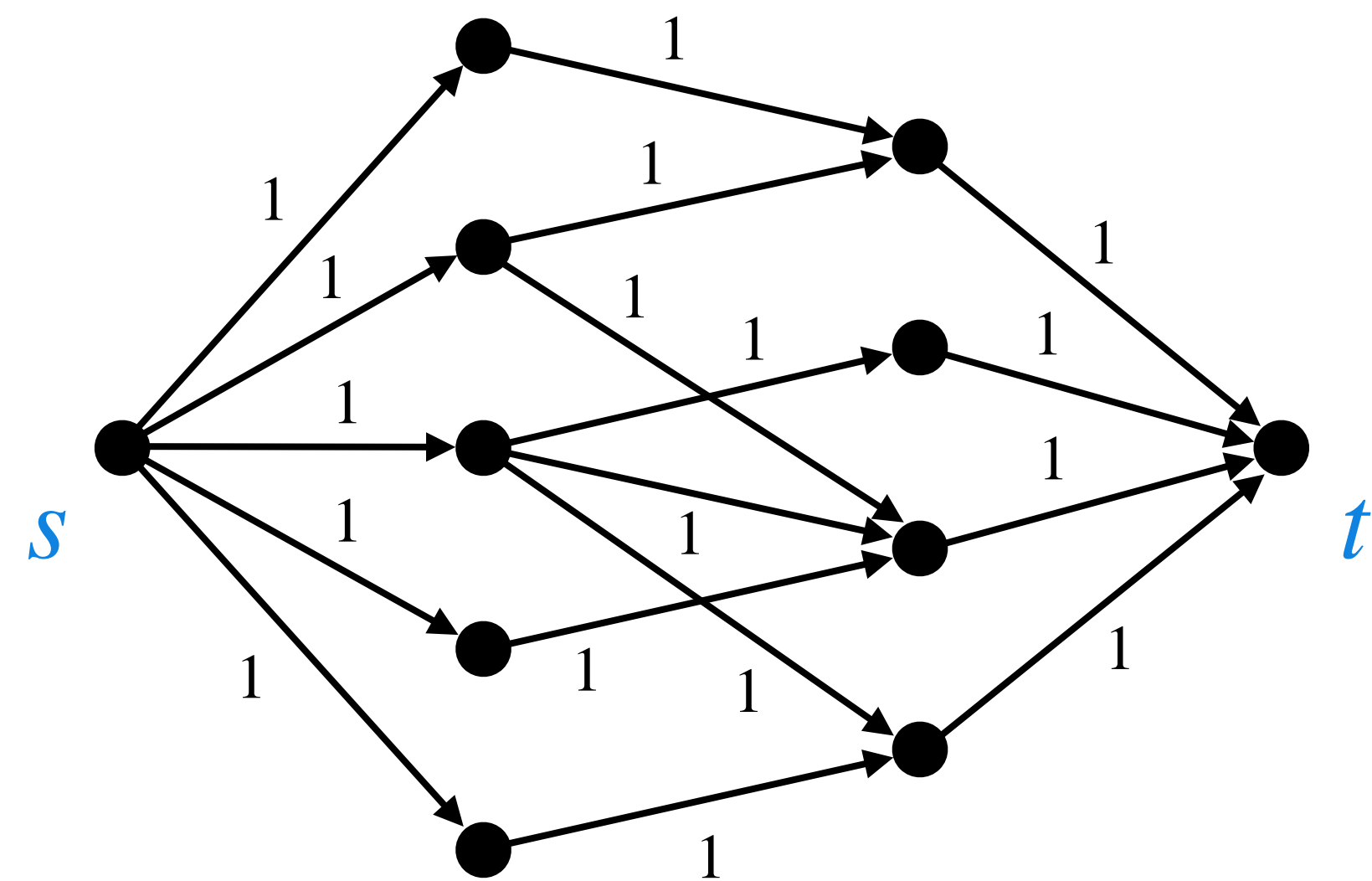


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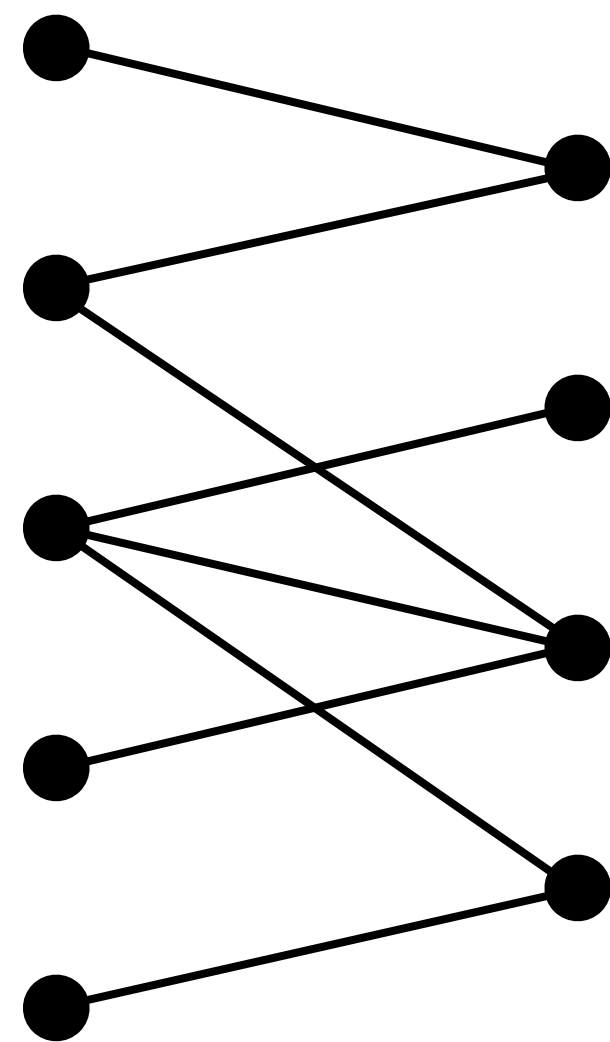
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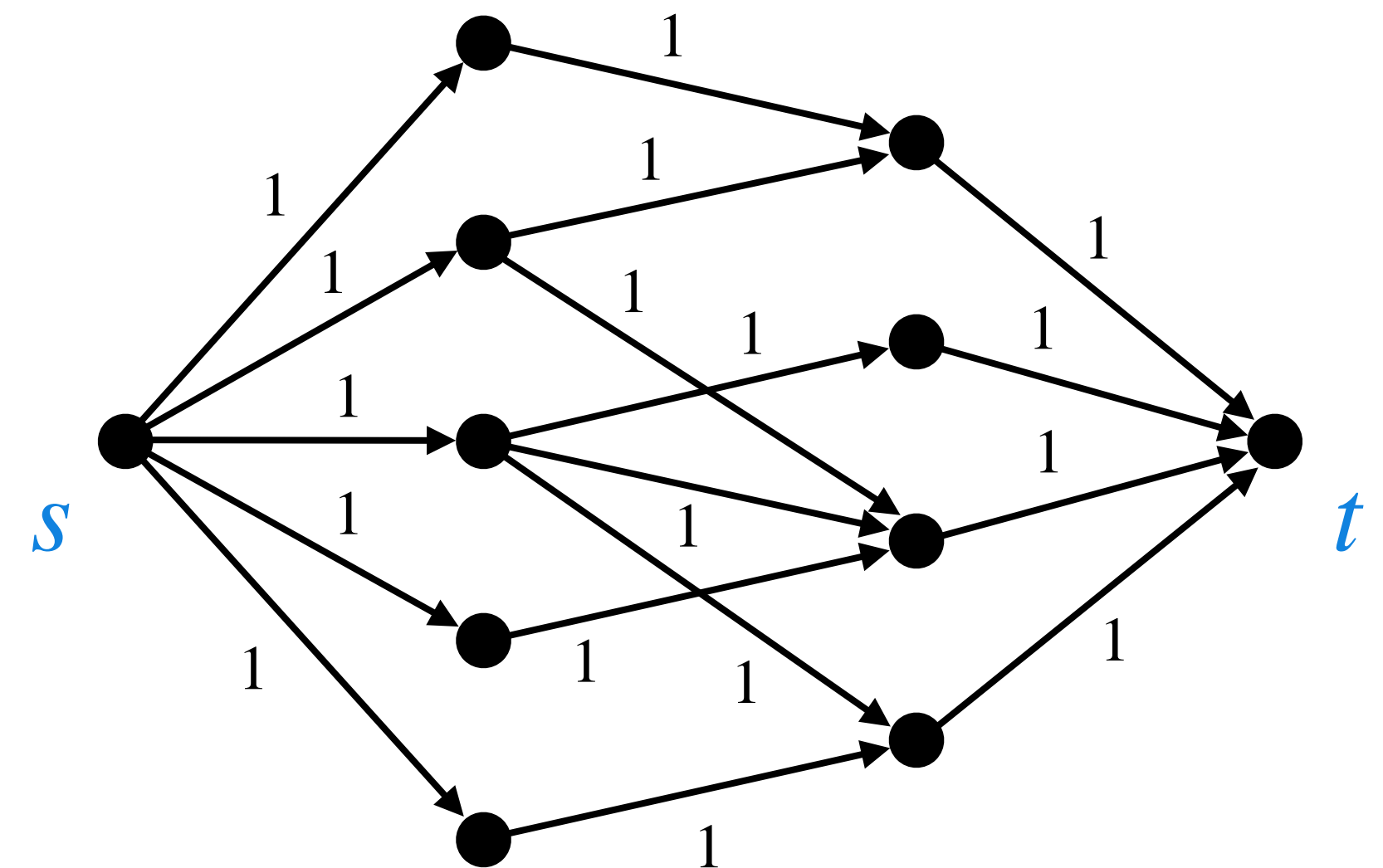
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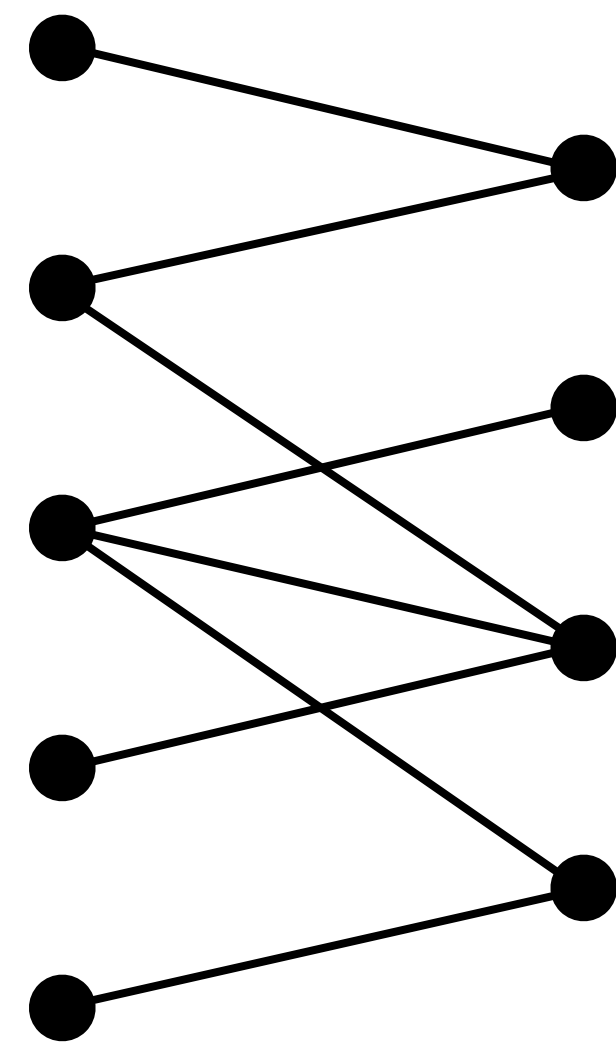


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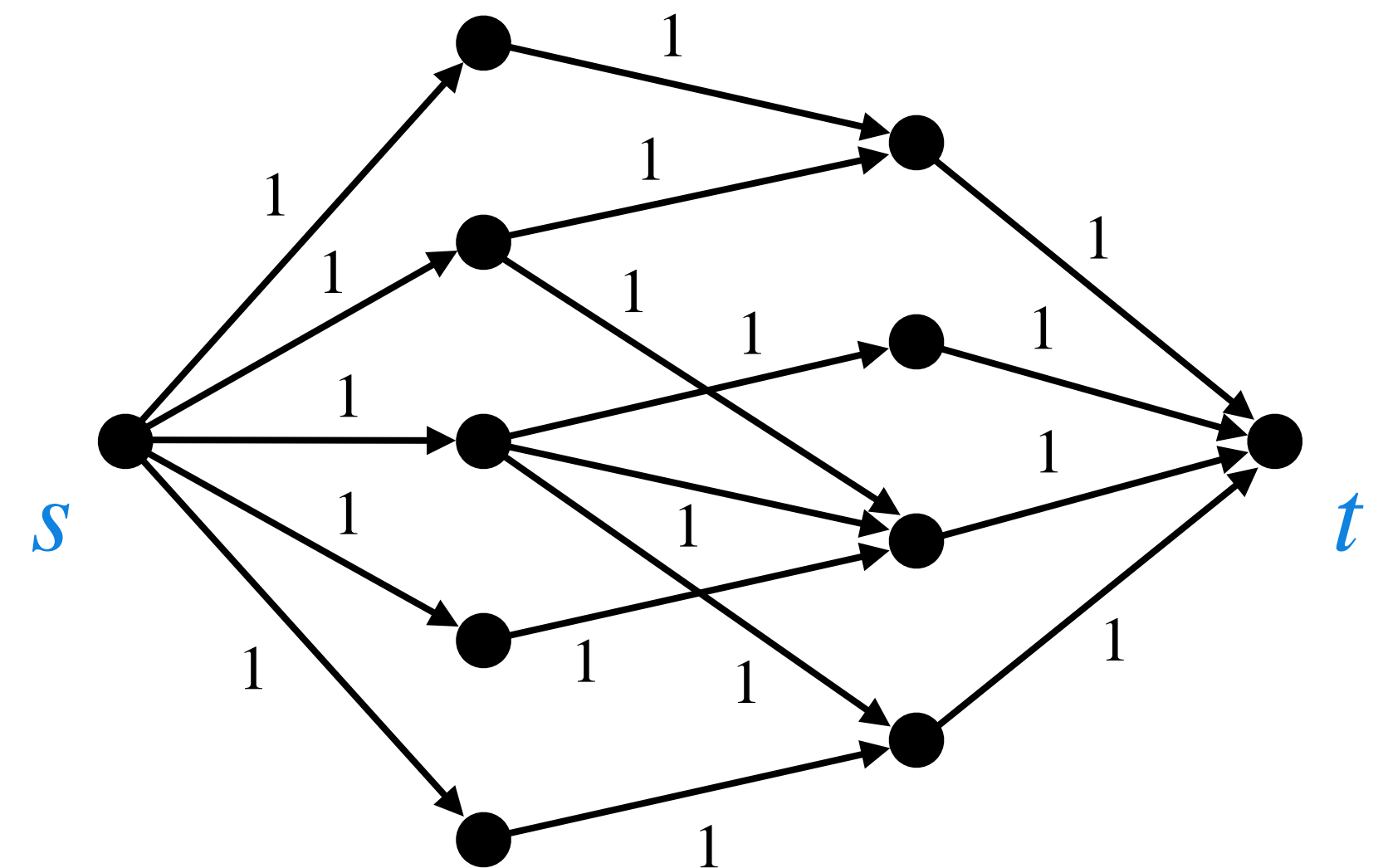
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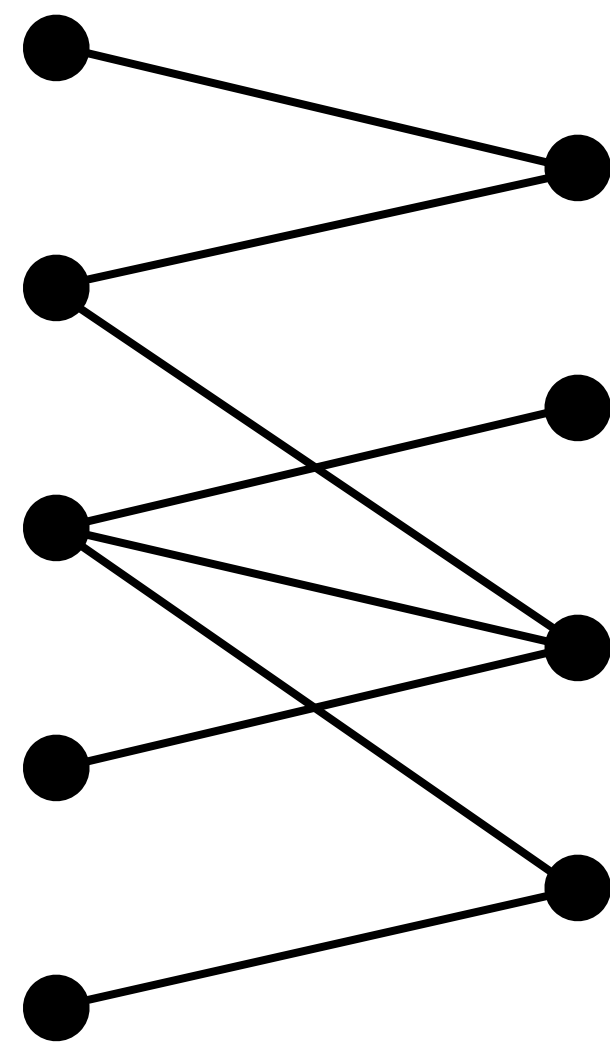
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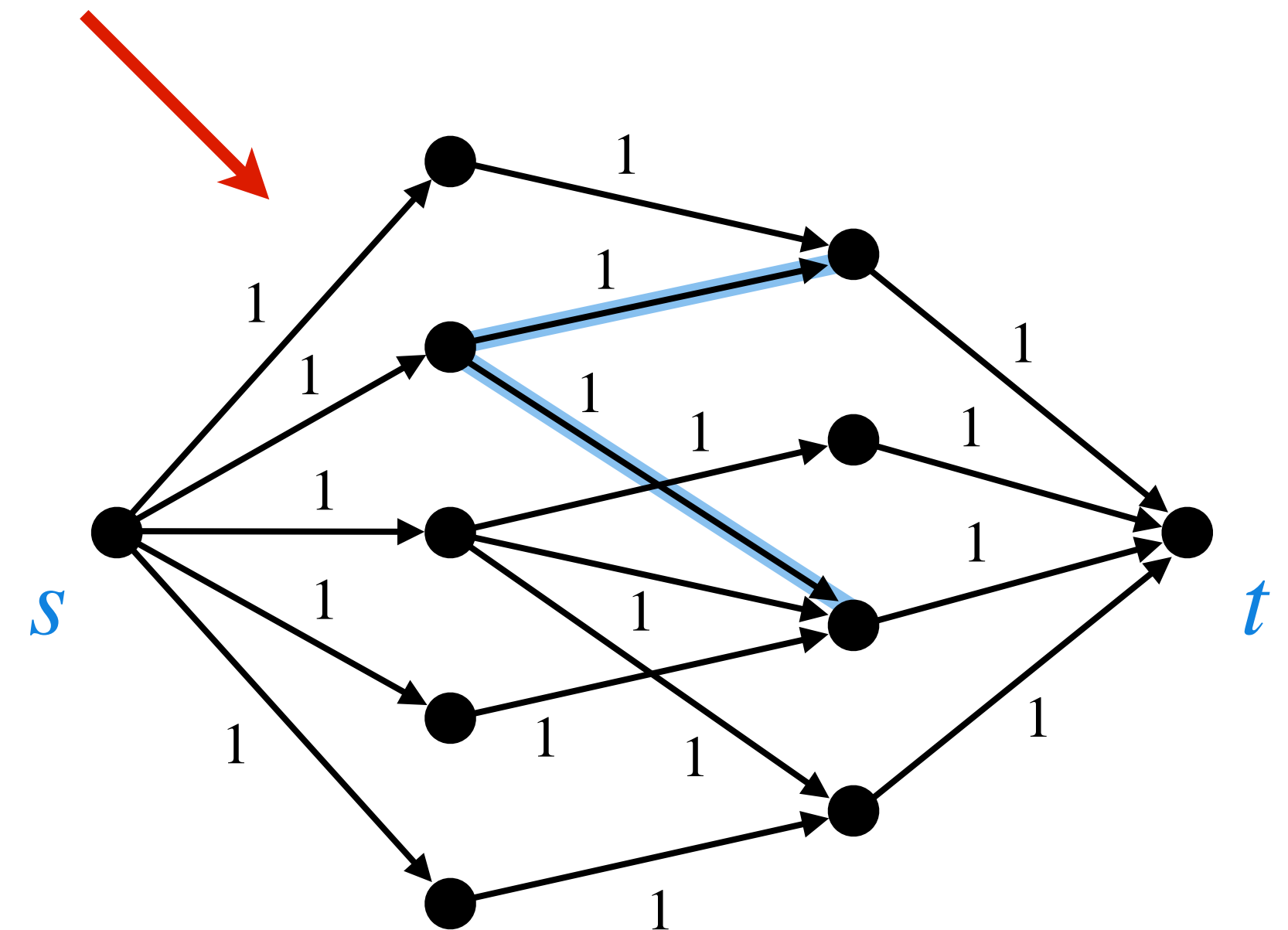
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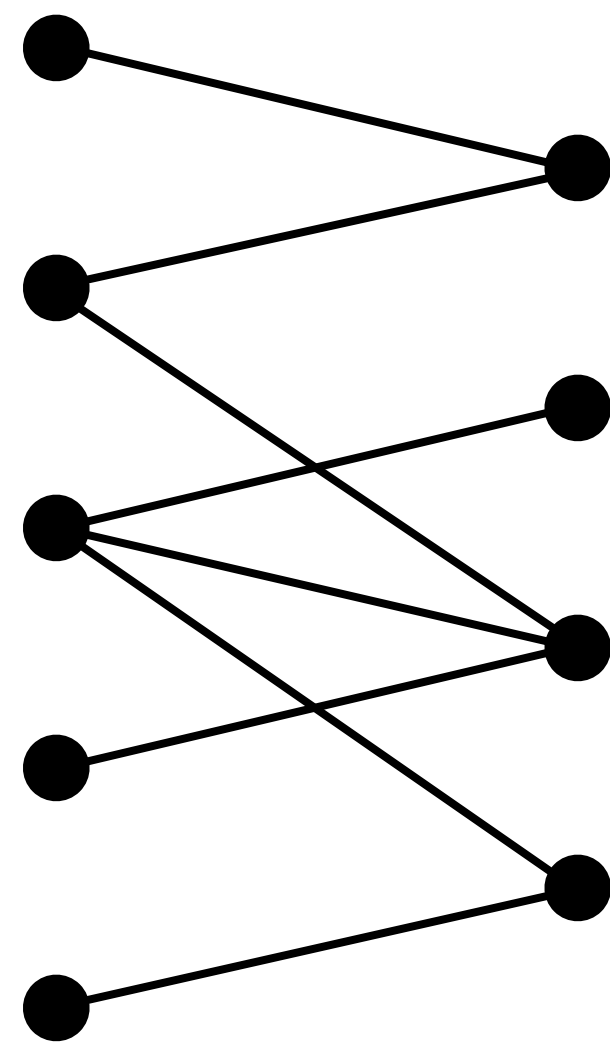
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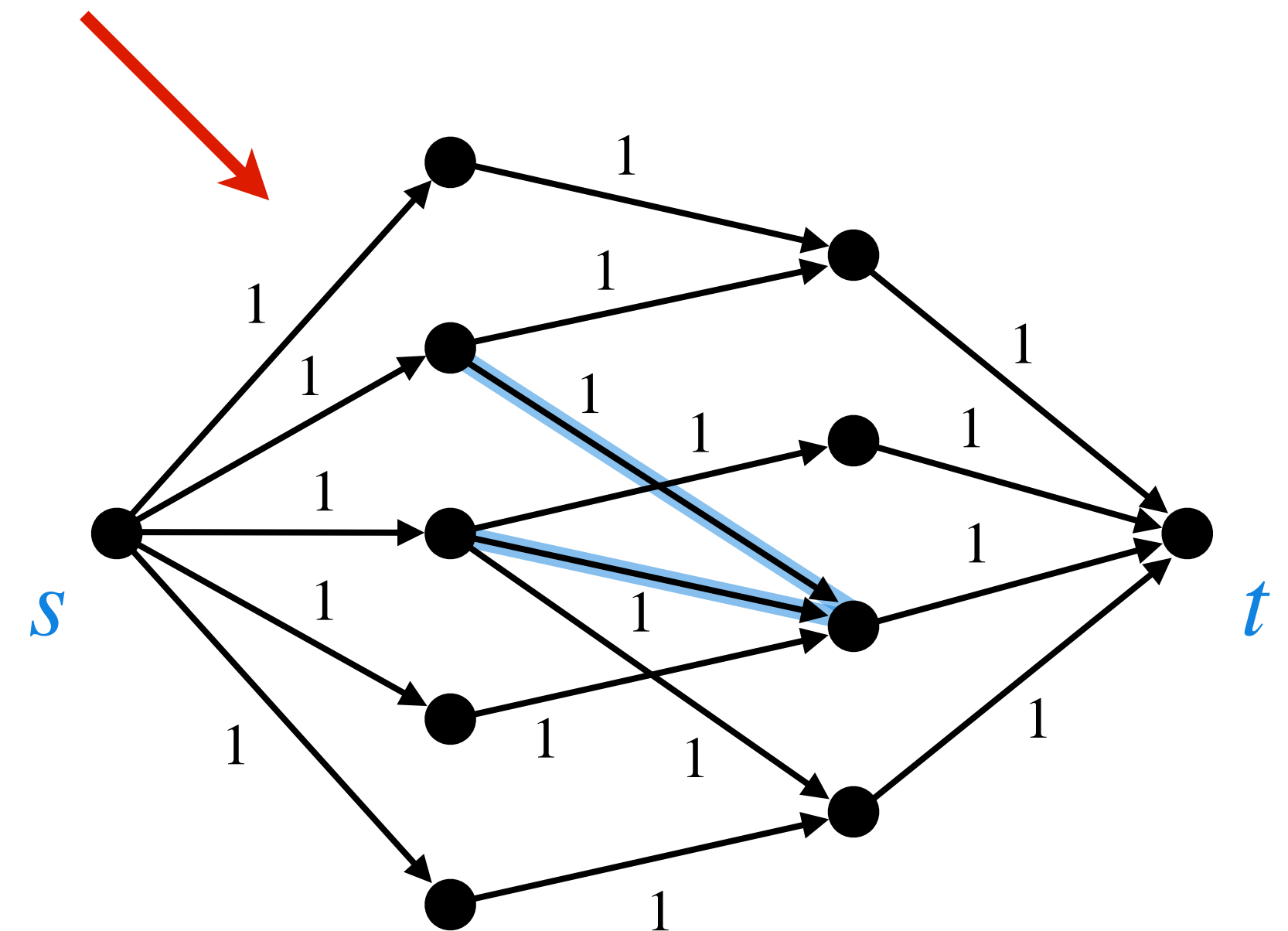
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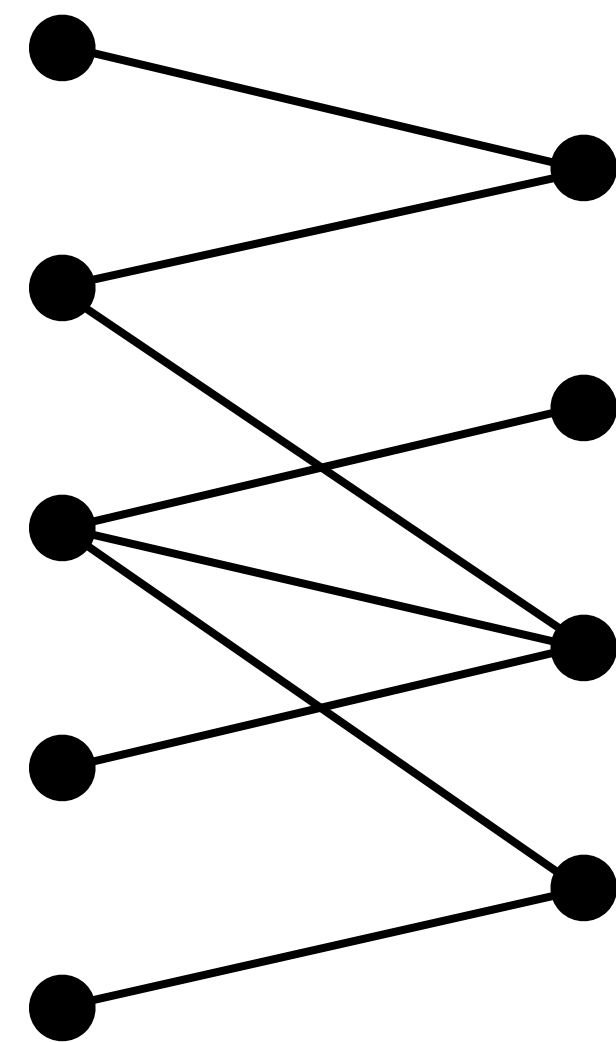


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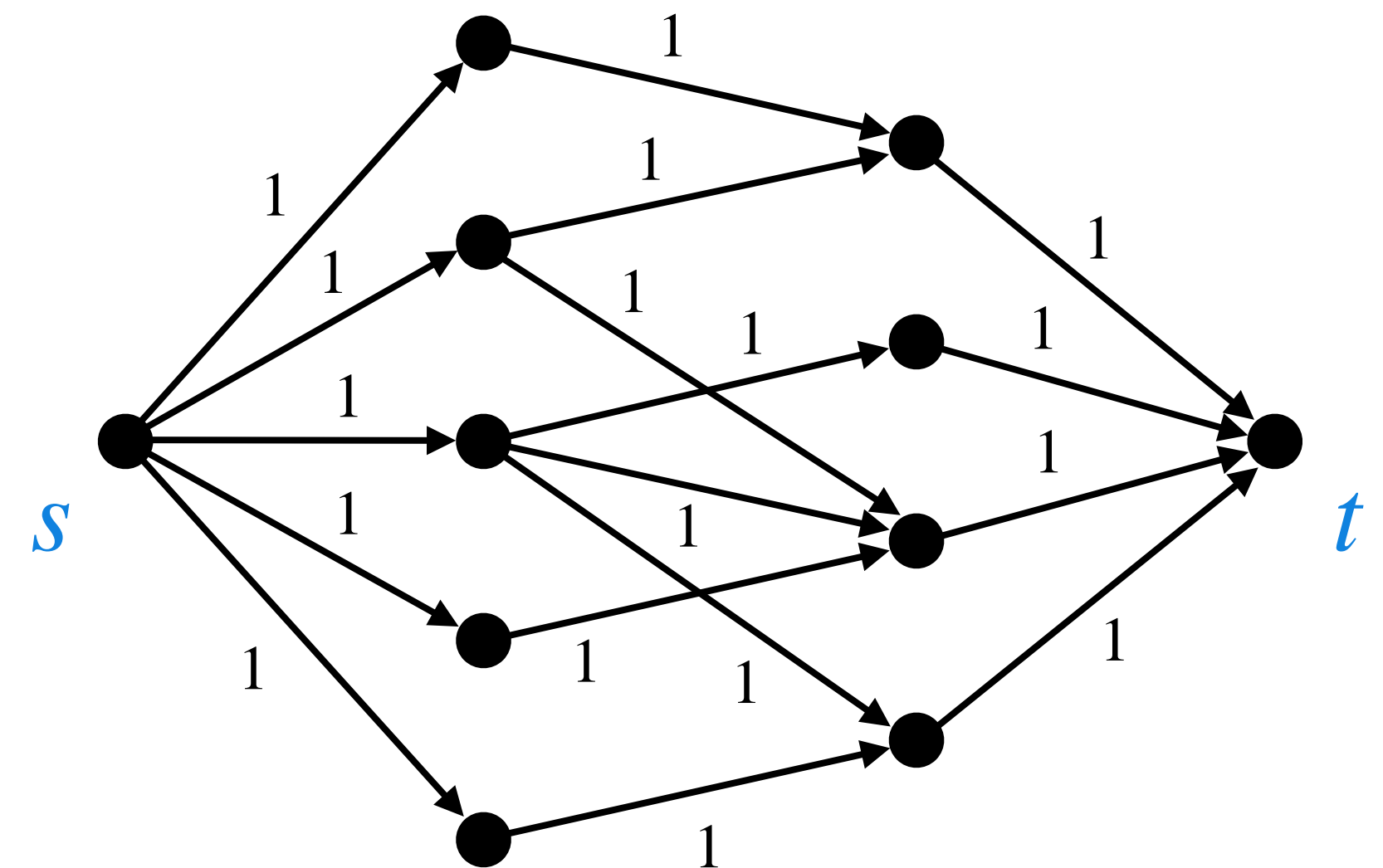
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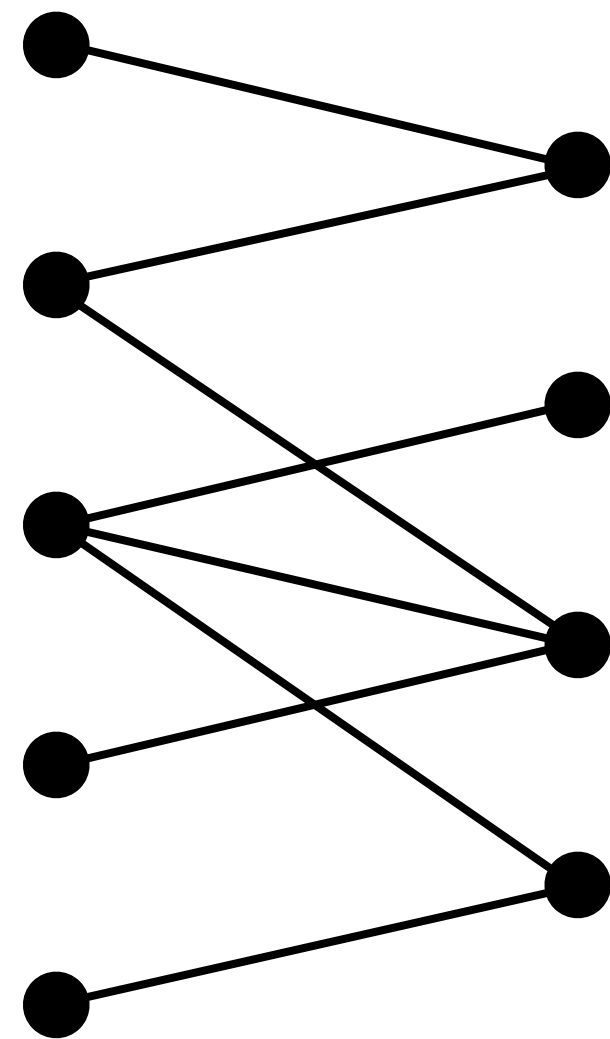


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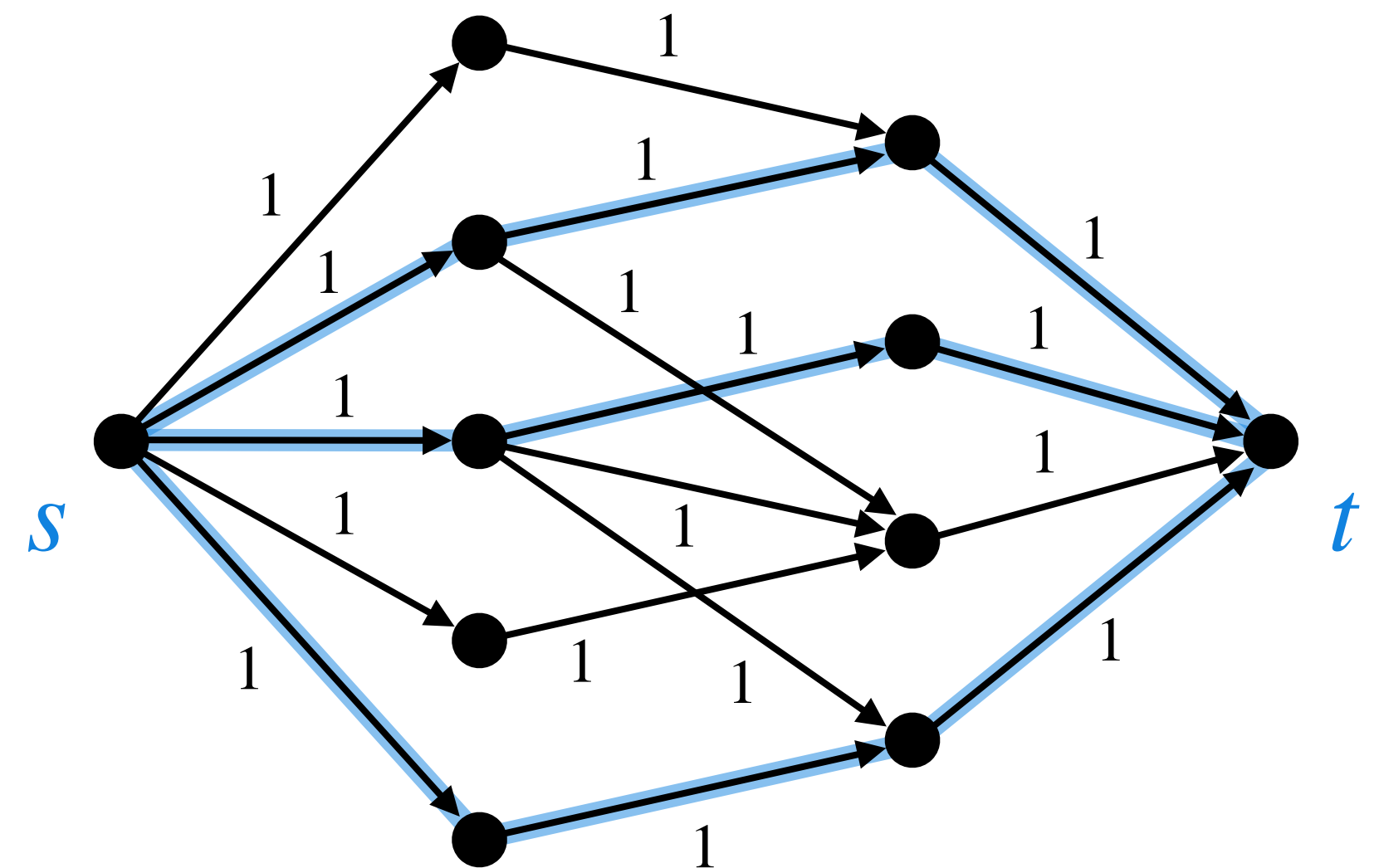
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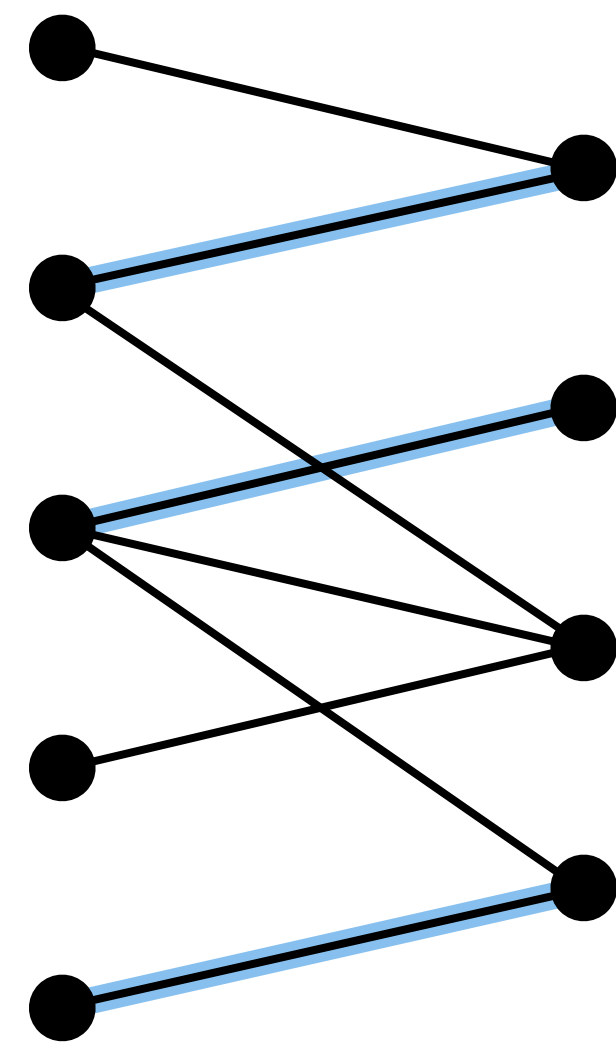


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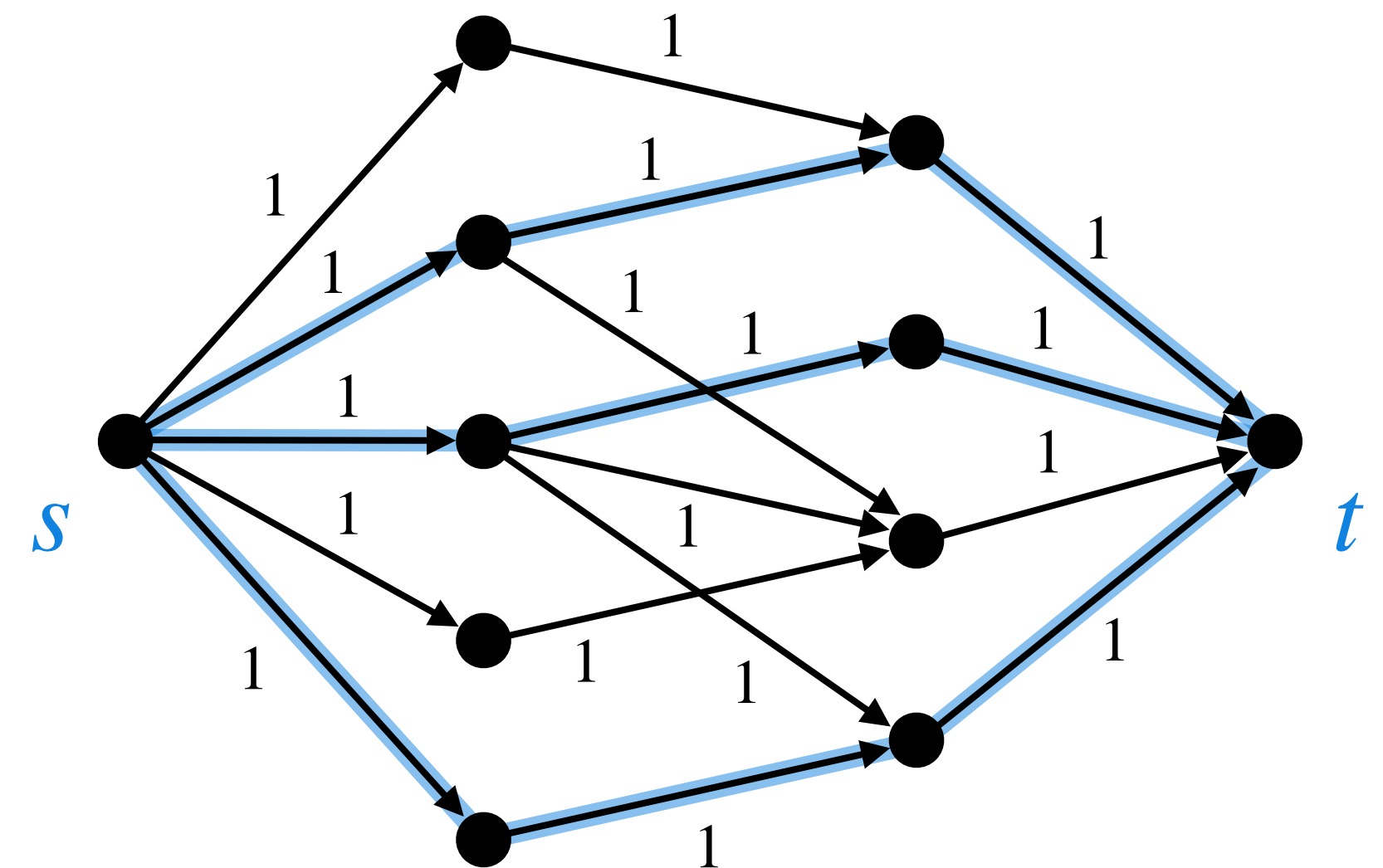
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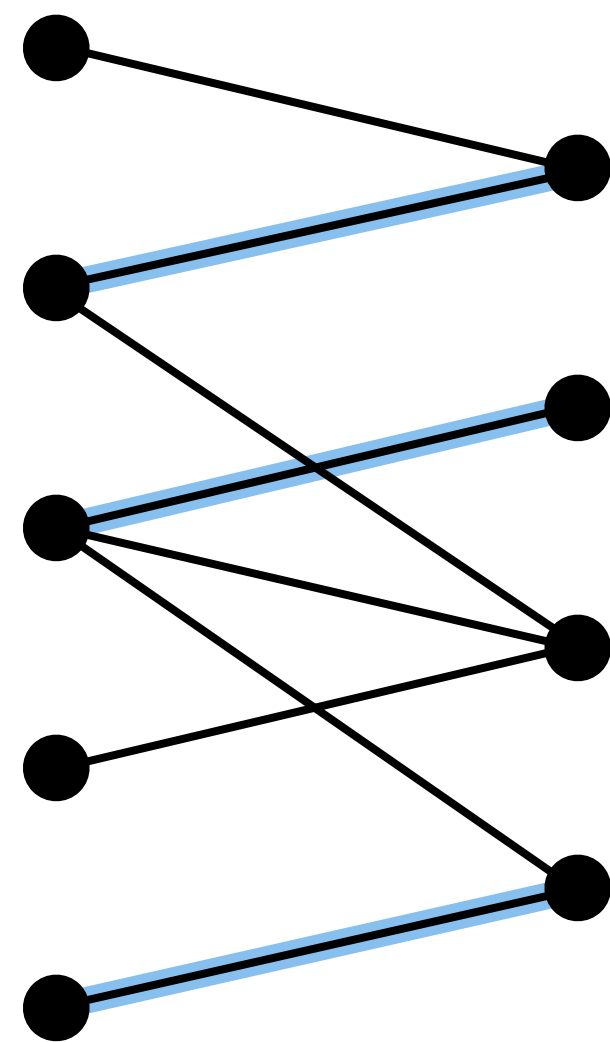
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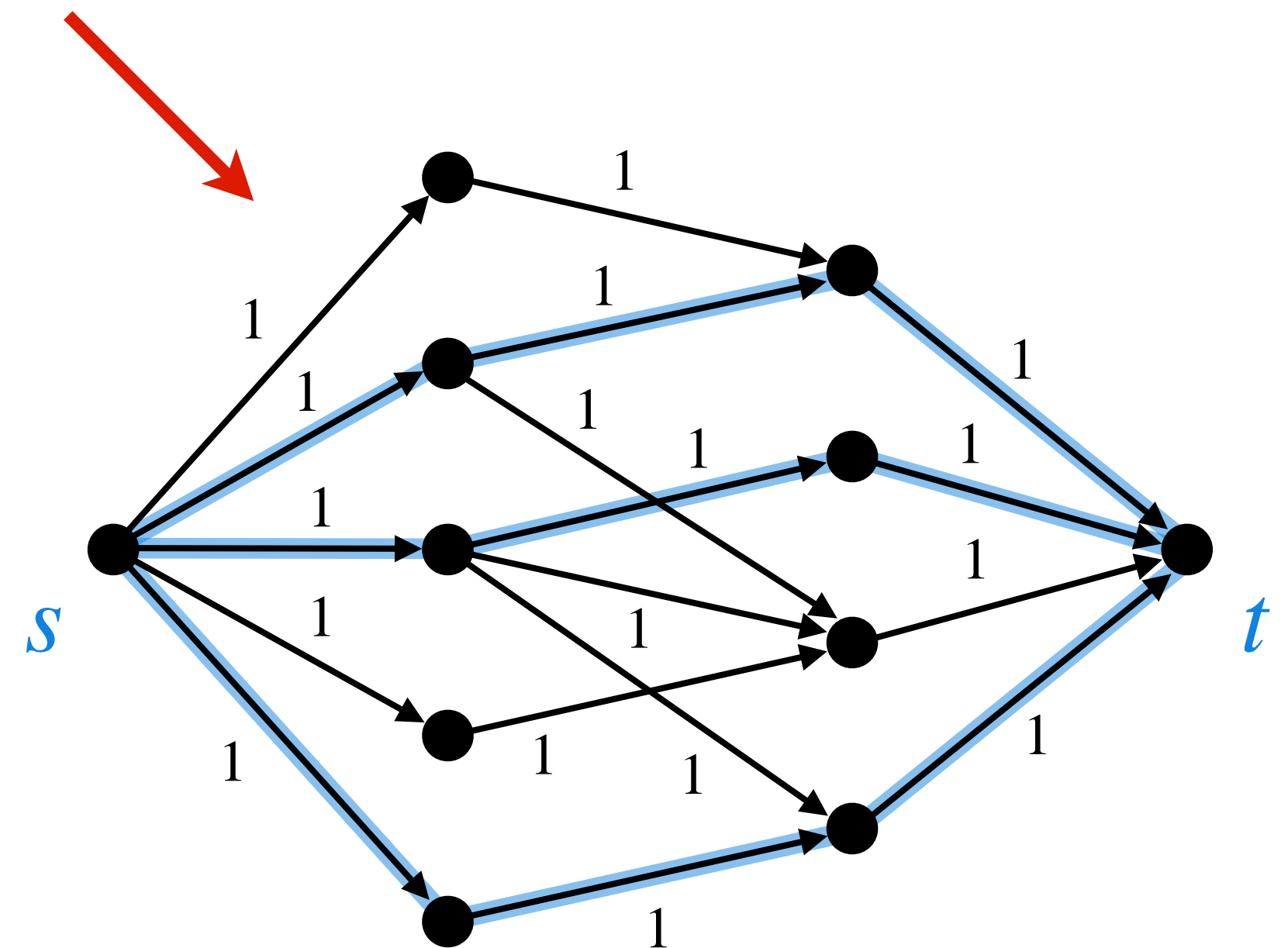
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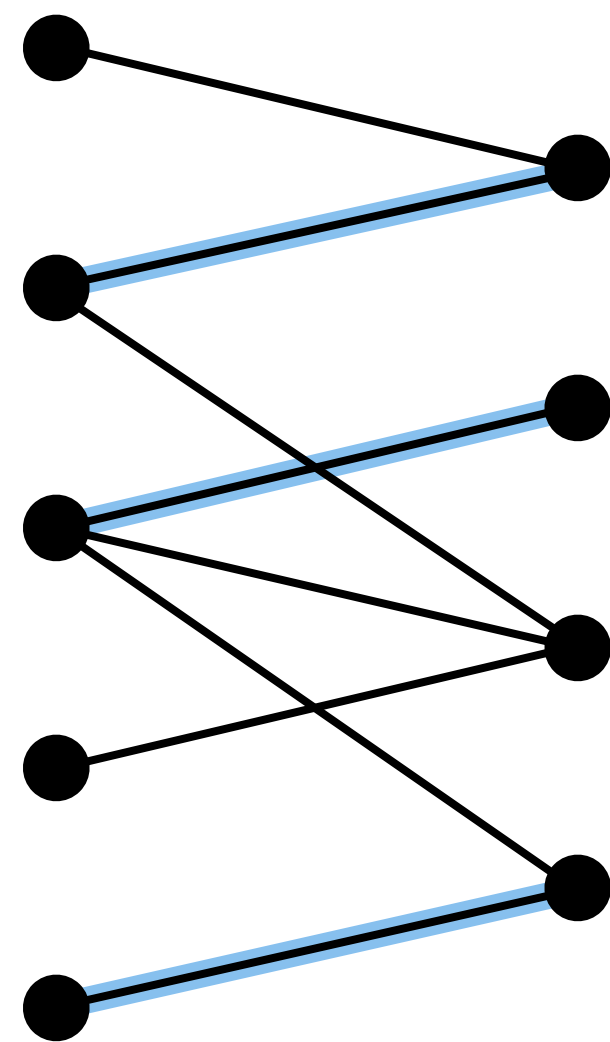
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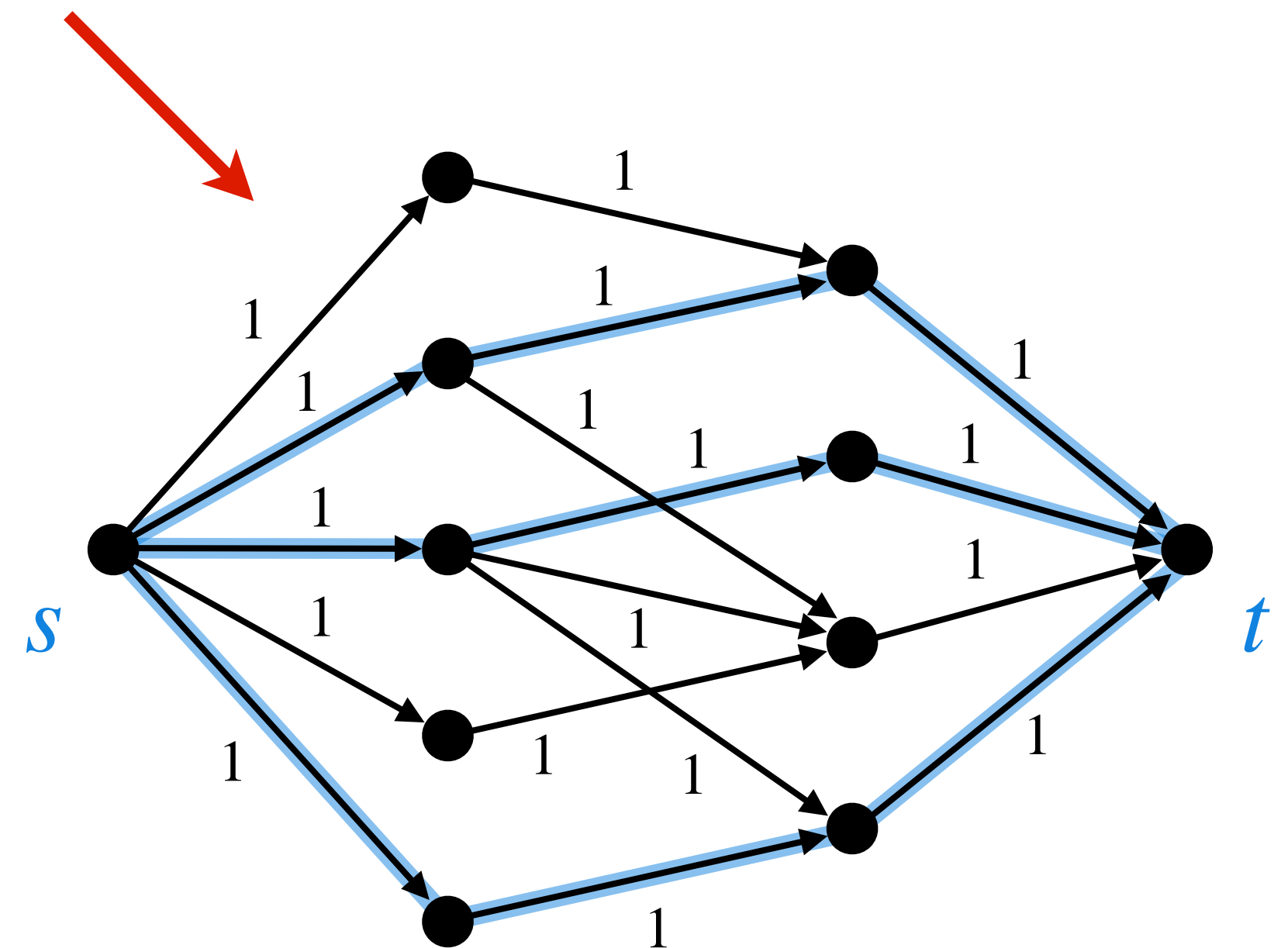
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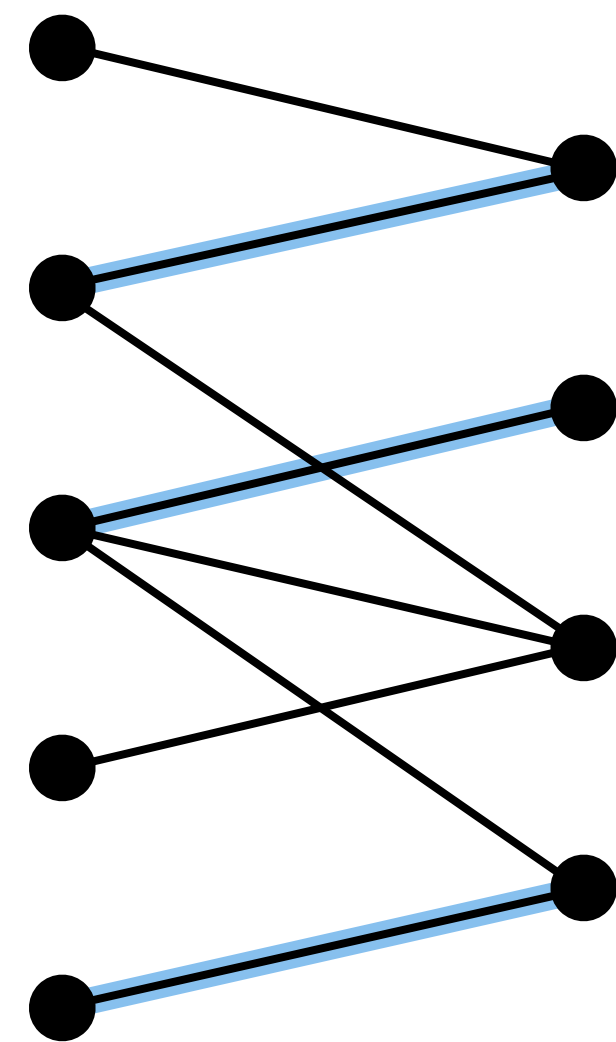


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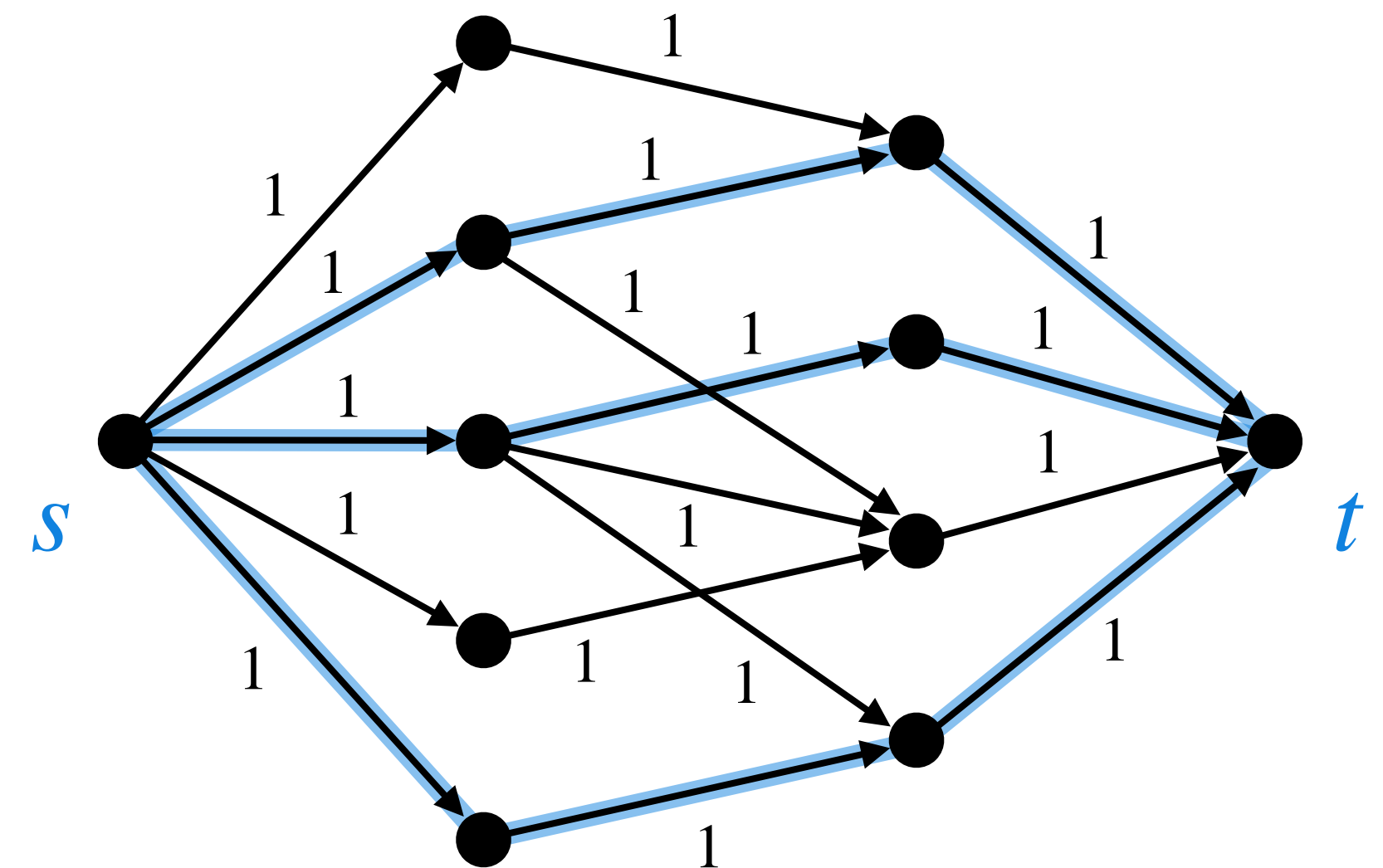
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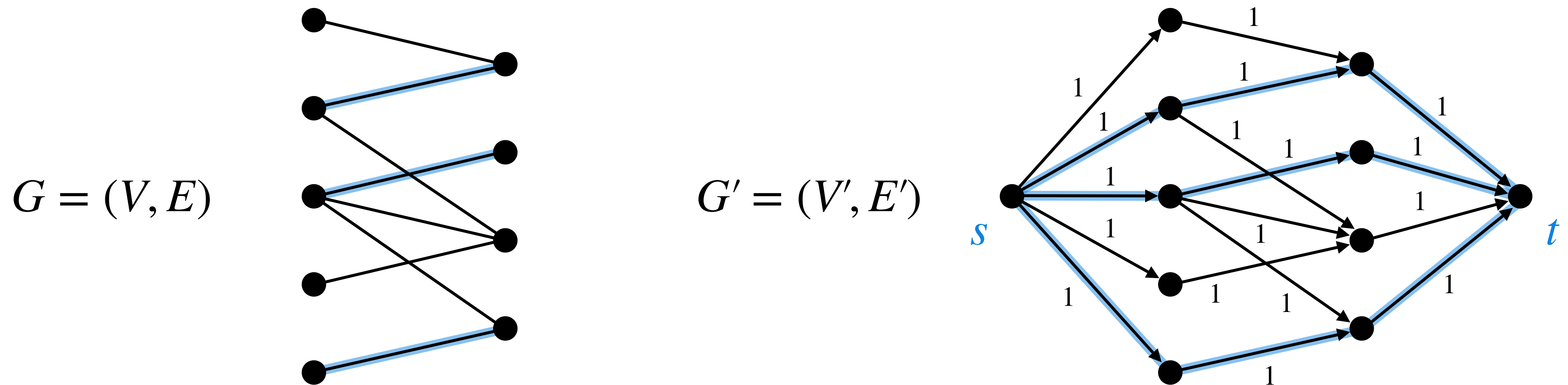
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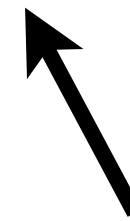
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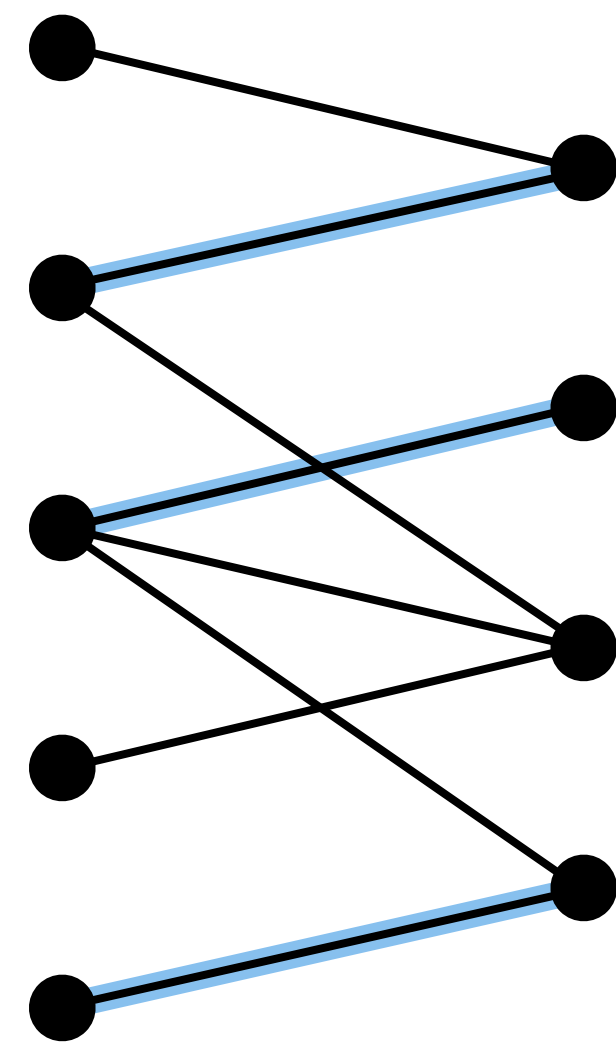
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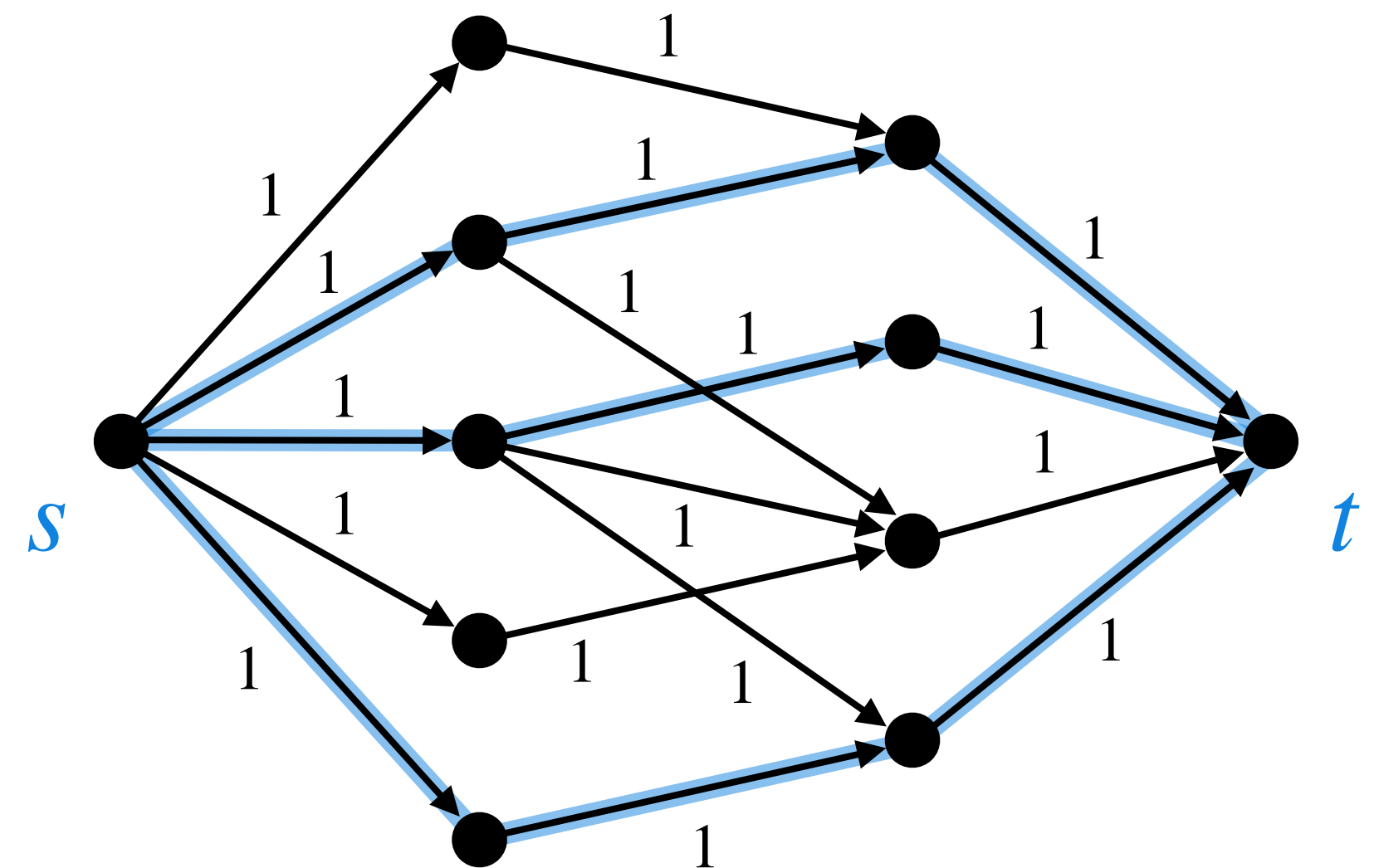
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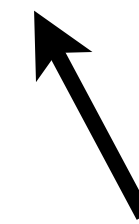
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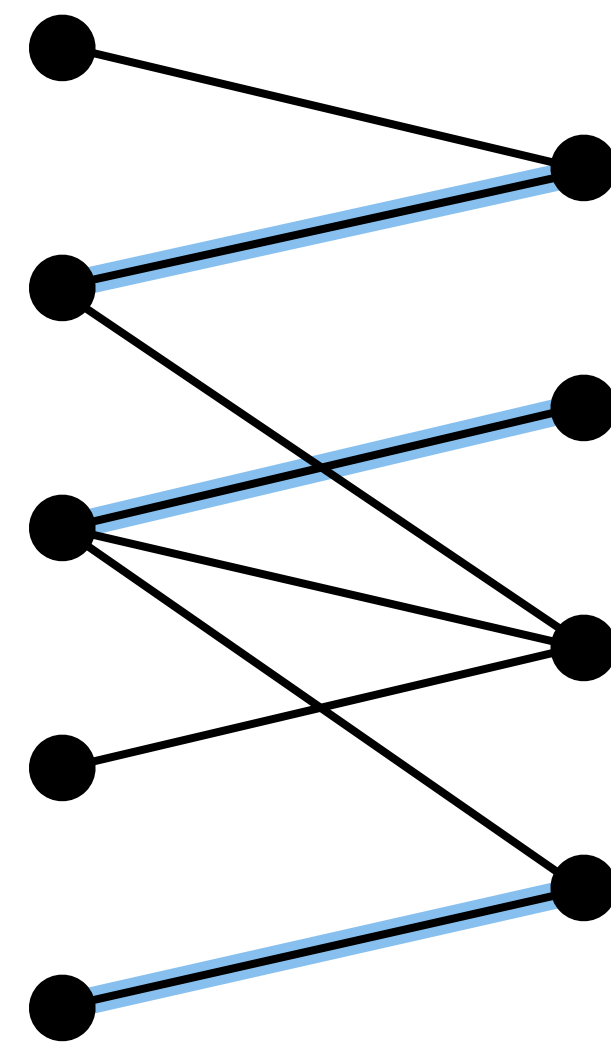
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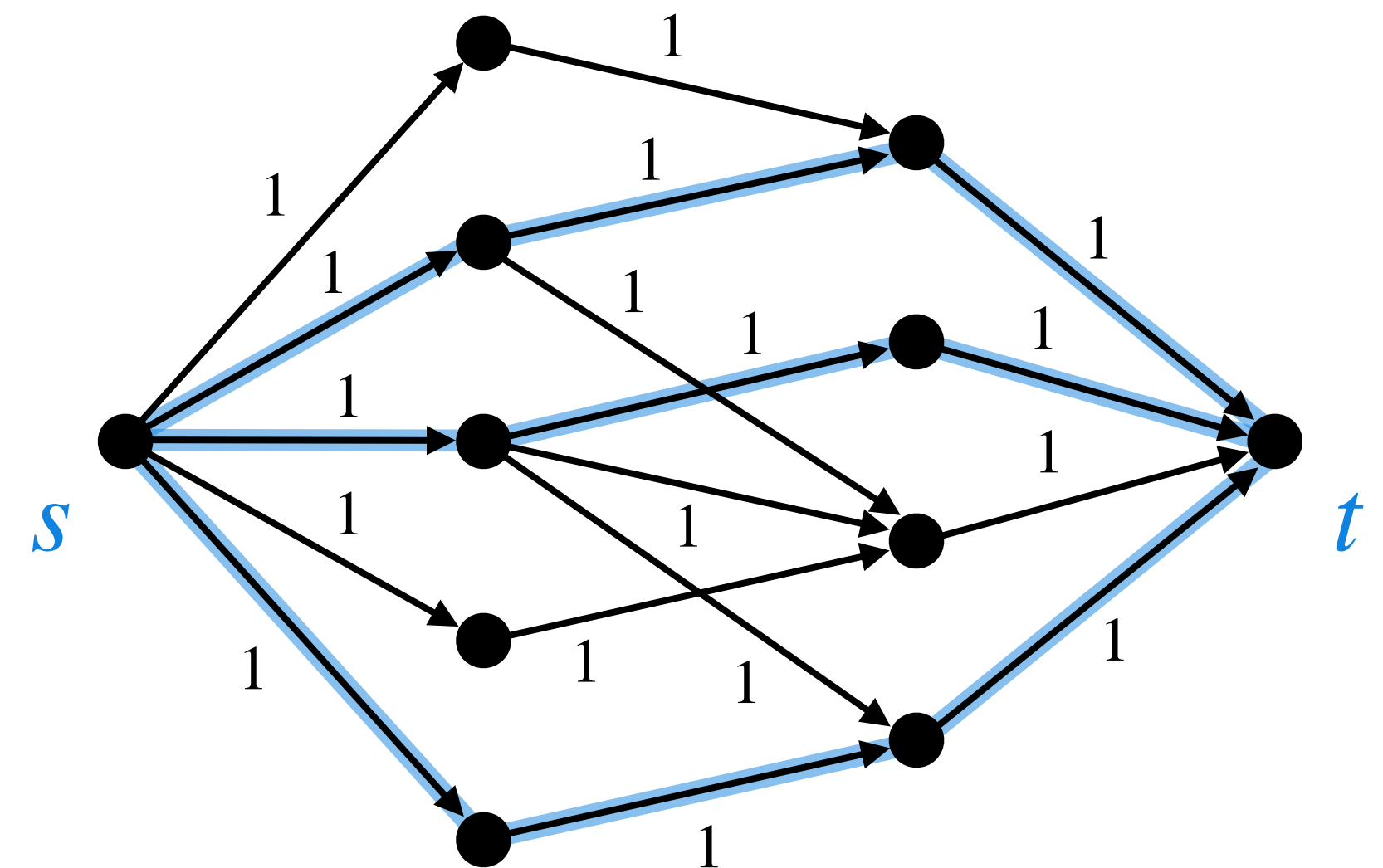


Prove it yourself why M will be a matching of size $|f|$.

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Claim: Let $G = (V, E)$ be a bipartite graph with partition (L, R) and let $G' = (V', E')$ be its corresponding flow network. Let f be a maximum integer-valued flow in G' . Then, the matching M in G corresponding to f is a maximum matching.

Proof: We know that $|M| = |f|$.

Suppose M' not M is the maximum matching. Hence, $|M'| > |M|$.

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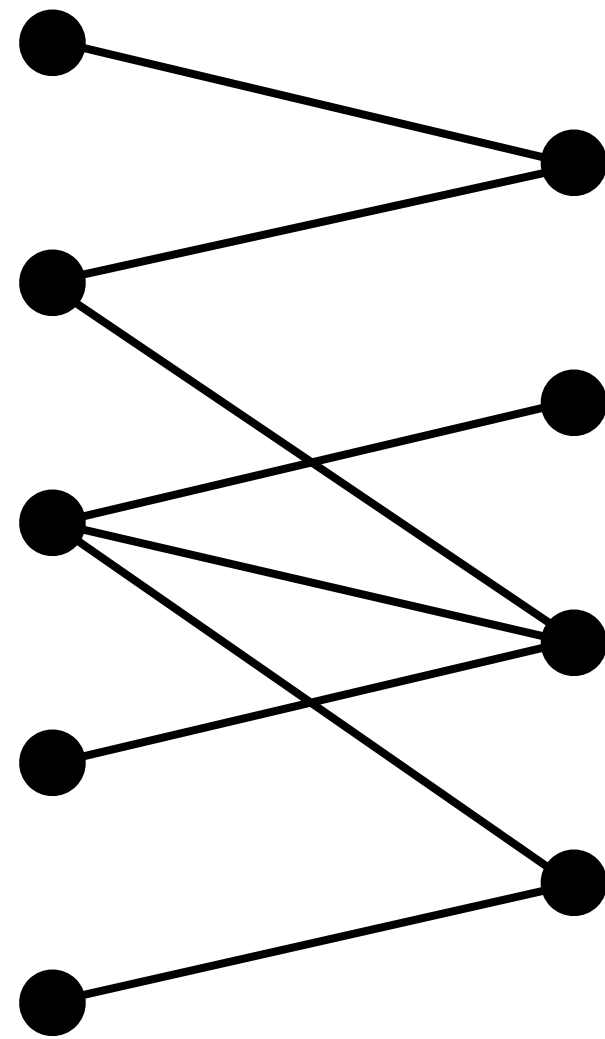
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What if the flow produced has fractional values?

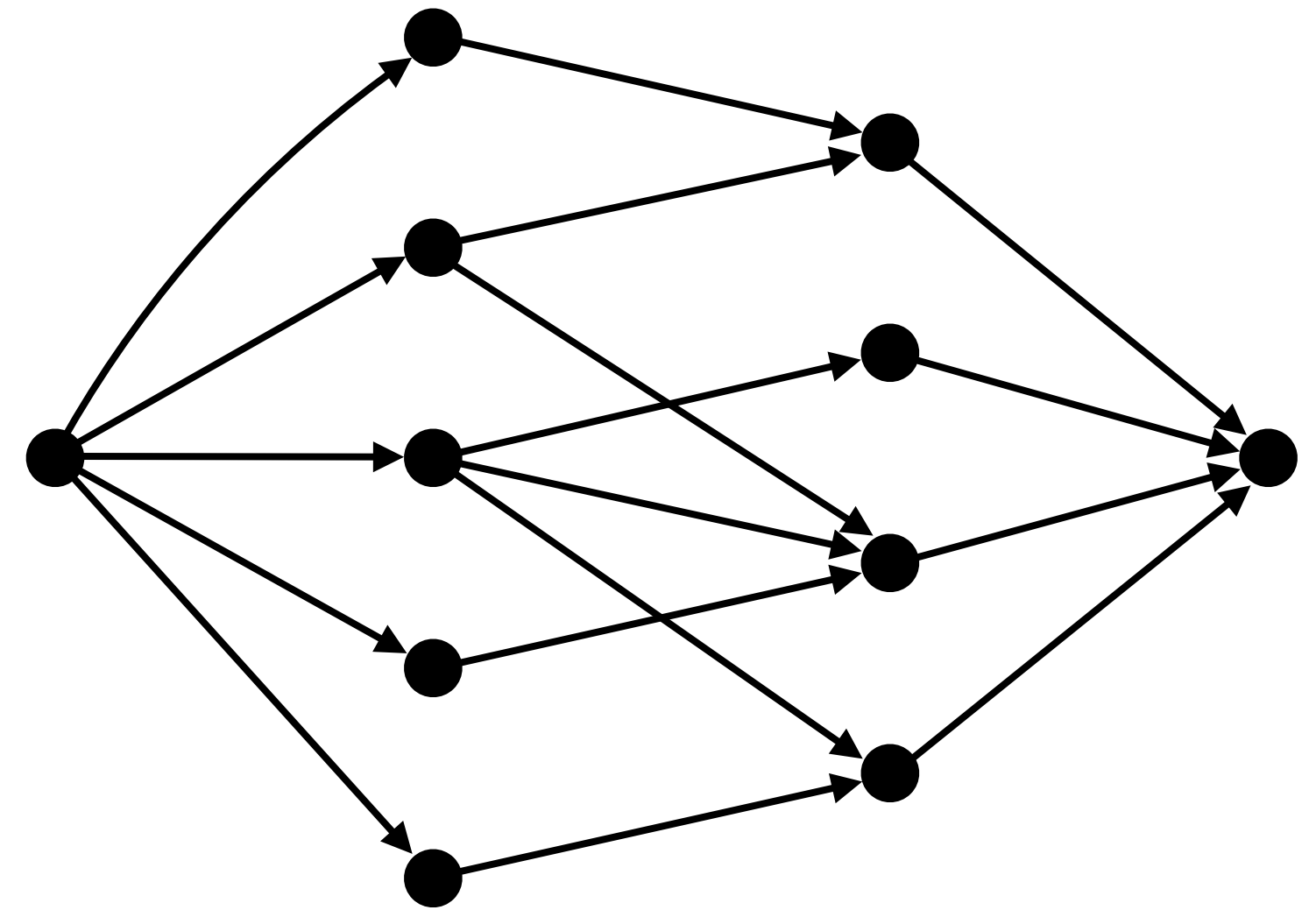


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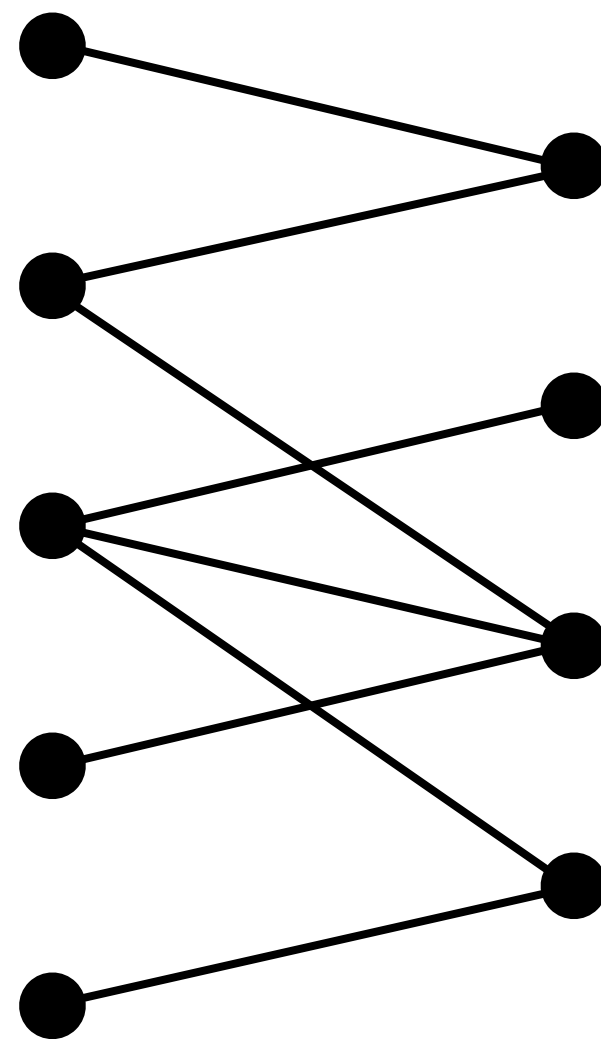


$$G = (V, E)$$



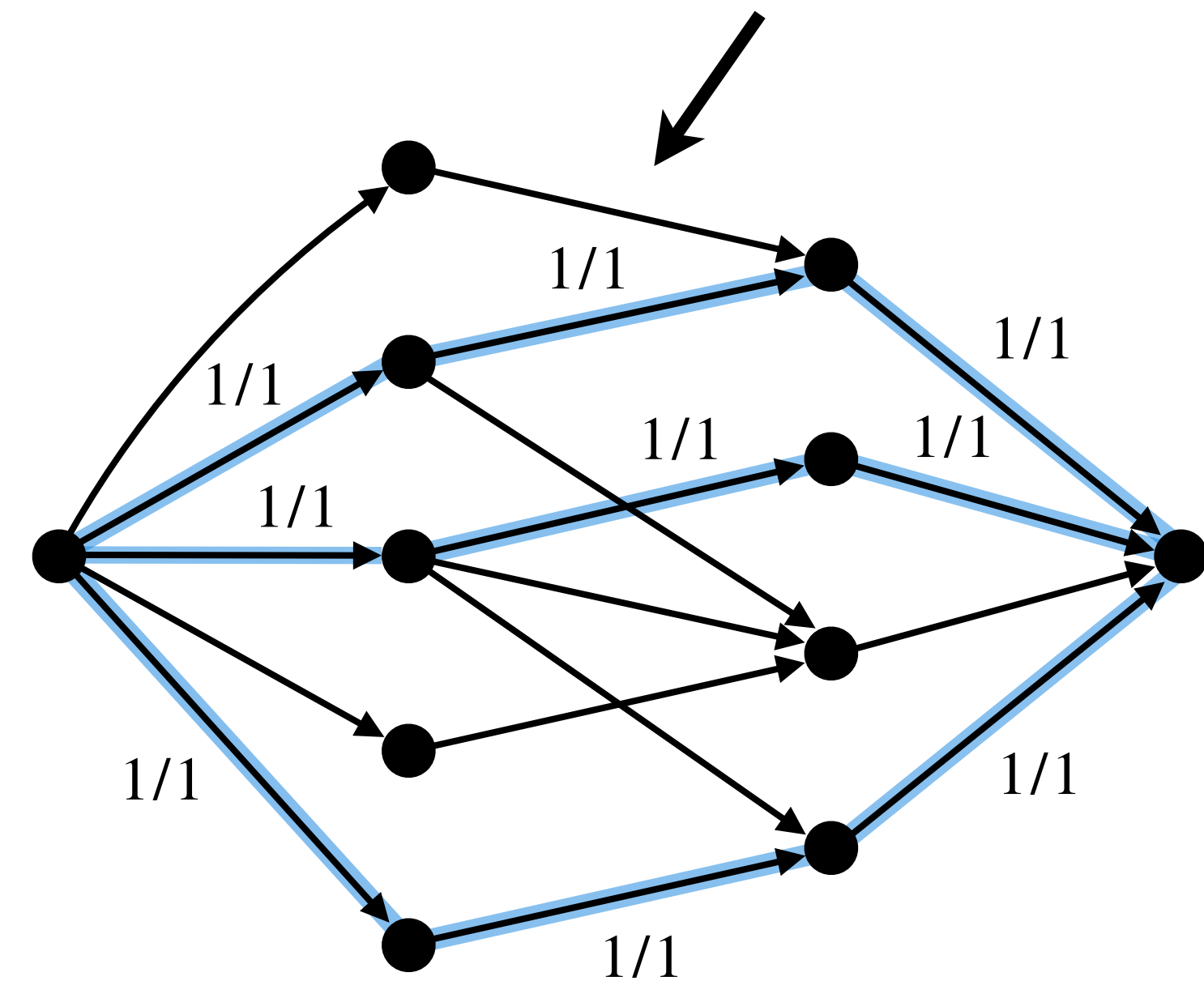
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Solving *Max-Bipartite-Matching* via *Max-flow*



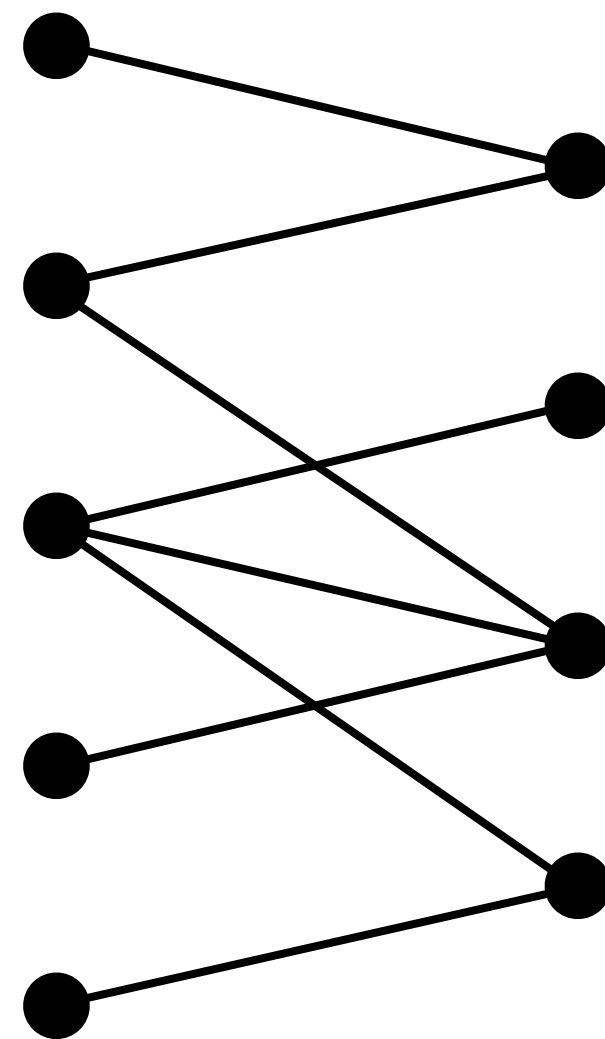
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Integer-valued flow produced by Ford-Fulkerson



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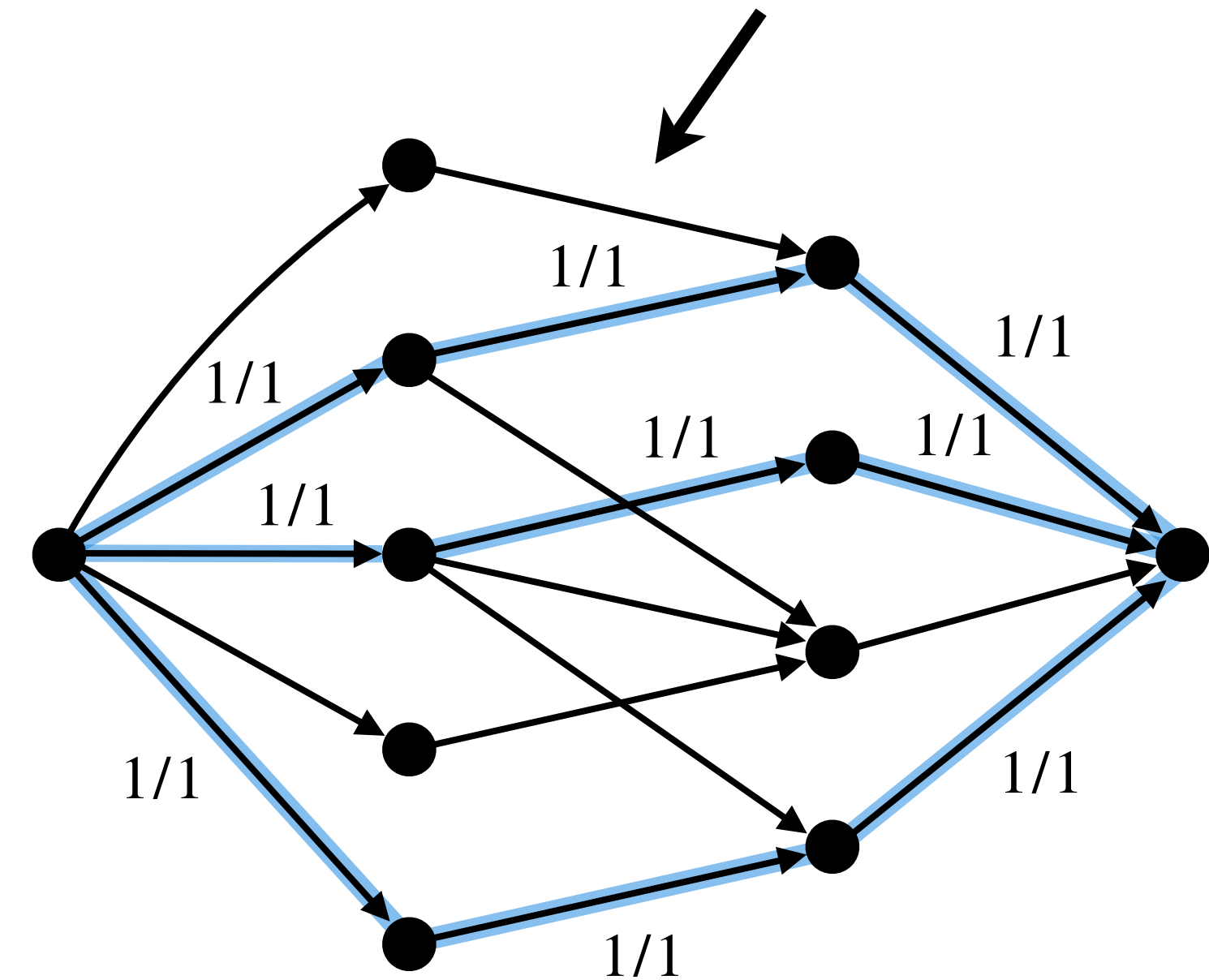


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Corresponding matching

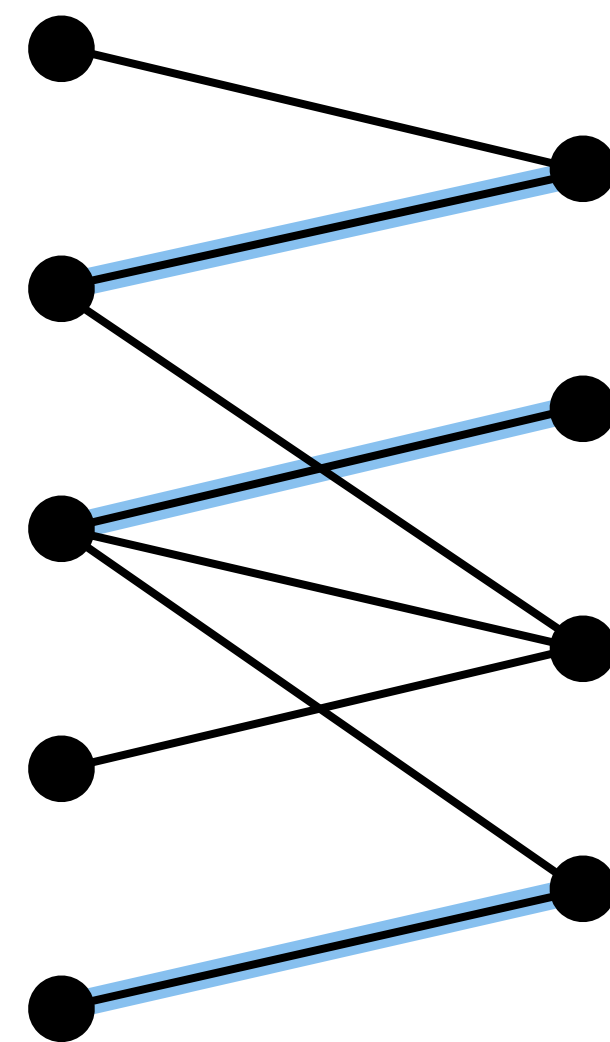


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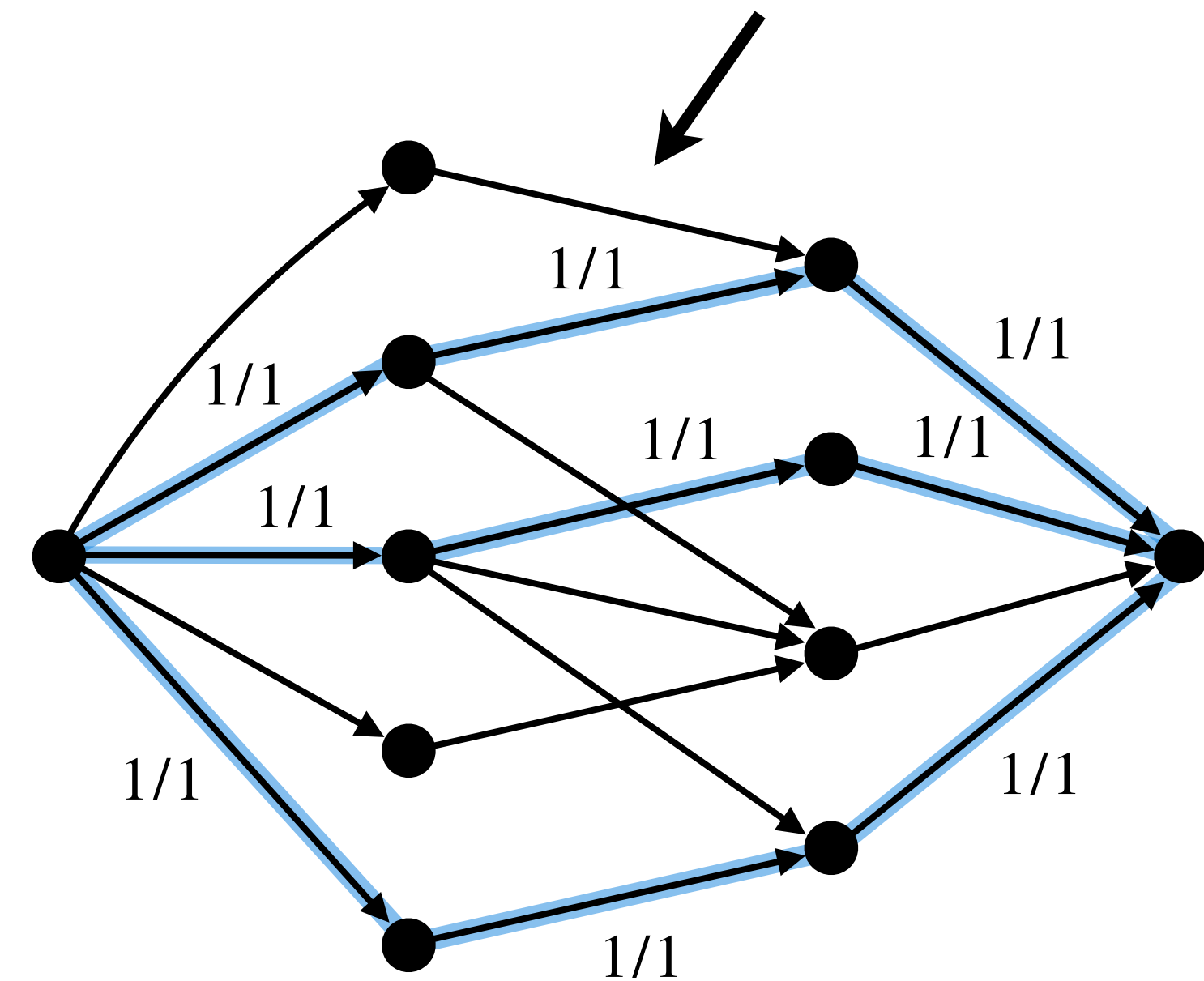


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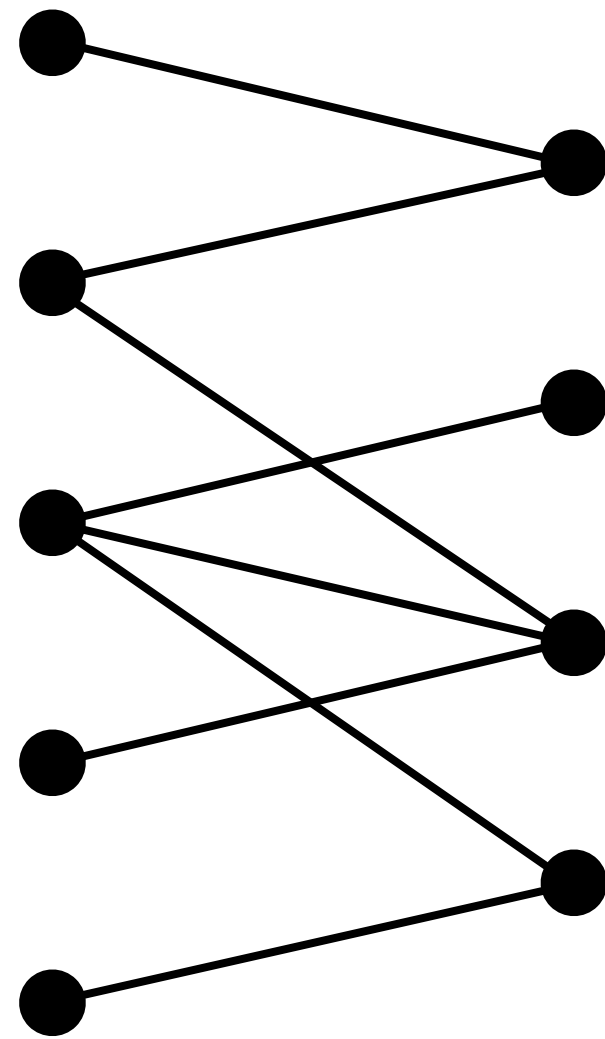
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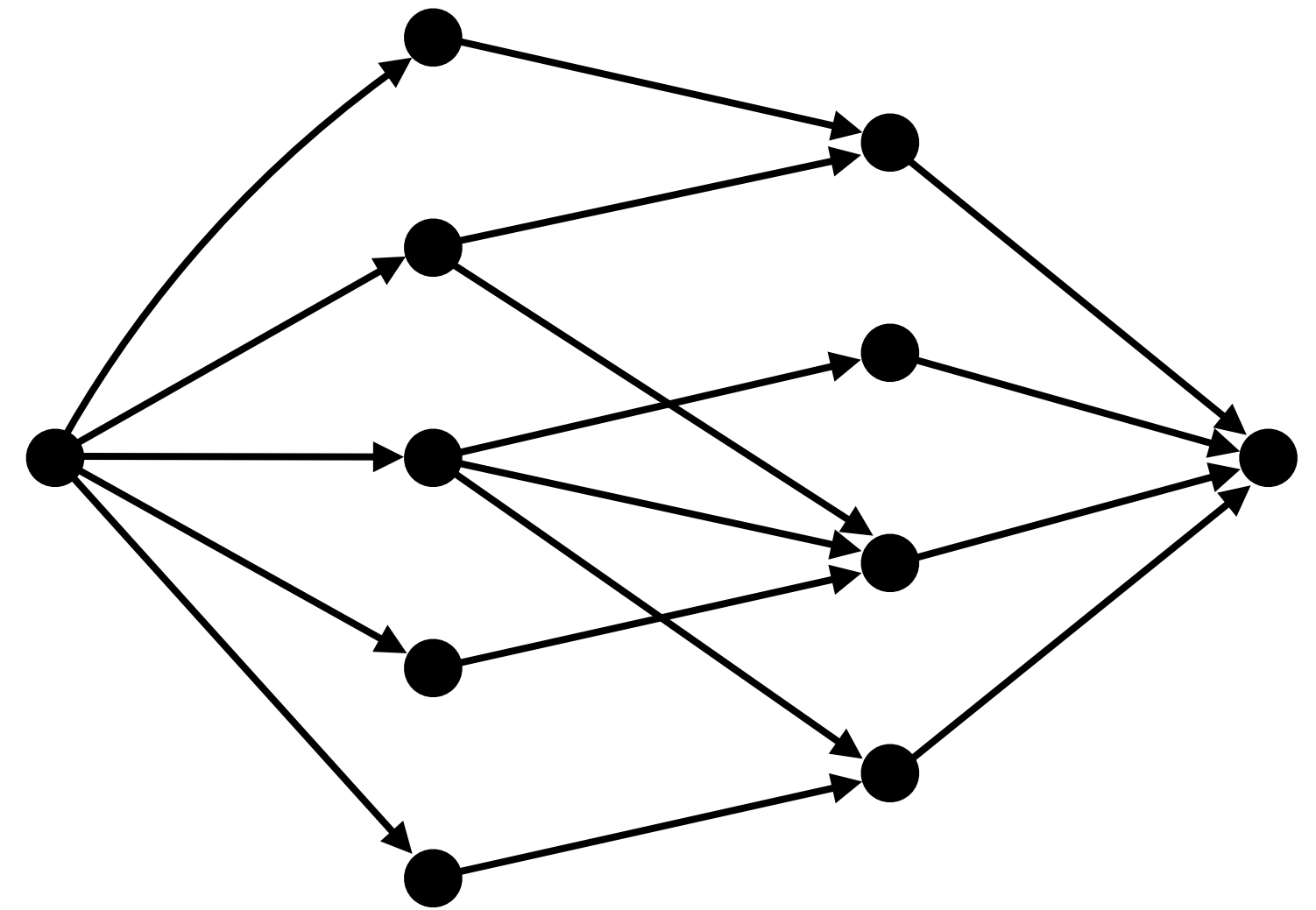
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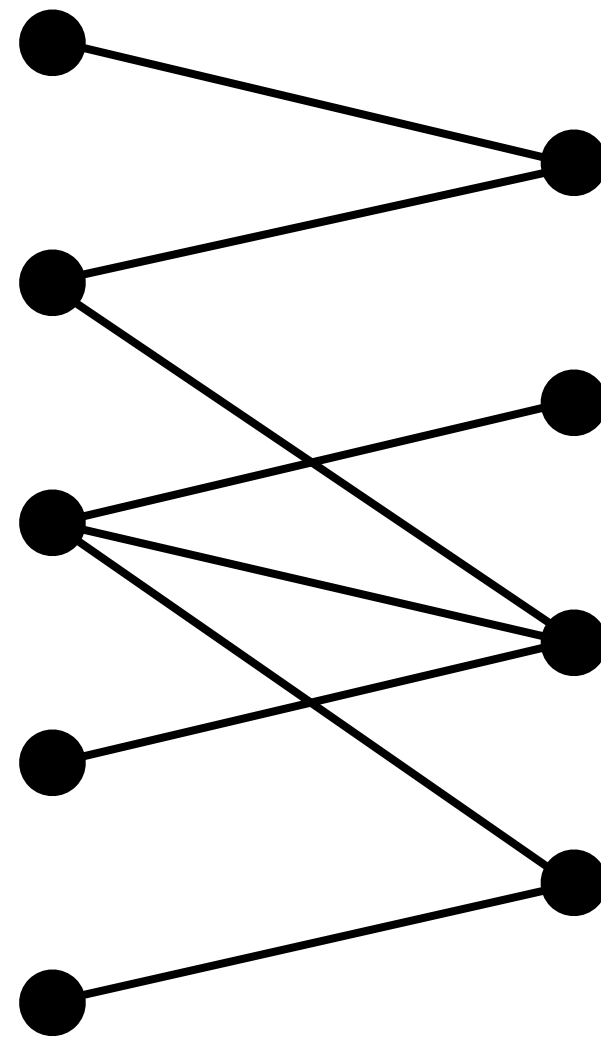


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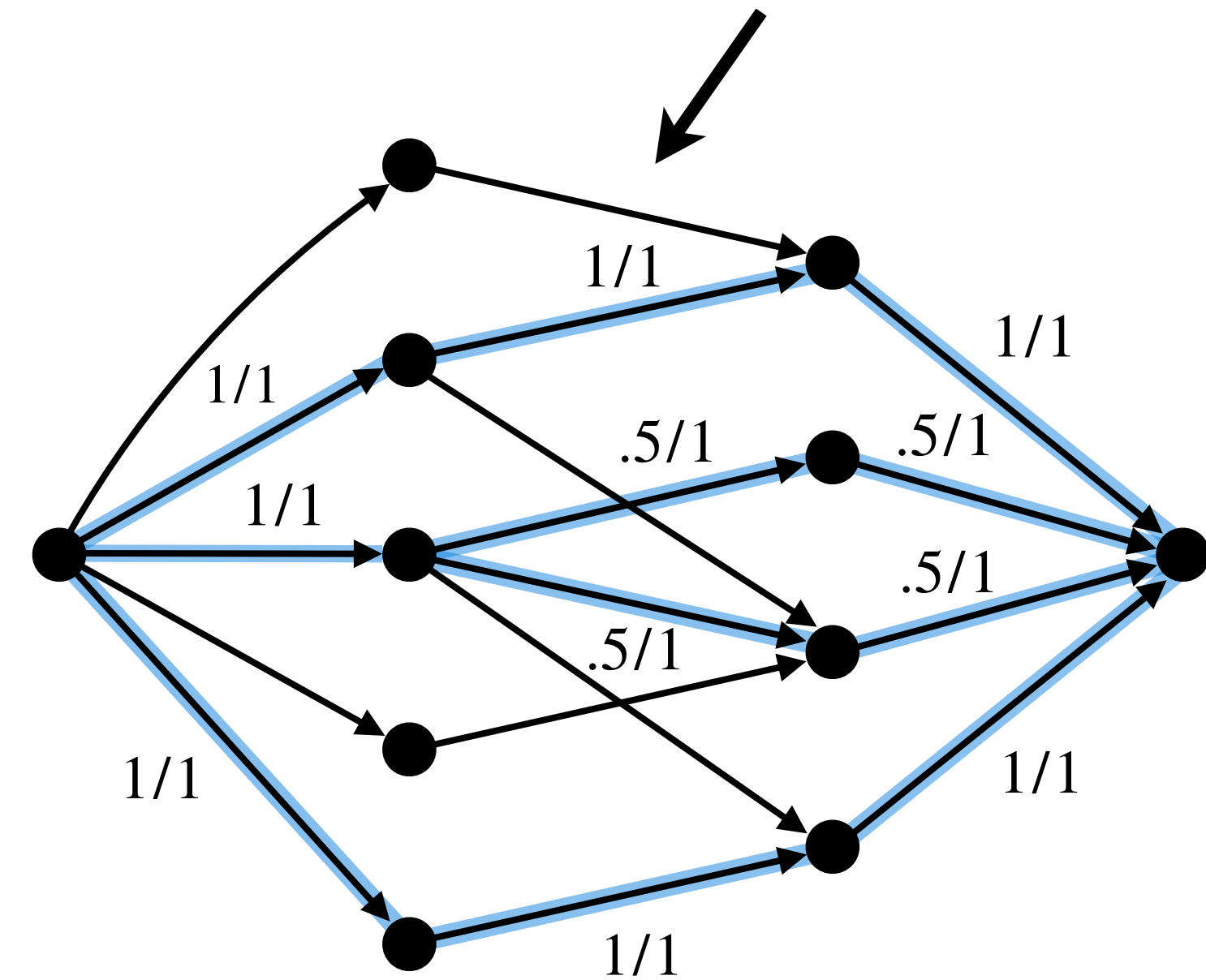
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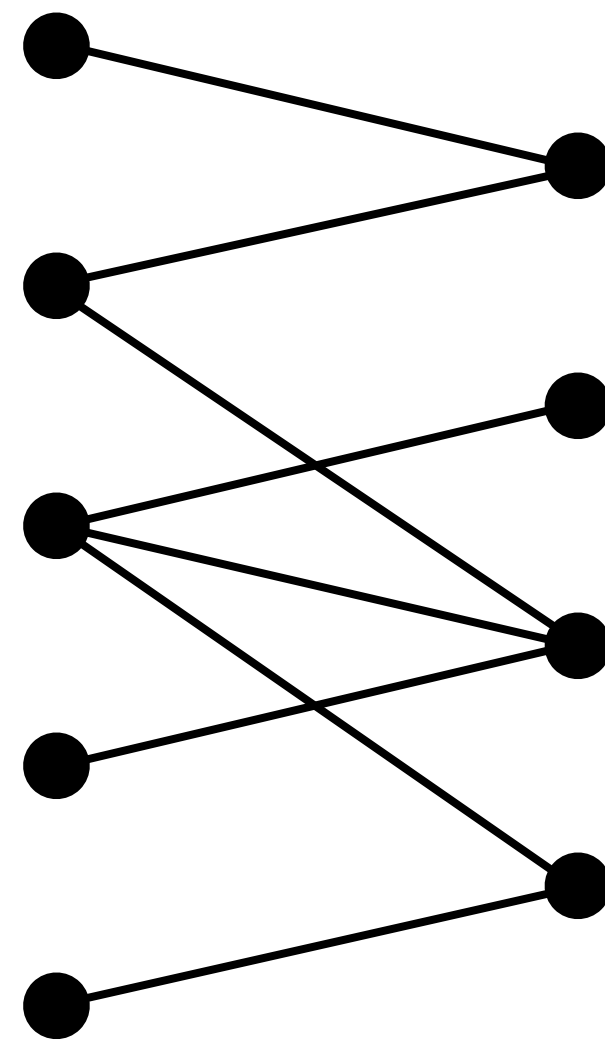
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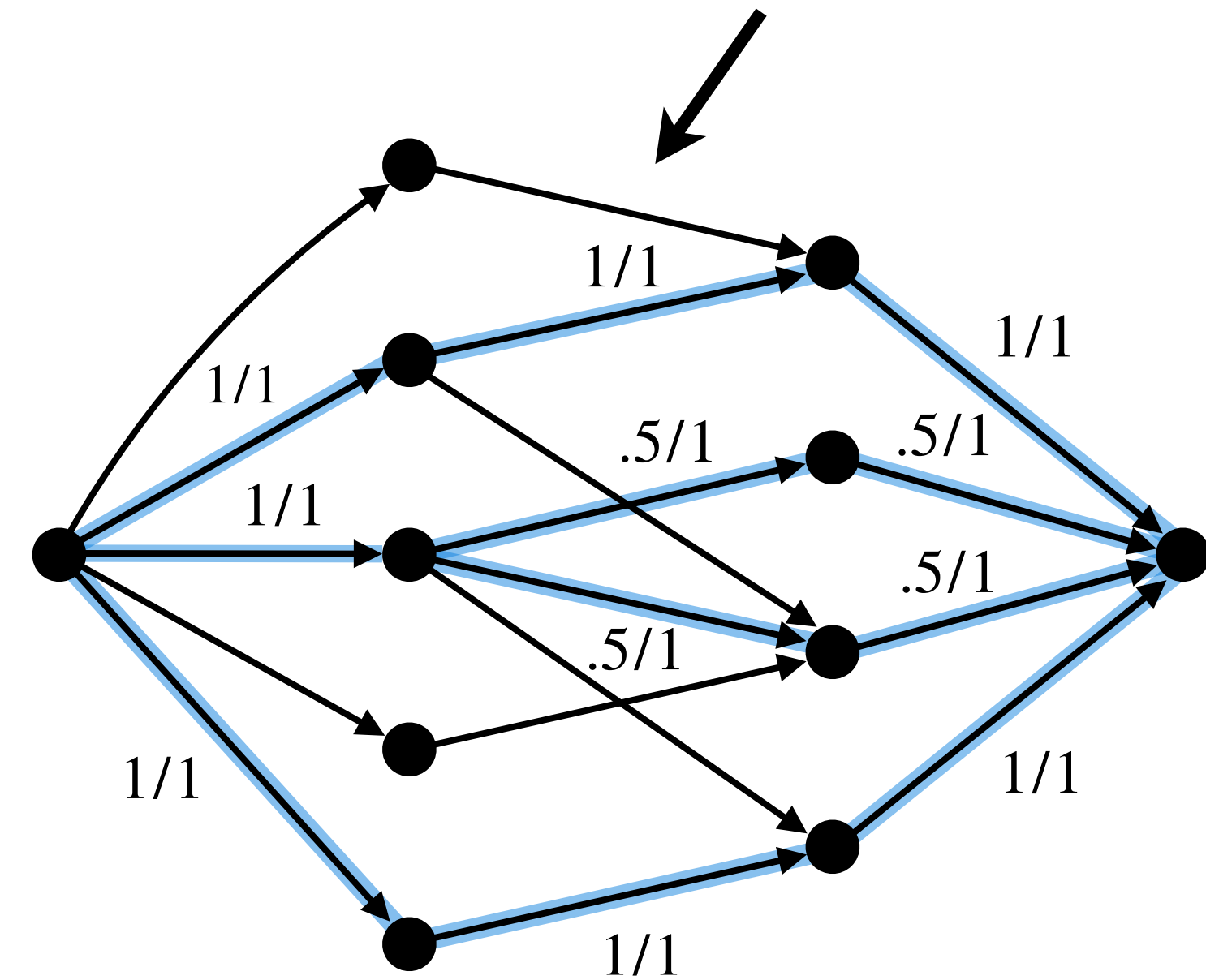


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Time Complexity: $O(|E| \cdot |f^*|)$, when $|E| \geq |V|$.